GEK-36445

INSTRUCTIONS

3S7513TC300A12

ELECTROHYDRAULIC CONTROL

FOR

BOILER FEEDPUMP TURBINE

GENERAL ELECTRIC

OCTOBER, 1973

TABLE OF CONTENTS

	PAGE
INTRODUCTION	3
RECEIVING, HANDLING AND STORAGE	3
DESCRIPTION	3
Speed Control	3
Speed Indication	4
Speed Reference or Set Point	5
High Pressure/Low Pressure Stop Valve Indication	5
Low Speed Switch	5
Power Supplies	6
Series Regulator	6
MONITORING	6
Power Supply Alarm	6
Ground Alarm (24 VDC)	6
HSS/LSS Alarm	7
Cable Check Alarm	7
Lamp Test	7
Test Switch	7
INSTALLATION AND ADJUSTMENT	7
OPERATION	7
MAINTENANCE	8
TROUBLESHOOTING	8
RENEWAL PARTS	8
METER RELAY LAMP REPLACEMENT	8
Pt. No. 4868K32P001 Lamp	8
Pt. No. 4872K12G700 Lamp & Holder	8
Reassembly	9
DRAWING LIST	10

These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company.

3S7513TC300A12

ELECTROHYDRAULIC CONTROL FOR BOILER FEEDPUMP TURBINE

INTRODUCTION

The electronic portion of an Electro-hydraulic Control system for a boiler feedpump turbine drive is described in these instructions.

Any adjustment or maintenance of this equipment should be performed either by qualified turbine personnel or supervised by such personnel.

This equipment is an electronic speed governor whose speed set point can be varied by an automatic boiler controller or by an operator from a remote control unit. Electronic redundancy and mechanical overspeed trip are used to provide a high degree of reliability and safety.

The elementary diagram and alignment diagram for this unit are included in these instructions. Since both the number and arrangement of auxiliary components vary with particular application, it is necessary to refer to the drawings furnished with each equipment to determine the correct connections and auxiliary components furnished in each instance.

RECEIVING, HANDLING AND STORAGE

Receiving and Handling

Immediately upon receipt, the equipment should be carefully unpacked to avoid damaging the apparatus. Particular care should be exercised to prevent small parts being mislaid or thrown away in the packing material.

As soon as the equipment is unpacked it should be examined for any damage that might have been sustained in transit. If injury or rough handling is evident, a damage claim should be filed immediately with the transportation company and the nearest General Electric Sales Office should be notified promptly.

Storage

If the equipment is not to be used as soon as it is unpacked, it should be stored in a clean dry place and protected from accidental damage. Particular care should be exercised to avoid storing the equipment in locations where construction work is in progress. If the equipment is to be placed in long term storage, the interior cabinet temperature should be 5°C above ambient.

DESCRIPTION

Speed Control

The speed control loop will be described first. Starting at zone B6 and B7 of the Alignment Drawing, the coil of the electrohydraulic servo valve receives a low power signal from the electronic unit and controls the flow of oil to a double acting hydraulic cylinder which positions the turbine steam control valve. Turbine speed is sensed by three magnetic sensors mounted near the periphery of a shaft driven gear. Pulses whose frequency is directly proportional to turbine speed are generated as each gear tooth passes a sensor. Primary and secondary speed channels in the electronic control produce a DC voltage proportional to the frequency of the sensor outputs. Outputs of two speed channels are fed into high value gate number

one, which rejects the lower of the two speed signals and passes on the higher valued speed signal. This redundancy permits one channel to fail without significantly affecting turbine speed. The secondary channel is adjusted for an output, 1% less than the primary channel. It is therefore uncontrolling, except in case of failure of the primary channel or if a "Backup Amplifier Test" switch in the cabinet is actuated which increases the output of the secondary channel to 1% greater than the primary channel. A test meter in the cabinet can be used to determine the controlling channel.

The speed signal from high value gate one is summed with the speed set point to obtain a speed error. See zone B3 of the alignment diagram. The speed error is passed through a stability compensation and error amplifier, which includes a regulation adjustment to obtain a valve lift demand signal. This is summed with a valve position signal to obtain a valve position error which is passed through the servo amplifier to the servo valve. Increasing the speed set point increases valve lift demand, valve position error, and valve opening. Speed will increase until the rise matches the set point increase.

The valve position signal is obtained from a Linear Variable Differential Transformer (LVDT) whose core is positioned by the control valve. Three kilohertz excitation for the LVDT is generated in the electronic cabinet. A demodulator in the cabinet produces a DC signal proportional to the AC output of the LVDT which varies with core position.

A control valve closing bias is applied directly to the servo coil of the control valve in the tripped condition. Before the EHC can reset the feedpump turbine, a valve closing signal from the servo amplifier (2SA) must be present to operate the Close Signal Check (CSC) relay and remove the bias signal that had been previously applied directly to the servo valve coil. Relay circuitry also requires that the electrical connectors must be plugged into the servo valve, the LVDT, and the speed sensors before starting. The LVDT signal is biased to close the control valve if the LVDT connector is removed with the turbine running. The servo valve contains a 5% mechanical bias which will close the control valve in the absence of an electrical signal.

Speed Indication

At zone B9 of the alignment diagram, a third magnetic sensor and circuitry identical to the previously described speed signal is used for speed indication. An analog signal is obtained for speed indication on the EHC Remote Control Panel, local panel, and customer use.

A clamp signal from the 10% speed voltage comparator holds the turbine to less than 10% speed if both the primary and secondary speed channels show less than 10% speed.

At 105% speed a voltage comparator operates a relay that unlocks the lock out valve in the emergency trip exerciser scheme. At 110% speed another voltage comparator operates the 110SX relay to indicate that 110% speed has been reached. A contact of this relay is provided for the customer.

Speed Reference or Set Point

Motor driven LVDTs are used to generate the manual and automatic speed references. The core position is varied by a two-phase stepping motor which moves in discrete steps and locks in position when not running. Raise and Lower pushbuttons on the operator's remote control panel cause the manual reference to run from the LSS stop to the HSS stop in 80 to 90 seconds. Raise Fast and Lower Fast pushbuttons will run the manual reference from stop to stop in approximately 10 seconds. The automatic speed reference circuits contain interface circuitry for compatibility with a specific boiler controller. The auto speed reference is adjusted to run from the LSS to the HSS in 4 seconds. The automatic and manual speed references are fed into a low value gate which rejects the higher reference and passes on the lower reference to control turbine speed. A relay operated from this gate indicates which of the two references is controlling.

An overspeed test can be performed after first running both manual and automatic references to the high speed stop. The overspeed test pushbutton is then actuated causing an additional speed reference to be generated which increases linearly with time at approximately 12% speed per minute. This is added to both manual and automatic references. If turbine speed rises to 120% of rated, a speed signal will override the ramp generator and prevent further speed increase. Upon release of the test pushbutton, the reference ramps back to normal at the same rate as it increased.

High Pressure/Low Pressure Stop Valve Indication

The EHC cabinet contains circuitry for driving HPSV and LPSV position indicators located on the control panel supplied with the EHC cabinet. The signals for these position indicators are obtained from LVDTs mounted on their respective stop valves. This function is for position indication only and is not part of the control circuitry for the feedpump. Refer to zone C7, C8 of the alignment drawing.

Low Speed Switch

An electronic Low Speed Switch (LSS) is provided in the EHC cabinet to automatically, if selected, engage the turning gear on decreasing turbine speed less than 2 rpm.

If at turbine speeds greater than 200 rpm, pulses are lost from the LSPU shown at zone A9 of the alignment drawing, the LSS will prevent the LSSX relay from energizing and prevent automatic engagement of the turning gear by the EHC.

The LSS provides an alarm under the following conditions:

1. If the turbine is at standstill, 0 rpm, and then is increased in speed and pulses are not detected from the LSPU, the LSS 2K contacts will energize the LSS Alarm X relay when 200 rpm is reached.

2. If the pick-up pulses are detected from the LSPU but lost before 200 rpm is reached, the LSS 2K contacts will energize the LSS Alarm X relay when 200 rpm is reached.

3. If pick-up pulses are lost from the LSPU at turbine speeds greater than 200 rpm, the LSS 2K contacts immediately energize the LSS Alarm X relay.

4. Once the LSS Alarm X relay is energized, the automatic engagement of the turning gear is prevented

until pulses from the LSPU are obtained.

Power Supplies

Two redundant power supplies, referred to as Line A and Line B, are provided. They are the full wave rectifier, capacitor filtered variety. Each receives its 120 VAC from the customer's power.

The 120 VAC is fed to a transformer that has three secondary windings. The output of each of the rectified, filtered supplies is connected to three identical series regulators that are adjusted for 30 VDC, 24 VDC and -22 VDC.

The outputs of the series regulators on the Line A supply are adjusted slightly higher so they are carrying load and the Line B supply is the backup power supply.

Series Regulator

Operation of the series regulators is described as follows. Output of the voltage sensing bridge consisting of 15R, 16R, 17R, 1P and 2ZD is fed to 7Q and 8Q. An increase in output voltage raises the voltage on 8Q, increasing its conduction and lowering conduction of 7Q. Decreased current in 7Q decreases current in 4Q, 5Q, and 6Q dropping the output voltage until balance is restored. Excessive current output through 10R turns 3Q on, 2Q off, and 1Q on, grounding the base of 7Q. With 7Q off, series regulators 4Q, 5Q, and 6Q are off and the output goes to zero. When 1C discharges, 3Q is turned off and output is restored. The circuit will oscillate between off and on until the cause of the high current is removed.

MONITORING

Located in the front of the cabinet are two meter panels that contain several alarms, meters, indicating lights and test meter that will allow checking or monitoring of the power supplies and most amplifiers in the cabinet.

Power Supply Alarm

For example, an alarm will occur if the 30 volt Line A power supply should exceed the 95% or 105% set point of the meter relay in the cabinet. There will be three lights that will be illuminated. First, the Power lamp on the control panel supplied with the cabinet will change from a steady glow to a flashing indication, the Alarm lamp in the cabinet will be illuminated and the illuminated pushbutton above the Line A 30 volt meter will indicate either a high or low voltage.

The three Alarm lamps can be cleared after the malfunction is corrected by depressing the lighted pushbutton above the Line A 30 volt meter. Identical functions are provided for both the Line A and Line B 30 V, 24 V, and -22 V power supplies and 3 KHz 6 V RMS power supply in the EHC cabinet. If any of these power supplies are not at their proper voltage level when the 120 V 60 Hz is applied, startup of the feedpump will be prevented.

Ground Alarm (24 VDC)

If a ground were to occur on the positive or negative side of either the Line A or Line B 24 volt power supply, the Power lamp on the control panel would flash,

the +24 V GND/-24 V GND lamp in the cabinet would be illuminated, and the Alarm lamp in the cabinet would be illuminated. These lamps will be cleared when the ground is removed. Depressing the +24, -24 V lamp in the cabinet is not required to clear the lamp.

HSS/LSS Alarm

HSS and LSS lamp indication is provided both in the cabinet and on the control panel for both the Manual and Auto Speed reference travel limits. Lamp indication for the Auto Speed reference travel limits is provided in the cabinet only. Contacts of the AHSS and ALSS relays are provided for customer AHSS and ALSS travel indication. These lamps will be cleared when the respective speed reference is driven off the limit.

Cable Check Alarm

The Cable Check will prevent startup of the feedpump if the cables are not properly connected. The Power lamp on the control panel will flash, the Cable Check lamp in the cabinet will be illuminated, and the Alarm lamp in the cabinet will be illuminated until the cable is properly reconnected. If a cable becomes disconnected during operation, just an alarm will occur.

Lamp Test

A lamp test can be performed on the indicating lamps in the cabinet by depressing the Alarm pushbutton. There aren't any lamps in the Backup Speed Amplifier Test pushbutton.

Depressing the Power pushbutton on the control panel will test the lamps on the control panel. There aren't any lamps in the Lower Fast, Lower, Raise, Raise Fast pushbuttons.

Test Switch

The amount of current being supplied by each series regulator can be observed by the Test Switch and meter in the cabinet. When measuring series regulator current the meter is calibrated to read approximately 5 amps full scale.

The voltage output of amplifiers 1 thru 12 can be observed by using the Test Switch and meter. When measuring amplifier outputs the meter is calibrated to read 10 volts full scale. The amplifier numbers and function are shown on the alignment drawing and elementary diagram.

INSTALLATION AND ADJUSTMENT

The electronic control cabinet should preferably be mounted in a clean dry air-conditioned environment. Ambient temperature should not exceed 104°F unless otherwise noted on the outline drawing for a specific unit.

Equipment adjustment and the required test equipment are covered in detail in an alignment diagram and alignment instructions furnished with each unit.

The location of sub assemblies within the cabinet is shown on the elementary diagram furnished with each control. Printed circuit board locations are shown on sheets 1 and 2 of the elementary diagram.

OPERATION

Refer to the instruction book for the turbine.

MAINTENANCE

No periodic maintenance is required. Outage time due to component failure is best minimized by maintaining a complete set of spare parts. Elementary diagrams of the printed circuit boards are included in this instruction book. The assembly drawings showing component location on the printed circuit boards are furnished with each order except where component identifications are marked on the boards, in which case assembly drawings are not furnished.

TROUBLESHOOTING

Most troubleshooting problems can be quickly isolated to a particular section of the control by use of the built-in test meter, the meter relays, and alarm lights. These used in conjunction with the alignment diagram indicate the operating condition of each section. An oscilloscope is recommended for pinpointing a defective component or components. Each printed circuit board can be removed from the rack for servicing without disconnection due to the use of terminal boards.

RENEWAL PARTS

When ordering renewal parts, address the nearest Apparatus Sales Office of General Electric Company and supply the following information.

a. Catalog number stamped on the part and a complete description of the part, including its use and location.

b. Complete nameplate data appearing on the assembly of which the part is a component.

c. If possible, data on original order on which equipment was first supplied, including all numerical references.

Since operation of a steam turbine-generator is dependent upon operation of the feedpump control, it is suggested that an adequate stock of spare parts be carried on hand to minimize outage time due to component failure.

METER RELAY LAMP REPLACEMENT

Type 195 Meter Relays have been made with two versions of lamps; one without leads attached to the lamp (GE Pt. No. 4868K32P001) and the other with leads attached to the lamp and lamp holder assembly (GE Pt. No. 4872K12G700). These replacement parts can be ordered directly from: the nearest General Electric Sales Office or Utility & Process Automation Products Department, General Electric Company, 40 Federal Street, West Lynn, Mass. 01910.

In the event of lamp failure, the applicable lamp or lamp assembly can be replaced when the control unit is detached from the rear of the indicator set-point unit. Then, the lamp-access plate (see Figure) must be removed (screws are beneath adhesive labels on some models) to expose the lamp holder which can be pulled out.

Pt. No. 4868K32P001 Lamp

When installing the bayonet-type Pt. No. 4868K32P001 (without leads), twist the lamp in the holder assembly until the lamp filament will be in the horizontal position when the lamp holder is replaced in the indicator set-point unit.

Pt. No. 4872K12G700 Lamp & Holder

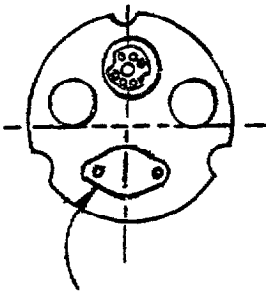
The Pt. No. 4872K12G700 lamp has been fixed in the lamp holder at the factory such that the filament position will be in the correct position when the keyed assembly

is inserted in the indicator set-point unit. Make sure the lamp-lead clips are engaged by the screws when the lamp access cover is replaced.

Note: If relay operation is not within ±2% of set-point index, the lamp holder position must be rotated 180°.

Reassembly

After the new lamp or lamp assembly has been installed, make sure the lamp access plate is replaced (and screws are tight) before control unit is reassembled on rear of indicator set-point unit.



LAMP ACCESS PLATE

DRAWING LIST

Description	Drawing Number
Diode Board, 1DB	44A300243
Reference Low Value Gate, 2LVG	44A300281
Compensation and Regulation, 1CR	44A301400
Overspeed Integrator, 1OSI	44A302692
Signal Low Value Gate, 1LVG	44A304763
Alignment Instructions	44A306316
Series Regulator Board #1, 1REC	44A306392
Renewal Parts	44A335416
Cable Wiring Diagram, Signal & Low Voltage, 1CO	44B300423
Control Panel Outline	44B306127
Low Speed Switch Power Supply, LPS	44B306144
Cable Wiring Diagram Power, 2CO & 3CO	44B306148
Series Regulator Board #2, 1SR	44B306197
Detector Indicator, 1DI	44C300609
Operational Amplifier, 10A	44C300610
Differential Transformer Drive Manual Speed Ref.	44C300882
Analog Output, 1AO2	44C301216
Analog Output, 3AO2	44C301622
Frequency to Voltage Converter, 2FVC	44C301682
Clock and Stepping Motor Drive, CL&SMD	44C303567
Servo Amplifier, 2SA	44C304444
3 kHz Oscillator, 6 OCS	44C304462
Differential Transformer Drive Auto Speed Ref.	44C304865
Signal Converter, 3SC	44C304898
Voltage Comparator, 1VC	44C305259
Elementary Drawing, Master	44C306404
Connection Drawing	44C306405
Low Speed Switch, LSS	44C306444
Boiler Control Interface, 7BCI	44C306492
Line A & Line B Power Supply	44C306526
Cabinet Outline	44D300043
Alignment Drawing	44D305957
Control Panel Connection Diagram	44D305974
Relay Panel Connection Diagram	44D305975
Summary	44X303674

GENERAL ELECTRIC

44A300243
CONT ON SHEET SH NO.

REV NO. 0

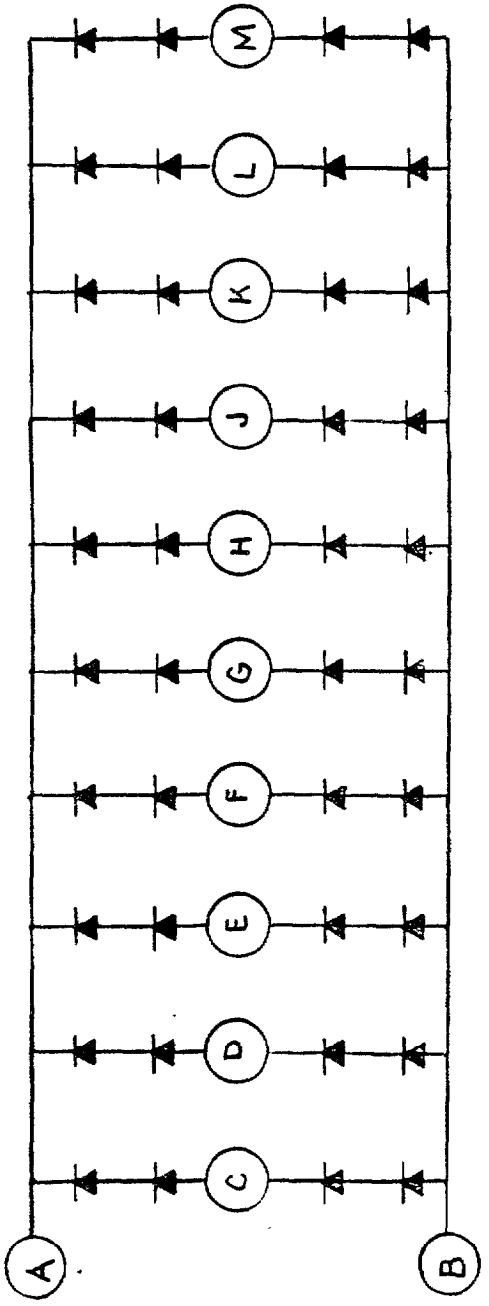
CONT ON SHEET SH NO.

TITLE

DIODE BOARD IDB

FIRST MADE FOR 44D236370-G01

REVISIONS



ALL DIODES 200 MA; 200 V.

11

DFB 12P

PRINTS TO

MADE BY R. Chambers 1/15/68

ISSUED MAR 12 1968

APPROVALS MBC

SPECIALTY CONTROL DIV OR DEPT.

WAYNESBORO, VA. LOCATION

44A300243
CONT ON SHEET SH NO.

CODE IDENT

FF-803-WF (2-67)
PRINTED IN U.S.A.

2

GENERAL ELECTRIC

44A300281

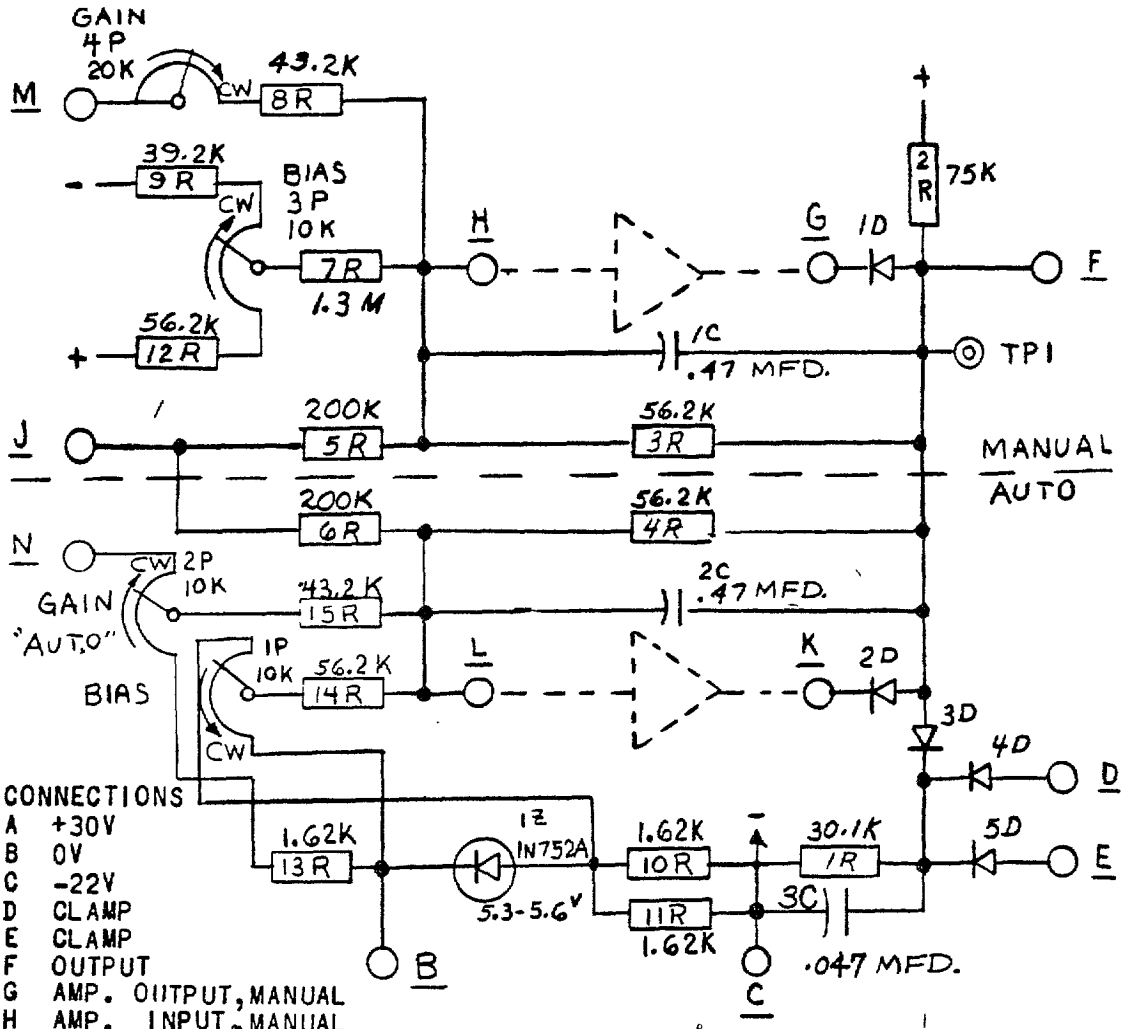
REV NO. (1) C. KENNEDY

CONT ON SHEET SH NO.

TITLE

REFERENCE LOW VALUE GATE (2LVG)

FIRST MADE FOR 44D236378-G01



CONNECTIONS

A	+30V	M	MANUAL INPUT
B	0V	N	AUTO INPUT (SIGNAL CONVERTER)
C	-22V		
D	CLAMP		
E	CLAMP		
F	OUTPUT		
G	AMP. OUTPUT, MANUAL		
H	AMP. INPUT, MANUAL		
J	OVERSPEED TEST INPUT		
K	AMP. OUTPUT, AUTO		
L	AMP. INPUT, AUTO		

CLAMP NOTE: ZERO VOLTS ON D OR E WILL HOLD OUTPUT (F) AT ZERO.

ADJUSTMENT:

1: WITH AUTO INPUT AT HSS AND MANUAL AT LSS, ADJUST BIAS FOR ZERO OUTPUT AT F.

2: WITH BOTH GAINS MAX. (CW) AND INPUTS AT HSS, ADJUST MANUAL GAIN FOR 5 VOLTS OUTPUT.

3: DECREASE AUTO GAIN UNTIL OUTPUT OF THE ASSOCIATED AMPLIFIER SWITCHES NEGATIVE TO POSITIVE.

4: AUTO BIAS IS SET CCW INITIALLY. FINAL SETTING PER APPLICATION REQUIREMENT.

REVISIONS

1	CHANGED 5R, 6R & 8R VALUES BUKOSKEY MAY 5 1969
2	C.A. Lewis OCT 19 1970
3	Dale Mod 11, 1971
11	
12P DFB	
FG-W	

PRINTS TO

MADE BY M Chambers Mod 11 1968

ISSUED APR 17 1968

APPROVALS RW

SPECIALTY CONTROL DIV OR DEPT

WAYNESBORO, VA LOCATION

44A300281

CONT ON SHEET SH NO.

FF-003-WF (8-66)
PRINTED IN U.S.A.

CODE IDENT NO

GENERAL ELECTRIC

44A301400 CONT ON SHEET SH NO.

REV NO.

44A301400 CONT ON SHEET SH NO.

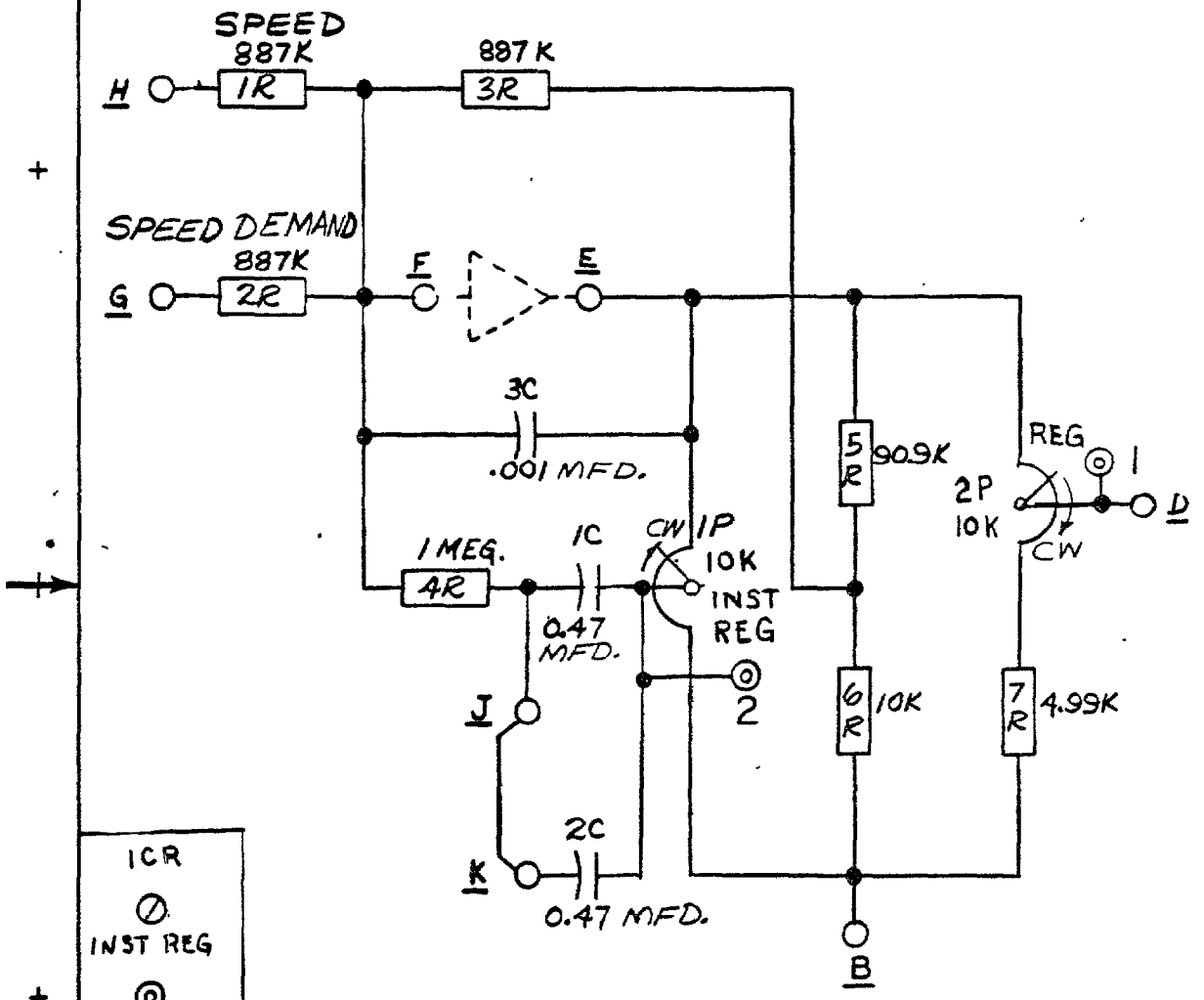
TITLE

COMPENSATION & REGULATION (ICR)

FIRST MADE FOR 44D236388-G01

REVISIONS

1	R.M. BUKOSKEY	APR 1 1969	
2	E.C. Allen	Nov. 17, 1971	
3	[illegible]	Aug. 28, 1972	
4	[illegible]	[illegible]	



ICR

INST REG

REG

11

PRINTS TO

MADE BY R.M. Bukoskey

ISSUED FEB 18 1969

APPROVALS R.W.

SPECIALTY CONTROL DIV OR DEPT.

WAYNESBORO, VA. LOCATION

44A301400 CONT ON SHEET SH NO.

CODE IDENT NO.

FF-803-WP (6-67)
PRINTED IN U.S.A.

GENERAL ELECTRIC

44A302692
CONT ON SHEET SH NO.

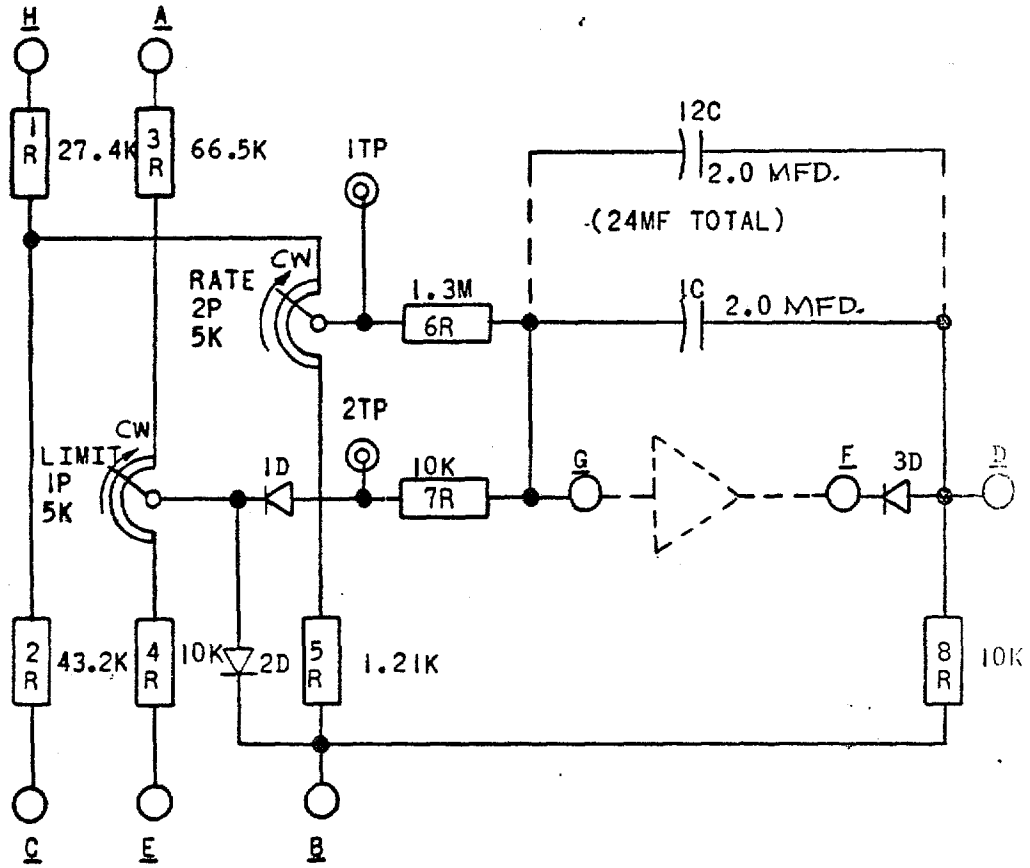
REV NO. 1 C. KENNEDY

44A302692
CONT ON SHEET SH NO.

TITLE
OVERSPEED INTEGRATOR
IØSI

FIRST MADE FOR 44D236373-G02

REVISIONS



CONNECTIONS

A + 30V
B 0V
C -22V
D OUTPUT
E SPEED SIGNAL
F AMPLIFIER OUTPUT
G AMPLIFIER INPUT
H POS. SUPPLY THROUGH RELAY
J N.C.

MADE BY L. Templeton 3-16-70
ISSUED APR 8 1970

APPROVALS R. Culver

Semiconductor Devices DIV OR DEPT.
Waynesboro, Va. LOCATION

44A302
CONT ON SHEET SH NO.

FF-308-WF (10-67)
PRINTED IN U.S.A.

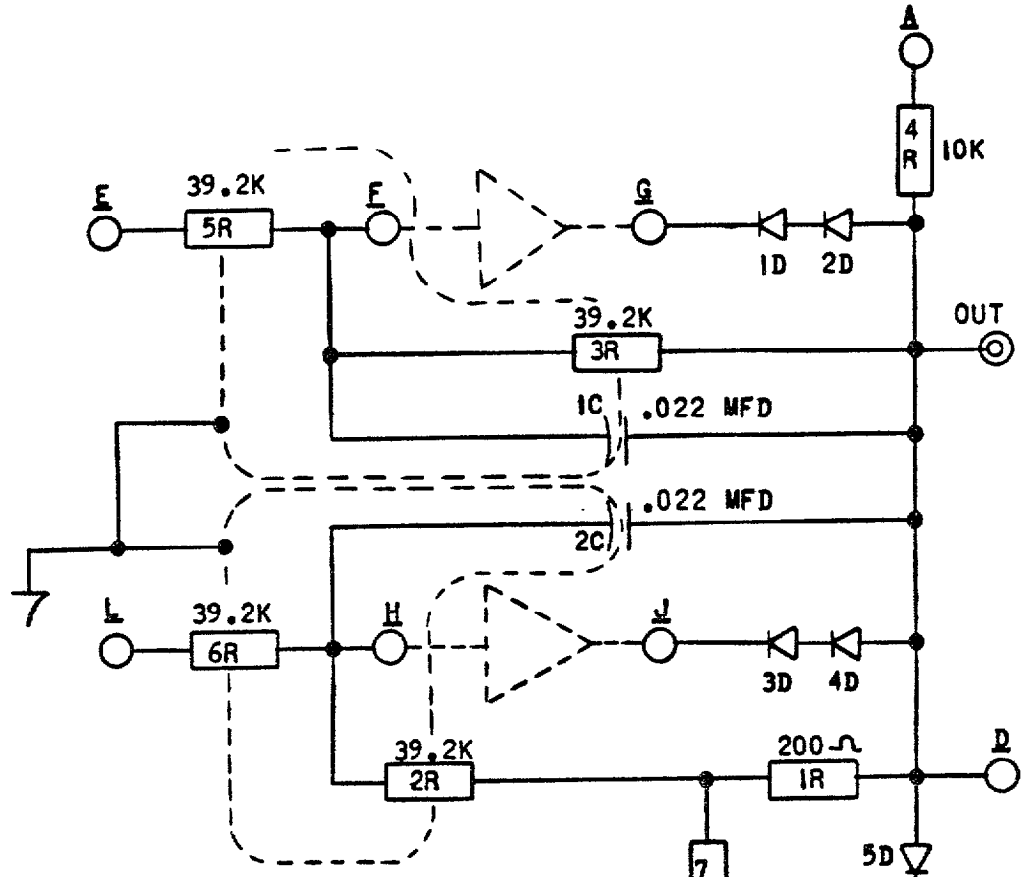
2-1

GENERAL ELECTRIC

44A304763 CONT ON SHEET SH NO. ILVG

REV NO.	TITLE
44A304763 CONT ON SHEET SH NO.	SIGNAL LOW VALUE GATE FIRST MADE FOR 44D236372-G02

REVISIONS



CONNECTIONS

A - +30V
B - ⏚
C -
D - OUTPUT
E - PRIMARY INPUT
F - PRIMARY AMP INPUT
G - PRIMARY AMP OUTPUT
H - BACKUP AMP INPUT
J - BACKUP AMP OUTPUT
K - BACKUP TEST INPUT
L - BACKUP INPUT

11
12P DFB
FG-W

PRINTS TO

MADE BY	APPROVALS	COMM. & CONT. DEVICES DIV OR DEPT	44A304763
H. Engelken 8-13-71	R. Culver	WAYNESBORO, VA. LOCATION	CONT ON SHEET SH NO.
ISSUED Aug. 16, 1971	D. L. Mazzoli		

FF-603-WF (2-62)
PRINTED IN U.S.A.

-1

REVISIONS

REFERENCE ALIGNMENT DIAGRAM . 44D305957

TEST EQUIPMENT

1. AUDIO OSCILLATOR: HEWLETT PACKARD 201C, 200CD; GENERAL RADIO 1211UC, OR EQUIVALENT. OUTPUT MUST BE 10 VOLTS RMS MINIMUM INTO 600 OHMS.
2. AC/DC DIGITAL VOLTMETER: 0.1 MV TO 100 V MAX.
3. DIGITAL FREQUENCY COUNTER: 6 VOLT RMS SENSITIVITY. 30 TO 5000 HZ.
4. TEN TURN POTENTIOMETER; NOT MORE THAN 5000 OHMS RESISTANCE.
5. TWO INPUT OSCILLOSCOPE, CALIBRATED SWEEP: X1 AND X10 PROBES.
6. OPTIONAL: BRUSH RECORDER SPECIAL TESTS ARE DESIRED.
7. VOLT - OHM - MILLIAMMETER
8. TWO 100K RESISTORS, TWO 1 MFD CAPACITORS (TO ADJUST OSCILLATOR OFFSET)

I. PRELIMINARY

1. CHECK INTERCONNECTION OF CONTROL CABINET, OPERATOR'S CONTROL DEVICES, TURBINE DEVICES, ETC.
2. MECHANICALLY ZERO TEST METER IN CABINET AND OPERATOR'S TACHOMETER AND OTHER INSTRUMENTS.

II. LINE "A" AND LINE "B" POWER SUPPLIES LOC. 12 AND 13

1. SET UP CONNECT 120V, 60 HZ TO LINE "A" POWER SUPPLY TERMINALS T24 AND T25. REMOVE POWER FROM LINE "B" TERMINAL T22 & 23.
2. DC SERIES REGULATORS, ADJUST POWER SUPPLIES AND METER CALIBRATIONS PER CHART. IF THE VOLTAGES LISTED CANNOT BE OBTAINED IT MAY BE NECESSARY TO ADJUST THE TAPS ON TRANSFORMER 1T, LOC. 12 & 13.

LINE "A" SUPPLY	TEST POINTS	ADJUST
+30V	BUS BARS	0308
-22V	BUS BARS	0316
24V	+F LOC. 0322; -G, LOC. 0324	0324

11
12P-DFB
FG-W
12P-FS

PRINTS TO

3. METER RELAY ALARMS VARY ALARM SET POINT OF LINE "A" POWER SUPPLY METER RELAYS TO CHECK ALARM OPERATION. SET ALARMS FOR 95% AND 105%.

4. CONNECT POWER TO LINE "B" POWER SUPPLY BY JUMPERING TERMINAL T24 TO T22 AND T25 TO T23, SO THAT POWER IS APPLIED TO BOTH SUPPLIES.

5. ADJUST THE INDIVIDUAL LINE "B" SERIES REGULATOR BOARDS SO THAT THEY ARE CARRYING ONLY 0.1 AMP AND THE LINE "A" SUPPLIES THE REMAINING LOAD. WHILE NOTING THIS CONDITION IS MAINTAINED MAKE FINAL ADJUSTMENTS IF NECESSARY PER CHART.

SUPPLY	LINE B	LINE A	TESTPOINTS
+30	0310	0308	BUS BAR
-22	0318	0316	BUS BAR
+24	0326	0324	LINE A, + F, 0322; -G, 0324 LINE B, + F, 0328; -G, 0326

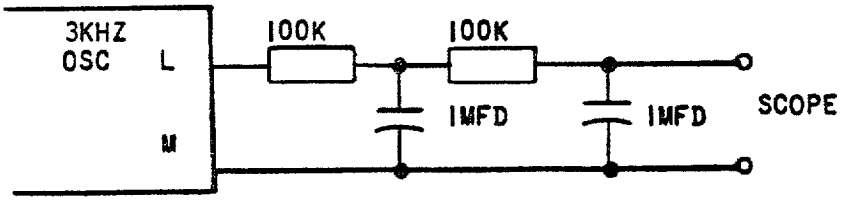
6. TEST AND SET LINE "B" METER RELAY ALARMS AT 95% AND 105%

III. 3KHZ OSCILLATOR, -0217

1. VERIFY OUTPUT FREQUENCY AT 0217 IS 3000HZ. ± 300 HZ. USING COUNTER AT TERMINALS L&M.

2. ADJUST VOLTAGE LEVEL FOR 6.000 ± .006V. SET METER FOR 100%. CHECK METER RELAY ALARMS AND SET AT 95 AND 105%.

3. OFFSET ADJUSTMENT, CONNECT A FILTER CIRCUIT AS SHOWN OR DIGITAL VOLTMETER DIRECTLY TO L&M.



3.1 REMOVE 3KHZ OSC BD FROM RACK AND ADJUST OFFSET FOR ZERO D.C. VOLTS± 1MV.

IV. SSPU SPEED INDICATION CIRCUITS

1. SET UP, REMOVE LEAD FROM T28 AND T29 AND CONNECT AUDIO OSCILLATOR. SET OSCILLATOR TO SSPU RATED HZ.

REVISIONS
CHG'D. III-3+3.1 D.K. MOYER Sept. 17, 1973
1
11
12P-DF3
FC-W
12P-T3
PRINTS TO

2. FREQUENCY TO VOLTAGE CONVERTER

2FVC LOC. 0224. ADJUST COARSE AND FINE POTS TO OBTAIN ⁄5.00V AT TERMINAL J.

3. VOLTAGE COMPARATORS IVC LOC., 0218, 0219, 0220, 0221. SET EACH VOLTAGE COMPARATOR (IVC) PER CHART ON ALIGNMENT DRAWING. RED LIGHT ON IVC SHOULD BE ON WHEN RELAY ENERGIZED. CHECK EACH IVC SETTING BY SLOWLY VARYING FREQUENCY ABOUT THE OPERATING POINT.

4. TACHOMETER 3AØ2 LOC. 0108 WITH OSCILLATOR AT ZERO FREQUENCY ADJUST 2A BIAS FOR ZERO INDICATION ON REMOTE CONTROL PANEL. SET OSCILLATOR TO RATED SSPU FREQUENCY AND ADJUST GAIN AND I OUT AS REQUIRED FOR CORRECT INDICATION OF TURBINE RPM.

4.1 REPEAT ZERO FREQUENCY AND RATED SPEED SETTINGS AND MAKE FINAL ADJUSTMENTS IF REQUIRED.

5. SECOND TACHOMETER 3AØ2 LOC. 0110 WHERE REQUIRED ADJUST 1A & 2A USING PROCEDURE IN STEPS 4 AND 4.1 ABOVE.

6. REMOVE OSCILLATOR FROM T28 AND T29 AND RECONNECT INCOMING LEADS.

V. SSPU SPEED MEASURING CIRCUITS

1. SET-UP JUMPER T4 TO T7 AND T5 TO T8. CONNECT OSCILLATOR AND COUNTER TO T4 AND T5.

2. FREQUENCY TO VOLTAGE CONVERTERS 2FVC, LOC. 0203 AND 0206 HIGH SPEED GATE ILVG, LOC. 0209.

2.1 PRIMARY AT RATED SHAFT SPEED PICKUP FREQUENCY, ADJUST 2FVC LOC. 0203 FOR -5.000 VOLTS AT OUTPUT TESTPOINT OF ILVG, 0209. (OBSERVE WITH TEST METER IN POSITIONS 6 AND 7 THAT AMPLIFIER 7 IS MORE POSITIVE THAN 6; IF NECESSARY ADJUST 2FVC, 0206, TO OBTAIN THIS CONDITION).

2.2 BACKUP DEPRESS BACKUP SPEED AMPLIFIER TEST SWITCH AND ADJUST 2FVC, 0206, FOR -5.05 VOLTS AT OUTPUT TP, 0209.

3. REMOVE TERMINAL POINT JUMPERS AND OSCILLATOR AND COUNTER LEADS.

VI. CONTROL VALVE RAM SERVO LOOP

1. CHECK LVDT CORE PER LEVER DIAGRAM.

REVISIONS
1 | CHG'D. LOC. IN III-4+5 D.K. MOYER Sept. 17, 1973

11
12P-DTB²
FG-W
12P-FS²
PRINTS TO

REVISIONS

CHG'D. LOC. IN VI-3.1 D.K. MOYER Sept. 17, 1973

1

2. DETECTOR INDICATOR 1DI, LOC. 0216. WITH CONTROL VALVE FULL OPEN ADJUST NULL OF 1DI FOR 0.00 ± 0.01 VOLTS AT OUT TP. NOTE VOLTAGE READING ON ALIGNMENT DIAGRAM. WITH VALVE IN CLOSED OVERTRAVEL POSITION RECORD OUT TP VOLTAGE ON ALIGN. DIAG. ADJUST ZERO FOR 0 INDICATION OF CONTROL VALVE POSITION ON OPERATOR'S CONTROL CONSOLE. RETURN VALVE TO FULL OPEN POSITION AND ADJUST CALIBRATE FOR 100% POSITION INDICATION.

3. DETERMINE GAIN OF DETECTOR CIRCUIT IN VOLTS PER INCH BY DIVIDING THE DIFFERENCE IN OUT TP VOLTS AT EXTREME VALVE POSITIONS BY THE TOTAL VALVE TRAVEL IN INCHES, AND RECORD THIS FIGURE ON THE ALIGNMENT DIAGRAM.

3.1 ANALOG CONTROL VALVE POSITION SIGNAL

1AØ2 LOC. 0106. WITH VALVE CLOSED ADJUST 1A BIAS FOR ZERO POSITION ON OPERATORS METER.

3.2 WITH VALVE OPEN ADJUST 1A GAIN AND I OUT FOR FULL OPEN INDICATION.

3.3 REPEAT 3.1 AND 3.3 AS NECESSARY.

4. MAKE ELECTRICAL CONNECTIONS TO SERVO VALVE.

5. SERVO AMPLIFIER, 2SA, LOC. 0214.

5.1 PS10 TRIP CHECK FOR APPROX. -0.7 VOLTS AT "I" TP UNDER TRIP CONDITION.

5.2 CLOSE SIGNAL CHECK WITH TURBINE TRIPPED, DISCONNECT DEMAND TERMINAL E AND LIFT TERMINAL D OF 2SA. CONNECT TEST POT WITH NEGATIVE OUTPUT TO TERMINAL D AND SET POT FOR ABOUT -3 VOLTS AT LIFT TP. THE OUTPUT OF AMPLIFIER 9 ON TEST METER POSITION 9 SHOULD BE ABOUT -10 VOLTS. TURN BIAS ADJUSTMENT CCW UNTIL AMP. 9 GOES TO ≠10 VOLTS. THERE SHOULD BE LITTLE OR NO CONTROL OF 9 AT INTERMEDIATE VOLTAGES. LEAVE 9 AT ≠10 VOLTS AND RESET TURBINE. SLOWLY REDUCE TOWARDS 0 THE TEST POT SETTING AND NOTE THAT 9 SWINGS FROM ≠10 TO A MINIMUM VALUE OF APPROX. -6.2 VOLTS. NOW APPLY MORE NEGATIVE VOLTAGE FROM THE TEST POT AND NOTE THAT 9 HAS A MAXIMUM VALUE OF APPROX. ≠6.2 VOLTS.

5.3 GAIN REFER TO ALIGNMENT DRAWING TO OBTAIN LIFT TEST POINT VOLTAGE REQUIRED TO PRODUCE 1 VOLT AT I TEST POINT, WHEN GAIN IS PROPERLY ADJUSTED.

11
12P-DF8
FG-W
12P-FS

PRINTS TO

5.31 VARY THE TEST POT SETTING UNTIL I TEST POINT IS 0.000 VOLTS AND NOTE LIFT TEST POINT VOLTAGE. CHANGE THIS VOLTAGE WITH THE TEST POT BY PLUS LIFT TEST POINT VOLTAGE FROM STEP 5.3, ADJUST GAIN UNTIL I TEST POINT READS MINUS 1.000 VOLTS.

5.32 RECHECK ZERO POINT BY RETURNING TEST POT SO THAT ZERO VOLTS IS ONCE AGAIN AT I TEST POINT AND TEST POT VOLTAGE IS THE SAME AS NOTED IN 5.31.

5.33 ADJUST TEST POT FOR MINUS LIFT TEST POINT VOLTAGE AND NOTE THAT I TEST POINT IS PLUS 1.000 VOLT.

5.4 BIAS WITH TEST POT ADJUSTED FOR 1.05 TIMES THE CONTROL VALVE CLOSED OVERTRAVELED IDI OUT TEST POINT VOLTAGE TO SIMULATE THE RAM AGAINST ITS INTERNAL STOP AND WITH ZERO VOLTS AT DEMAND TEST POINT, SET BIAS FOR - .035 ±.01V AT I TEST POINT. DISCONNECT TEST POT AND RECONNECT THE LEAD REMOVED FROM D. RE-CONNECT TERMINAL E.

6. AFTER REMOVING THE ELECTRICAL CONNECTOR, MOUNT THE SERVO VALVE AND VERIFY THAT THE MECHANICAL BIAS IS IN THE VALVE CLOSING DIRECTION.

VII. REGULATION AND COMPENSATION ICR, LOC. 0212

1. REGULATION CONNECT THE OUTPUT OF THE TEST POT TO SUPPLY A NEGATIVE VOLTAGE TO TERMINAL E. ADJUST REGULATION TO OBTAIN VALVE TRAVEL FROM NO LOAD LIFT TO FULL LOAD LIFT FOR 3.00 VOLTS CHANGE IN THE OUTPUT OF THE TEST POT.

2. COMPENSATION VERIFY THAT JUMPER J-K IS NOT CONNECTED.

3. INSTANTANEOUS REGULATION SET TEST POT FOR -1 VOLT OUTPUT AND ADJUST INST. REG. FOR VOLTAGE EQUAL TO .112 VOLTS AT INST REG. T.P.

REVISIONS
1 D.K. Moyer Sept. 19, 1973
11
12P-DFB
FG-W
12P-FS
PRINTS TO

REVISIONS

CHG'D. LOC. 1N VIII-3,4,4 D.K. MOYER Sept. 17, 1973
1

VIII. MANUAL SPEED REFERENCE (STAND-STILL CHECKS)

1. THIS REFERENCE IS OBTAINED FROM THE UPPER OF THE TWO MOTOR DRIVEN LVDT'S. WHEN POWER IS APPLIED IT SHOULD RUN TO LSS AND LIGHT THE LSS INDICATING LIGHT.

2. SLOW REFERENCE SPEED, CL & SMD, LOC. 0118. SHUT OFF HIGH PRESSURE OIL SUPPLY OR DISCONNECT SERVO VALVE BEFORE RAISING MANUAL REFERENCE. ACTUATE RAISE PUSHBUTTON AND TIME TRAVEL TO HSS LIGHT INDICATION. TIME SHOULD BE 90-90 SECONDS.

3. FAST REFERENCE SPEED, CL & SMD, LOC. 0118. ACTUATE LOWER FAST PUSHBUTTON AND TIME MOTOR TRAVEL FROM HSS LIGHT TO LSS LIGHT. IT SHOULD BE 8 TO 10 SECONDS.

4. BIAS AND GAIN, MANUAL REFERENCE, 2LVG, LOC. 0116. DISCONNECT K OF 2LVG. RAISE MANUAL REFERENCE OFF LSS AND BACK INTO IT AT SLOW RATE WITH LOWER PUSHBUTTON. ADJUST MANUAL BIAS FOR 0.000 VOLTS AT OUTPUT TP, 0113. RAISE TO HSS AT SLOW RATE AND ADJUST MANUAL GAIN FOR OUTPUT VOLTAGE AT TP 0116 AS SPECIFIED BY ALIGNMENT DIAGRAM.

11
12-3FB
FG-W
12-FB

PRINTS TO

IX. AUTO SPEED REFERENCE, (STAND-STILL CHECK)

1. BOILER CONTROLLER INTERFACE, 7BCI, LOC. 0122.
CONFIRM THAT THE INTERCONNECTING CABLE BETWEEN THE FEEDWATER CONTROLLER AND EHC CABINET IS A SHIELDED CABLE, AND THE SHIELD OF THE CABLE IS CONNECTED TO COMMON ON THE FEEDWATER CONTROLLER AND TO T34 ON THE EHC CABINET.

   THE ADJUSTMENT OF IP (CMR) IS TO ELIMINATE ANY NOISE THAT MAY BE PICKED UP BETWEEN THE FEEDWATER CONTROLLER AND THE EHC CABINET.

2. USING A SCOPE, DETERMINE IF NOISE IS PRESENT BETWEEN TERMINALS B AND D OF 7BCI, LOC. 0122. IF A NOISE SIGNAL (PROBABLY 60 HZ OR 120 HZ) EXISTS, SET FEEDWATER CONTROLLER SIGNAL FOR LSS AND ADJUST IP FOR MINIMUM NOISE AT IA OUT.

3. IF THERE ISN'T ANY NOISE SIGNAL BETWEEN TERMINALS D AND B OF 7BCI LOC. 0122, THEN DISCONNECT CUSTOMER'S INCOMING SHIELDED CABLE AT EHC TERMINALS T32, T33, AND T34 AND TEST EHC AS FOLLOWS. PLACE A JUMPER BETWEEN T32 AND T33 AND CONNECT OSCILLATOR BETWEEN T32 AND T34. SET OSCILLATOR AT 6 V RMS, 60 HZ AND ADJUST IP FOR MINIMUM AT IA OUT. REMOVE OSCILLATOR AND JUMPER AND RECONNECT CUSTOMER LEADS TO T32, T33 AND T34.

4. WITH FEEDWATER CONTROLLER AT LSS, ADJUST 2P BIAS FOR OVDC AT IA OUT.

5. SIGNAL CONVERTER 3SC, LOC. 0125

   WITH FEEDWATER CONTROLLER AT LSS, CHECK THAT 2LS LIMIT SWITCH CONTACTS BETWEEN ITB(D)& ITB(E) ON AUTO SPEED REF. LOC. 0005 ARE CLOSED. IF NECESSARY, REDUCE INPUT FROM FEEDWATER CONTROLLER UNTIL 2LS LIMIT SWITCH CLOSES. JUMPER SIGNAL FAILURE HOLD CONTACTS T77 AND T78 WITH 2LS LIMIT SWITCH CLOSED AND FEEDWATER CONTROLLER SIGNAL AT LSS. CONNECT VOLTMETER TO ITP OF 3SC LOC. 0125 AND CHECK FOR ZERO VOLTS. IF NECESSARY, OBTAIN ZERO VOLTS BY WITHDRAWING 3SC FROM RACK AND ADJUSTING 5P.

6. REMOVE T77-T78 JUMPER AND SET FEEDWATER CONTROLLER SIGNAL TO HSS. CHECK THAT ILS LIMIT SWITCH CONTACTS BETWEEN ITB(D) AND ITB(C) ON AUTO SPEED REF. LOC. 0005 ARE CLOSED. IF NECESSARY, ADJUST IP CW ON 3SC LOC. 0125 AND/OR 2P GAIN UNTIL ILS LIMIT SWITCH CLOSES. WITH FEEDWATER CONTROLLER AT HSS AND ILS LIMIT SWITCH CLOSED, JUMPER SIGNAL FAILURE HOLD CONTACTS T77-T78 AND ADJUST 2P GAIN FOR MINUS 5.00 V ±.01 VDC AT ITP.

REVISIONS

1	CH'D. LOC. IN II-1,2,3 D.K. MOYER Sept. 17, 1973
11	
12P-DFB	
FG-W	
12P-FS	

PRINTS TO

7. WITH T77-T78 STILL JUMPERED, ADJUST IP B.C. REF. FOR ZERO VOLTS AT 2TP. REMOVE JUMPER.

8. WITH FEEDWATER CONTROLLER AT LSS CHECK FOR ZERO VOLTS AT ITP AND WITH FEEDWATER CONTROLLER AT HSS CHECK FOR ZERO VOLTS AT 2TP.

9. ADJUST 3P DEADBAND FOR MINUS 2VDC (0.1% DEADBAND) AT 3TP. DISCONNECT WIRE AT OUTPUT TERMINAL F OF 3SC LOC. 0125.

10. ADJUST "BOILER CONTROLLER" INPUT TO LSS (-9±1VDC AT 2TP). CONNECT AN OSCILLOSCOPE TO TERMINAL F AND CHECK FOR A PULSE WIDTH OF 1 MSEC. IF NECESSARY, WITHDRAW BOARD AND ADJUST FREQUENCY CONTROL POTENTIOMETER 4P. RECONNECT WIRE TO TERMINAL F. AUTO SPEED REFERENCE SHOULD DRIVE FROM LSS TO HSS IN 4 SECONDS OR LESS.

11. AUTO SPEED REFERENCE STOPS, LSS, 2LVG, LOC. 0116 DISCONNECT G, 0116 AND WITH AUTO REFERENCE AT LSS ADJUST SC BIAS, 0116 FOR OUTPUT TEST POINT VOLTS PER ALIGNMENT DIAGRAM.

12. HSS, 2LVG, LOC. 0116. WITH AUTO REFERENCE AT HSS ADJUST SC GAIN, 0116 FOR OUTPUT TEST POINT VOLTS PER ALIGNMENT DIAGRAM. RECONNECT G.

13. LOCAL/REMOTE CONTROL LIGHT, IVC, LOC. 0117. ADJUST REFERENCE, 0117 FOR 6.5 VOLTS AT L TP. CHECK THAT LOCAL CONTROL INDICATION IS OBTAINED WHEN AUTO REFERENCE IS AT LSS AND MANUAL REFERENCE IS LOWERED SO THAT 2LVG OUTPUT IS LESS THAN APPROXIMATELY 1 VOLT.

14. AUTO SPEED REFERENCE INDICATION, IDI LOC. 0128. ADJUST ZERO AT LSS AND CAL AT HSS TO OBTAIN 0 AND 100 INDICATIONS ON OPERATORS POSITION INDICATOR.

X. OVERSPEED TEST INTEGRATOR, 10SI, LOC. 0112.

1. RATE CONNECT VOLTMETER TO OUTPUT TEST POINT 2LVG, 0116. WITH MANUAL AND AUTO REFERENCE AT HSS JUMPER T54 TO T58. VOLTMETER READING SHOULD INCREASE 0.6 ± 0.1 VOLTS PER MINUTE FOR A (12% ± 2%) SPEED INCREASE PER MINUTE. ADJUST RATE, 0109 AS REQUIRED.

2. LIMIT CONNECT OSCILLATOR AND COUNTER TO T4 AND T5. SET FREQUENCY FOR 120% SSPU FREQUENCY. OBSERVE OUTPUT OF AMPLIFIER 5 WITH TEST METER. JUMPER T54 TO T58 AND ADJUST LIMIT, 0109 FOR POINT AT WHICH AMPLIFIER SWITCHES FROM POSITIVE SATURATION TO ZERO VOLTS. CHECK BY REDUCING OSCILLATOR FREQUENCY AND SLOWLY INCREASING IT TO THE SWITCHING POINT FREQUENCY.

REVISIONS

1 | CHG'D. LOC. IN II-11, 12, 13, 14 AND IN I-1. D.K.MOYER Sept. 17, 1973

11

12P-DC2

FG-W

12P-FS2

PRINTS TO

XI. LP STOP VALVE POSITION INDICATOR IDI, LOC. 0102. WITH STOP VALVE FULLY CLOSED ADJUST DIFF. TRANS. NULL FOR -3 VOLTS AT OUT TP AND METER ZERO FOR 0 POSITION. ADJUST CAL. FOR 100% WITH VALVE FULL OPEN.

XII. HP STOP VALVE POSITION INDICATOR IDI, LOC. 0104. SAME AS XI.

XIII. LOW SPEED SWITCH POWER SUPPLY, LPS, LOC 0227

1. CHECK FOR 5.6 VDC AT TERMINAL D.
2. CHECK FOR 12VDC AT TERMINAL E.

XIV. LOW SPEED SWITCH LSS LOC. 0225

1. STANDSTILL CHECKS - CHECK THAT TERMINALS F&G ARE +5.6VDC WITH RESPECT TO B. THIS WILL CONFIRM ITDX & 200RPM RELAYS ARE ENERGIZED.
2. RELAY CONTACTS AT TERMINAL D AND E SHOULD BE CLOSED.
3. ITP SHOULD BE 5.6 VDC.
4. REMOVE WIRE FROM F AND JUMPER F TO B. RELAY CONTACTS AT H & K CLOSE. REMOVE JUMPER AND CONNECT WIRE TO F.
5. OPERATE TURNING GEAR TO OPEN CONTACTS AT H & K.

OPERATIONAL CHECKS

1. CHECK FOR 0-5 VOLT PULSE AT TERMINAL N WHEN TURNING GEAR OPERATED.
2. POSITIVE PULSES SHOULD BE PRESENT AT 2TP WHEN TURNING GEAR OPERATED.
3. ITP SHOULD BE ZERO VOLT WHEN TURNING GEAR OPERATED.
4. RELAY CONTACTS AT D & E SHOULD BE OPEN WHEN TURNING GEAR OPERATED.
5. WITH TURNING GEAR OFF AND TURBINE AT STANDSTILL RELAY CONTACTS AT D & E SHOULD BE CLOSED.

XV. SEE LEVER DIAGRAM FOR RUNNING CHECKS OF SPEED REGULATION.

REVISIONS

1 | CH'G'D. LOC. IN XIII | D.K. MOYER Sept. 17, 1973

11

12P-DFB

FG-W

12P-FS

PRINTS TO

GENERAL ELECTRIC

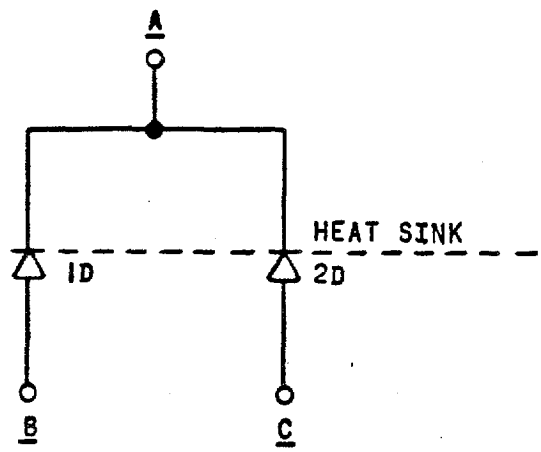
44A306392
CONT ON SHEET SH NO.

REV NO. 44A306392
CONT ON SHEET SH NO.

TITLE RECTIFIER 1 REC

FIRST MADE FOR 448317827-G01

REVISIONS



11
FG-W
12A-DF3

PRINTS TO

MADE BY Clarence Ellison 4/24/73
ISSUED D. Moyer April 26, 1973

APPROVALS R.G.S. R.J. Nelson

DRIVE SYSTEMS PROD. DEPT OR DEPT.
WAYNESBORO, VA. LOCATION

44A306392
CONT ON SHEET SH NO.

FF-803-WF (8-71)
PRINTED IN U.S.A.

CODE IDENT NO.

3

3S7513TC300A12 EHC BOILER FDPUMP

QTY	PART NUMBER	DESCRIPTION
4	3S7513KF300A5	LINE POWER PANEL
2	3S7513KF306A9	METER PANEL
2	3S7513KF306A11	METER PANEL ASM
2	3S7513KF307A13	RELAY PANEL ASM
10	44B331761-G01	PWB VOLT COMPARATR
2	44D236389-G02	PWB SERVO AMPLI
10	44D581585-G01	PWB DETECTOR IND
18	44D581589-G01	PWB OPER AMPLIFIER
2	44B331717-G01	PWB 3KHZ OSC
2	44D236379-G01	PWB ANALOG OUTPUT
4	44D236379-G03	PWB ANALOG OUTPUT
4	44D531918-G01	PWB CL MOTOR DRIVE
6	44D236365-G02	PWB 2FREQ VOLT CNV
2	44B331743-G01	PWB SIG CONV 3SC
2	44B331746-G03	PWB BOILR CON INTF
6	44D236370-G01	PWB DIODE BOARD
12	44B517827-G01	PWB SERIES REG 1
12	44D236371-G01	PWB SERIES REG 2
2	44D236373-G02	PWB OVERSPD INTEG
2	44D236388-G01	PWB COMP+REG ICR
2	44D515178-G01	DIFF TRANS DRV STD
2	44D515178-G02	DIFF TRANS DRV FST
2	44D236378-G01	PWB REF LOW VALUE
2	44B537328-G01	PWB LPF PWR SUPPLY
2	44B537327-G01	PWB LOW SPD SW LSS
6	44C231596-G01	ASM
4	K9774719L50	CAP 3000UF 50VDC

3S7513KF306A5 LINE POWER PANEL

QTY	PART NUMBER	DESCRIPTION
4	44B534726-G01	SUPPLY TRANSF
36	44A516724-M19	CAP 19000UF 75VDC
12	44A533453-D36	RES 360OHM 100W
12	44B214518-G01	REACTOR

REVISIONS	PRINTS
	11
	40

3S7513KF306A9 METER PANEL

QTY	PART NUMBER	DESCRIPTION
10	44A317759-018	SWITCH
40	GE387	LAMP
8	44A316746-001	Meter Relay, DC
2	44A316747-001	Meter Relay, AC

3S7513KF306A11 METER PANEL ASM

QTY	PART NUMBER	DESCRIPTION
2	128A4057H20	RES 200K OHMS 5 W
4	44A317759-028	SWITCH
4	44A317759-019	SWITCH
2	44A317759-012	SWITCH
6	44A317759-020	SWITCH
40	GE387	LAMP
4	44A316746-001	Meter Relay, DC

Lamps & holder replacement Part No. 4872K12G700 for meter relays may be ordered thru nearest G.E. Sales Office from Instrument Dept., G.E. Co. Lynn, Mass.

3S7513KF307A13 RELAY PANEL ASM

QTY	PART NUMBER	DESCRIPTION
44	3S2791G138D4	MINIATURE RELAY
4	3SAA13433A2	RELAY
2	12HMA1183	RELAY
2	44B5124510.083	TIME DELAY RELAY
2	44A2305560.001	SIGNAL FLASHER
72	44B2320190.004	RECT 1A 400V
4	44A517994.G01	COMP BD ASM
8	3S2791G138D3	MINIATURE RELAY
4	K9774723K40	CAP 50UF 450VDC
4	44B514080.G01	D C REACTOR
4	K5908303G1	T B ASM
56	9338545P2	RELAY SOCKET
20	44B2320611.001	RECT 200MA 200VDC

44D315178.G01 DIFF TRANS DRV STD

QTY	PART NUMBER	DESCRIPTION
4	44A316424.001	SPRING
2	44A516456.G01	TRANSFORMER ASM
2	K241259P1	SPRING
2	44A516738.001	STEPPING MOTOR DC
4	44A218484.001	MICRO SW, ASSEMBLY
2	44A517758.001	SPUR GEAR
2	44A515989.003	GEAR
2	44A515989.001	SPUR GEAR
2	44A515989.004	SPUR GEAR
2	44A515988.001	SPUR GEAR
2	44A515988.002	SPUR GEAR
12	44B232214.007	BEARING
2	K9774719L50	CAP 500UF 50VDC

REVISIONS	PRINTS
	11
	40

44A316456.G01 TRANSFORMER ASM

QTY	PART NUMBER	DESCRIPTION
2	44A517777.001	TRANSF LVDT

44D315178.G02 DIFF TRANS DRV FST

QTY	PART NUMBER	DESCRIPTION
4	44A316424.001	SPRING
2	44A516456.G01	TRANSFORMER ASM
2	K241259P1	SPRING
2	44A533467.001	STEPPING MOTOR DC
4	44A218484.001	MICRO SW, ASSEMBLY
2	44A517758.001	SPUR GEAR
2	44A515989.003	GEAR
2	44A515989.005	SPUR GEAR
2	44A515989.003	GEAR
2	44A515988.001	SPUR GEAR
2	44A515988.002	SPUR GEAR
12	44B232214.007	BEARING
2	K9774719L50	CAP 500UF 50VDC

44A316456-G01 TRANSFORMER ASM

QTY	PART NUMBER	DESCRIPTION
2	44A317777-001	TRANSF LVDT

3S7513KL308A10 CONTROL PANEL

QTY	PART NUMBER	DESCRIPTION
4	CR2940U207	CONTACT BLOCK
4	CR2940UA202C	PUSH BUTTON
2	44A317759-007	PUSHBUTTON SWITCH
8	44A317759-001	PUSHBUTTON SWITCH
4	44A317759-055	SWITCH PB ILL
2	44C392721-G01	DIODE BOARD
20	GE387	LAMP
4	CR2940UA200C	OPERATOR
2	6105700G43G	IND,LIGHT
2	44A317759-054	ILLUM PB SWITCH

REVISIONS	PRINTS
	11
	40

GENERAL ELECTRIC

44B300423

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

REV NO. 44B300423 CONT ON SHEET SH NO.

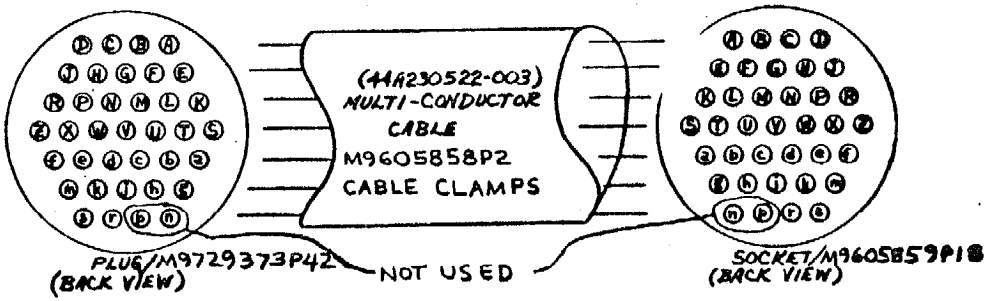
TITLE

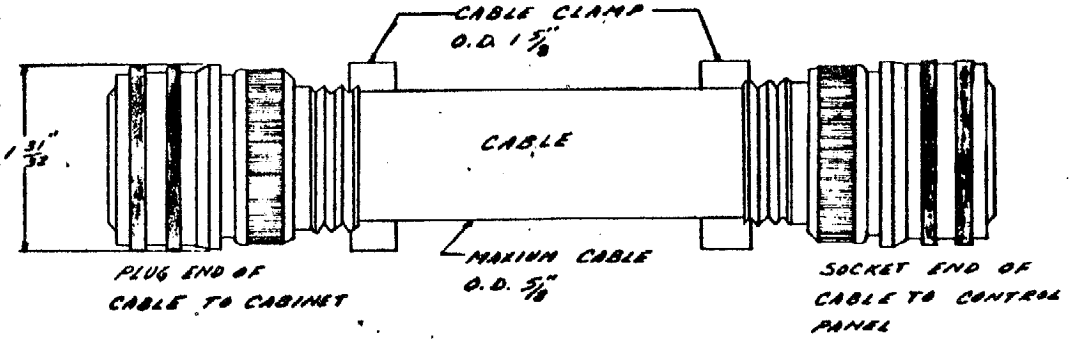
CABLE WIRING

FIRST MADE FOR 3S7513TC300A2

SIGNAL FLOW VOLTAGE

	PLUG TERMINAL	WIRE	SOCKET TERMINAL
SHIELDED CABLE	V	SHIELD	V
	M	WHITE/BLACK/VIOLET	M
	N	BLACK	N
	W	BROWN	W
SHIELDED CABLE	d	WHITE/BLACK/GRAY	d
	c	SHIELD	c
	U	RED	U
SHIELDED CABLE	T	WHITE/BROWN/RED	T
	L	SHIELD	L
	F	ORANGE	F
SHIELDED CABLE	G	WHITE/BROWN/ORANGE	G
	H	SHIELD	H
	P	WHITE	P
SHIELDED CABLE	X	WHITE/BLACK	X
	e	SHIELD	e
	k	WHITE/BROWN/YELLOW	k
SHIELDED CABLE	j	WHITE/BROWN/GREEN	j
	h	SHIELD	h
	b	WHITE/BROWN	b
SHIELDED CABLE	g	WHITE/RED	g
	a	SHIELD	a
	S	WHITE/ORANGE	S
	K	WHITE/YELLOW	K
	E	WHITE/GREEN	E
	A	WHITE/BLUE	A
	B	WHITE/VIOLET	B
	C	WHITE/GRAY	C
	D	WHITE/BLACK/BROWN	D
	J	WHITE/BLACK/RED	J
	R	WHITE/BLACK/ORANGE	R
	Z	WHITE/BLACK/YELLOW	Z
	f	WHITE/BLACK/GREEN	f
	m	WHITE/BLACK/BLUE	m
	s	WHITE/BROWN/BLUE	s
	r	WHITE/BROWN/VIOLET	r





NOTES:

1. USE PLASTIC SLEEVING OVER WIRE AND TERMINALS TO INSULATE CONNECTIONS FROM EACH OTHER.

REVISIONS		PRINTS TO
1.	Work Order 19 1968	11
2	W. Tompkins	122-0-EB
3	ADDED DATA: 3d. 13 J.K. MOYER 1968	EG-W

MADE BY / ISSUED	APPROVALS		
R.M. Bukosky MAY 22 1968	R.M.B	SPECIALTY CONTROL (DIV OR DEPT) WAYNESBORO, VA. (LOCATION)	44B300423 CONT ON SHEET SH NO.

PP-600-F (10-68) PRINTED IN U.S.A.

44B306127

GENERAL ELECTRIC

44B306127 CONT ON SHEET FNL. SH NO 1

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—

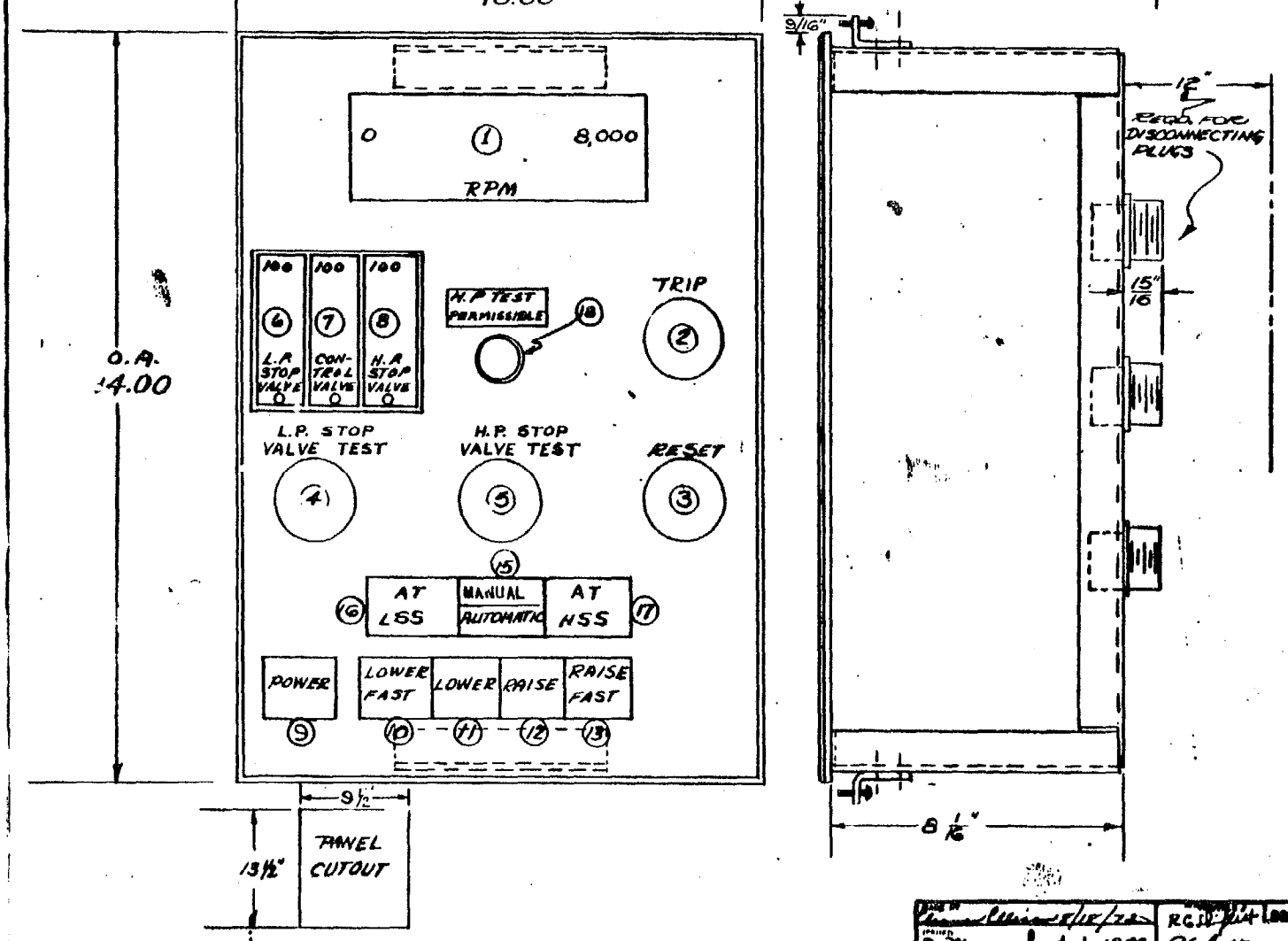
APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

REV LTR 44B306127 CONT ON SHEET FL. SH NO 1

TITLE (OUTLINE)
FEEDPUMP TURBINE CONTROL PANEL

FIRST MADE FOR 3S7513KL300A10 TURBINE 122808, 809

ARKANSAS POWER & LIGHT RUSSELVILLE



PARTS LIST

A. 1 INSTRUMENTS, GE TYPE 180.

B. 2,3 GE CR2940UA203C PUSHBUTTON WITH GUARD.

C. 4,5 GE CR2940UA202C PUSHBUTTON WITH GUARD.

D. 6,7,8 INSTRUMENTS, GE TYPE 185.

E. 9 ILLUMINATED PUSHBUTTON RED, POWER ON, CONDITIONS NORMAL, YELLOW (FLASHING) POWER ON, CONDITIONS ABNORMAL DEPRESS TO TEST ALL LAMPS.

F. 10-13, PUSHBUTTONS.

G. 16, 17 INDICATING LIGHTS, GREEN

H. 14 ILLUMINATED PUSHBUTTON.

I. 9-17, MASTER SPECIALTIES SERIES 12, VERTICAL BARRIER.

J. 18 - GREEN LIGHT "TEST PERMISSIBLE"

K. FINISH LIGHT YELLOW, GLIDDEN A99L.

L. 15, INDICATING LIGHT SPLIT FIELD
MANUAL CONTROL - GREEN
AUTOMATIC CONTROL - RED

REVISIONS			PRINTS TO	
1	Issued	Nov. 14, 1972	11	
2	R.A. Clemmer	Mar. 1, 1973	12P-DFB	
3	H. Tompkins	April 4, 1973	FG-W	
4	D.K. Meyer	May 29, 1973	12P-F	5

DRAWN [illegible] 8/15/72

APPROVED D. Meyer Sept. 1, 1972

R.G. Baker

DRIVE SYSTEMS PROD. DEPT. WAYNESBORO, VA.

44B306127 CONT ON SHEET FNL. SH NO 1

GENERAL ELECTRIC

44B306144

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING.—

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

REV NO.

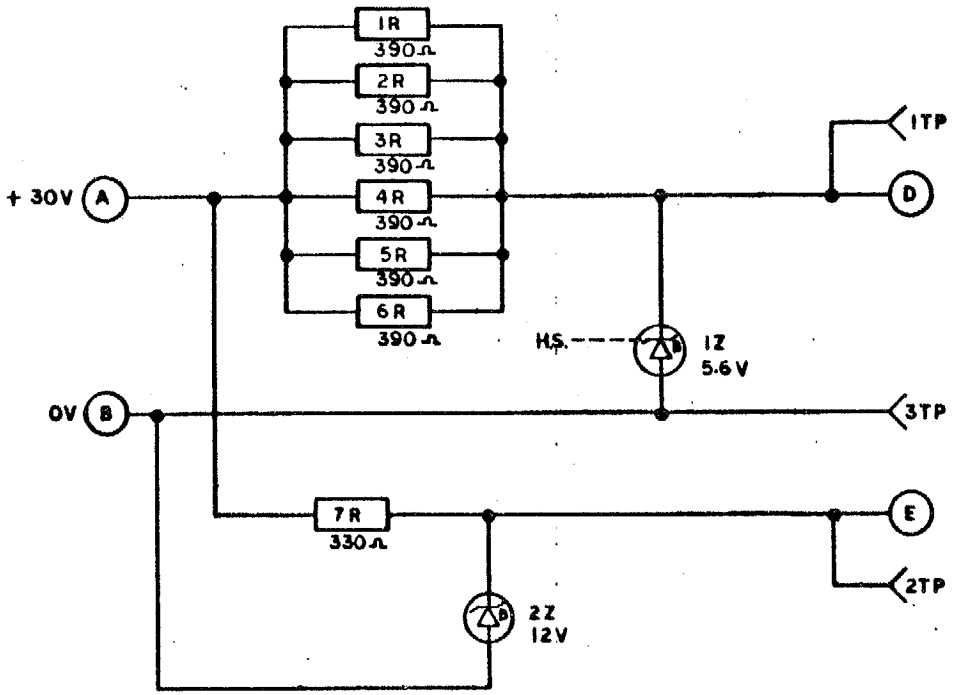
CONT ON SHEET SH NO.

TITLE

LSS PWR SUPPLY

FIRST MADE FOR

ELEM DIAGRAM



REVISIONS

PRINTS TO

11

MADE BY Guy Rousie 9-11-72

ISSUED P.K. Mayer Sept. 12, 1972

APPROVED

DRIVE SYSTEMS PROD. DEPT. OF (DIV.)

WAYNESBORO, VA. LOCATION

44B306144 CONT ON SHEET SH NO.

3

PP-646-F 80-70
PRINTED IN U.S.A.

GENERAL ELECTRIC

44B306148

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—					REV NO
APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENTIONS			
		FRACTIONS	DECIMALS	ANGLES	
	✓	±	±	±	

CONT ON SHEET — SH NO

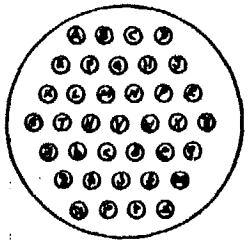
TITLE
POWER CABLE WIRING, ELECTRONIC BOILER FEED PUMP TRUBINE CONTROL

FIRST MADE FOR 3S7513TC300A10
2C0, 3C0 POWER CABLE

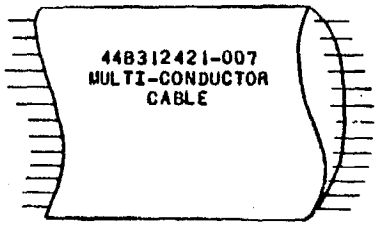
ASSEMBLY NOTES

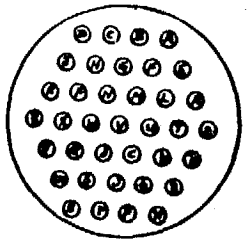
THE SPECIFIC COLOR OF WIRE CONNECTED TO A PIN IS NOT IMPORTANT, BUT THE SAME WIRE SHOULD BE CONNECTED TO THE SAME PIN AT THE OTHER CABLE END AND CONNECTOR.
EXAMPLE: IF A RED WIRE IS CONNECTED TO PIN "A" ON THE CONNECTOR PLUG, THE OTHER END OF THAT RED WIRE MUST BE CONNECTED TO PIN "A" OF THE SOCKET CONNECTOR AT THE OTHER END.

CONNECT SAME COLOR WIRE TO THESE PINS AT BOTH CABLE ENDS	DO NOT CONNECT ANY WIRES TO THESE PINS
A	B
C	D
T	G
E	L
F	M
	N
	P
H X	R
J Z	S
K	U
	V
	W

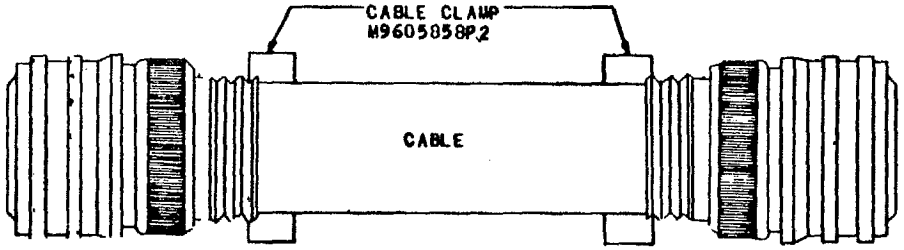


SOCKET
M9605859P18





PLUG
M9605859P42



SOCKET END OF
CABLE TO EHC REMOTE CONTROL PANEL.

PLUG END OF
CABLE TO EHC CABINET

REVISIONS	PRINTS TO
	11
	APPD
	FG-W

MADE BY: L. Tompkins 10-18-72; B. Mayer Oct. 20, 1972
APPROVED: Parker
COMML & CONT. DEVICES DIV OR DEPT, WAYNESBORO, VA LOCATION
44B306148 CONT ON SHEET — SH NO

3

44B306197

GENERAL ELECTRIC

44B306197

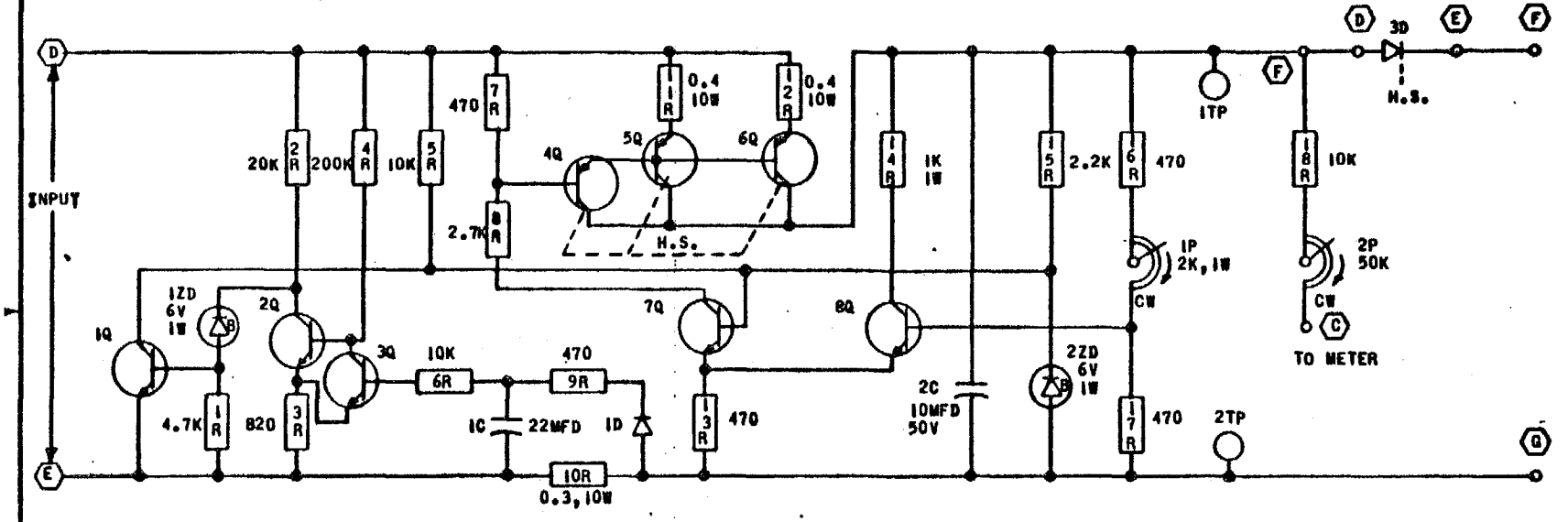
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

REV NO. 44B306197

TITLE: SERIES REGULATOR ISR

FIRST MADE FOR 44B2363[illegible]-G01



REVISIONS

PRINTS TO

11

FG-W

12P-DEB

MADE BY: [illegible] 4/29/73

ISSUED: D. Moyer April 26, 1973

APPROVALS: R G [illegible]

DRIVE SYSTEMS PROD. DEPT. OR DEPT. WAYNESBORO, VA. LOCATION

44B306197 CONT ON SHEET SH NO.

PF-606-P 8-72B PRINTED IN U.S.A.

3

GENERAL ELECTRIC

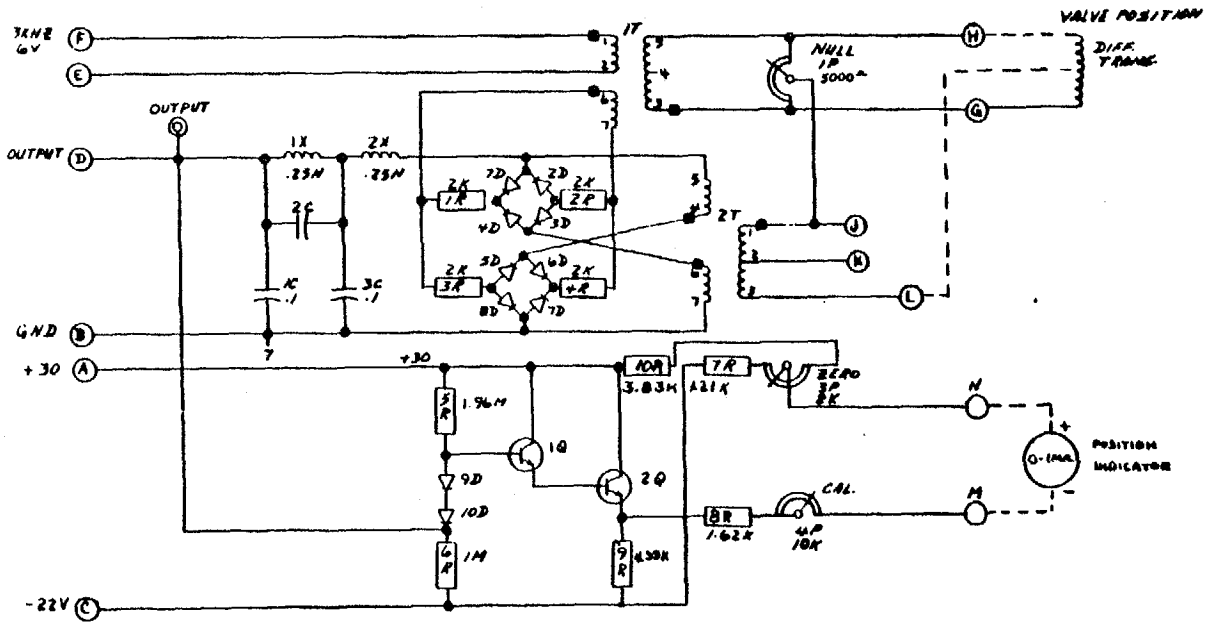
44C300609

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —

APPLIED PRACTICES	SURFACES	TOLERANCES ON NUMBERS: FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

TITLE: DETECTOR, INDICATOR 1DI

FIRST MADE FOR 44D381585-G01



NOTES:

1. CONNECTIONS

A	+30V	6	DIFFERENTIAL TRANSFORMER CONNECTIONS
B	GROUND		
C	-22V		
D	OUTPUT		
E, F	6V 3 KHZ		

OUTPUT METER 0-1MA

44C300609 REV NO. 0

REVISIONS			PRINTS TO
1	[illegible]	NOV 19 1968	M
2	[illegible]	MAY 7-8 1969	FG-M
3	[illegible]	OCT 19 1969	12P-DFB
4	[illegible]	DEC 19 1969	
5	[illegible]	MAY 6 1970	
6	[illegible]	OCT 19 1970	

DRAWN: Templeton JAN 8 1968
CHECKED: APR 17 1968
APPROVED: [illegible]
SPECIALTY CONTROL WAYNESBORO, VA.

44C300609

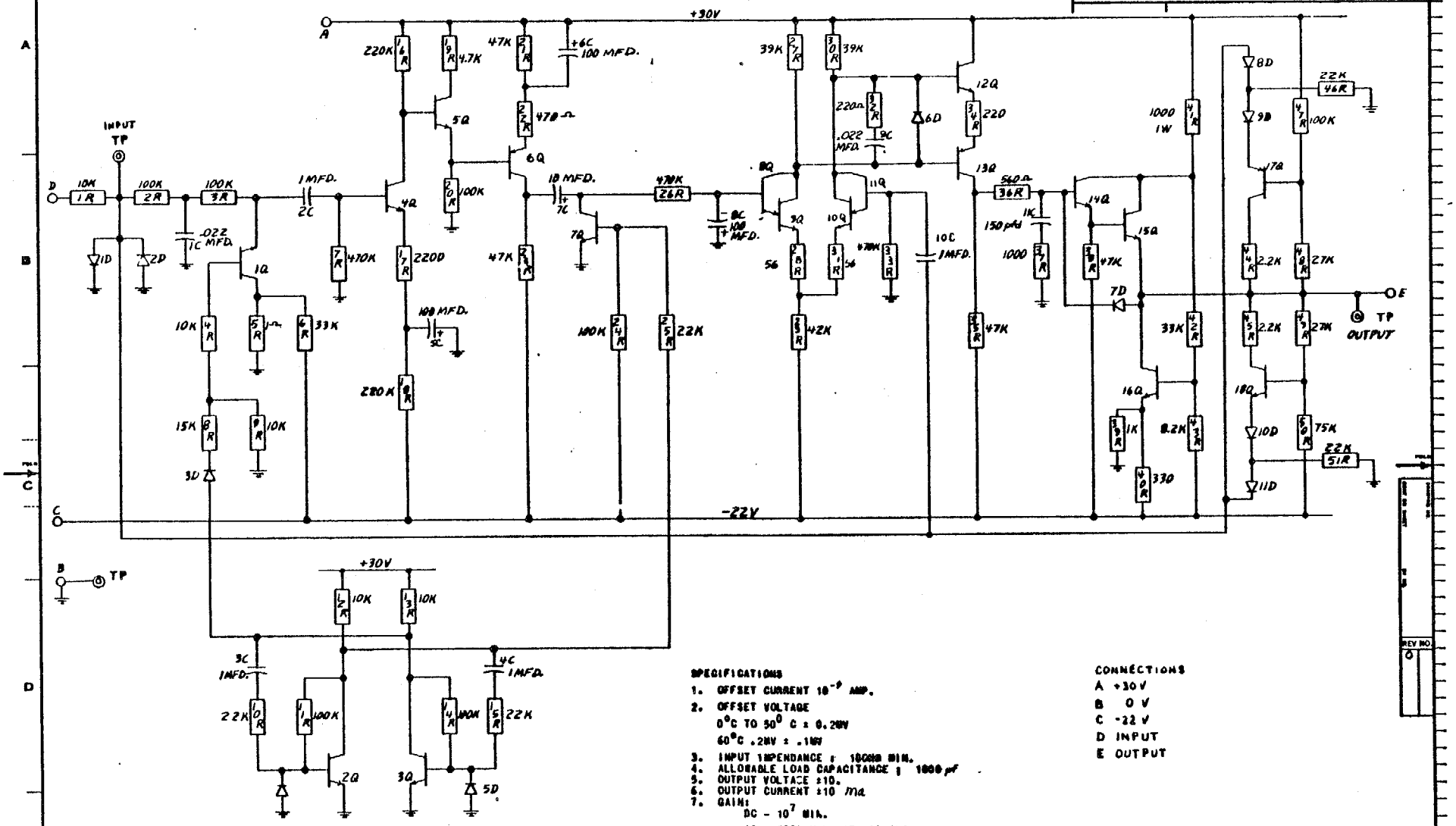
GENERAL ELECTRIC

44C300610

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING—
APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE
CHOPPER STABILIZED
OPERATIONAL AMPLIFIER
FIRST MADE FOR 44D381589-G01

1ØA



SPECIFICATIONS

1. OFFSET CURRENT 10^{-9} AMP.
2. OFFSET VOLTAGE
   0°C TO 50° C ± 0.2MV
   60°C .2MV ± .1MV
3. INPUT IMPEDANCE : 10GΩ MIN.
4. ALLOWABLE LOAD CAPACITANCE : 1000 μf
5. OUTPUT VOLTAGE ±10.
6. OUTPUT CURRENT ±10 ma
7. GAIN:
   DC - 10^7 MIN.
   AC - 5000 MIN. AT 100 CPS
   CHARACTERISTIC: BREAK FREQUENCY - 250 CPS
   20 db PER DECADE TO CROSSOVER.
8. CHOPPER STABILIZER RECOVERY TIME: 1 MINUTE
9. POWER REQUIREMENTS: 30 VOLTS - 25 ma
   -22 VOLTS - 20 ma
10. AMPLIFIER PROTECTION:
    A. OUTPUT CURRENT LIMITED TO APPROX 10 ma
    B. OUTPUT LIMITED TO 10 VOLTS
    C. ALLOWABLE INPUT VOLTAGE BEFORE DAMAGE - 70 VOLTS.

CONNECTIONS

A +30 V
B 0 V
C -22 V
D INPUT
E OUTPUT

REVISIONS		PRINTS TO
1	Now Th. Rev	11
2	Revised	12P-DEB

J. C. Glenn JAN 22 1968			SPECIALTY CONTROL	44C300610
ISSUED MAR 12 1968			WAYNESBORO VA	

GENERAL ELECTRIC

44C300882

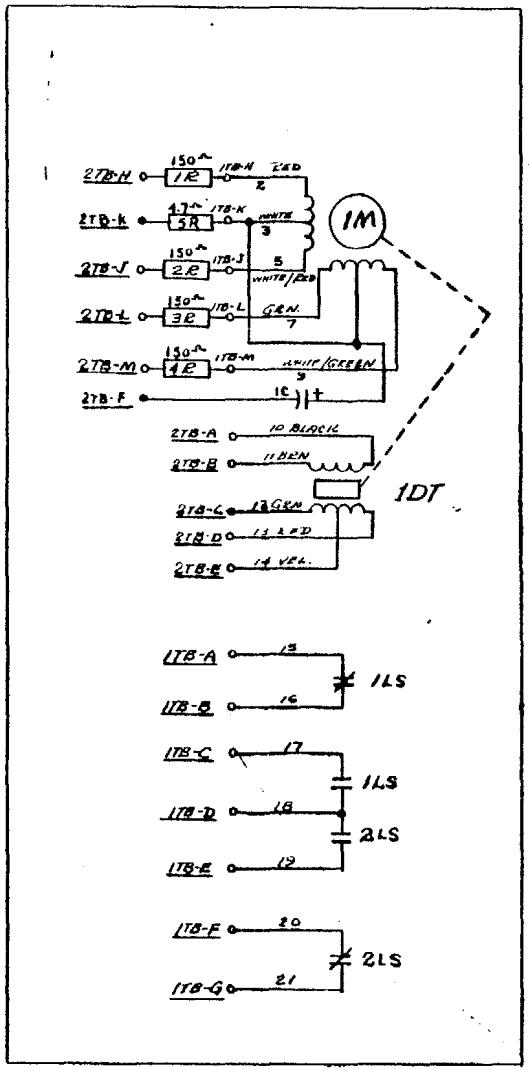
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—
APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE
DIFFERENTIAL TRANSFORMER DRIVE

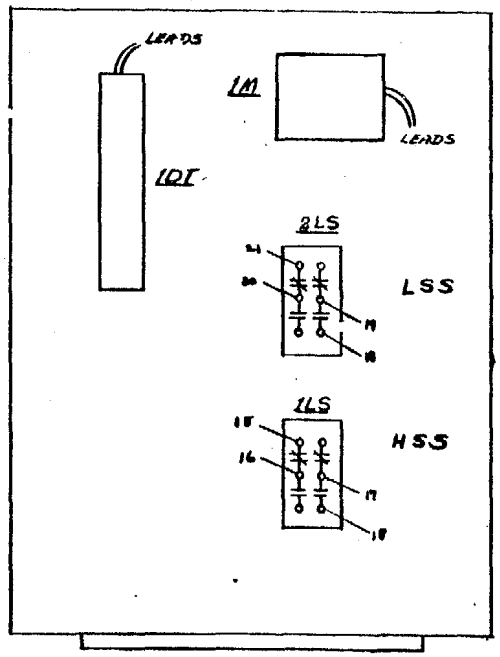
FIRST MADE FOR 44D315178-601

ELEM-CONN

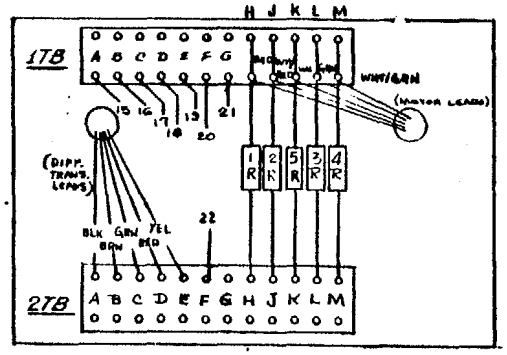
ELEMENTARY



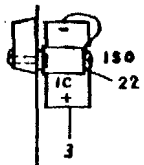
CONNECTION



TOP VIEW



END VIEW



WIRE TABLE

CIRCUIT	CONNECTION
15	1TB-A, 1LS
16	1TB-B, 1LS
17	1TB-C, 1LS
18	1TB-D, 1LS
18	1TB-D, 2LS
19	1TB-E, 2LS
20	1TB-F, 2LS
21	1TB-G, 2LS
22	IC−, 150, 2TB-F
3	IC+, 1TB-K

1M - MOTOR LEADS

CIRCUIT	COLOR	CONNECTION
2	RED	1TB-H
3	WHITE	1TB-K
5	WHITE/RED	1TB-J
7	GREEN	1TB-L
9	WHT/GRN	1TB-M

1DT - DIFFERENTIAL TRANS. LEADS

CIRCUIT	COLOR	CONNECTION
10	BLACK	2TB-A
11	BROWN	2TB-B
12	GREEN	2TB-C
13	RED	2TB-D
14	YELLOW	2TB-E

FACTORY NOTES:

1) USE #20 WIRE, MS7294-94 P16/B
2) RESISTORS 1R, 2R, 3R, 4R 150 OHM 10 WATT
   RESISTOR 5R 4.7Ω 10 WATT
3) GUM LABELS
   1TB
   2TB
   HSS
   LSS

REVISIONS			PRINTS TO
1	K EL MAY 8 1968		11
2	L HOLT JUN 26 1970	12P DFB	
3		FG-W	

44C300882

GENERAL ELECTRIC

44C301216

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—

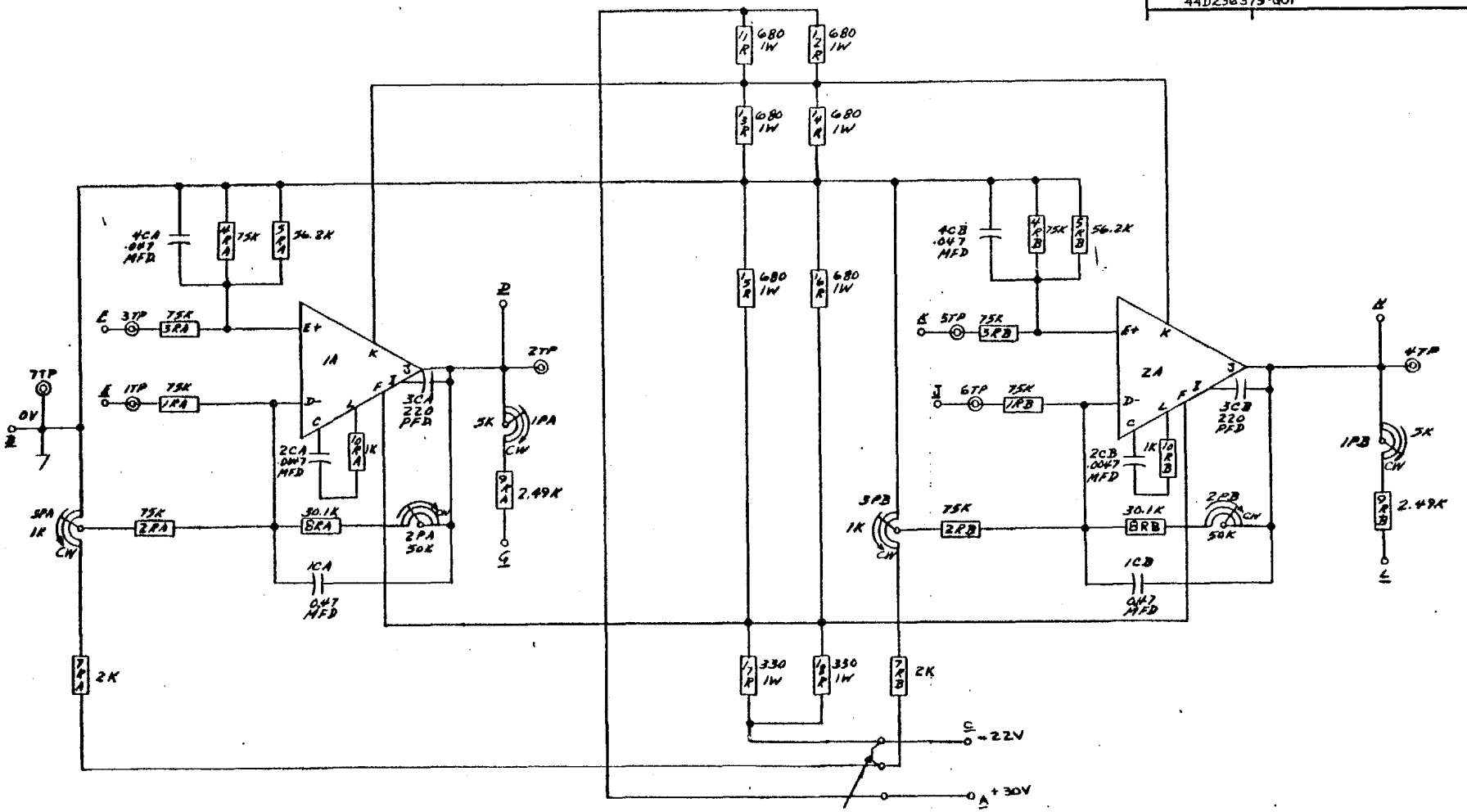
APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

TITLE

ANALOG OUTPUT 1A#2

FIRST MADE FOR

44D236379-G01



JUMPER POINTS PROVIDED ON THE PRINTED CIRCUIT BOARD SO RESISTORS 7RA & 7RB MAY BE CONNECTED TO EITHER +30V OR -22V.

CONNECTIONS

A = +30V
B = 0V
C = -22V
D = 1A VOLTAGE OUTPUT
E = 1A INPUT
G = 1A CURRENT OUTPUT
H = 2A VOLTAGE OUTPUT
J = 2A INPUT
L = 2A CURRENT OUTPUT

REVISIONS		PRINTS TO
1	[illegible] SEP 30 1969	
2	R. Mau OCT 9 1969	
3	[illegible] FEB 3 1970	
4	[illegible] NOV 17, 1971	

DRAWN	Templeton SEP 16 1968	ENGINEER	SPECIALTY CONTROL DEPT.	44C301216
ISSUED	SEP 24 1968	[illegible]	WAYNESBORO, VA LOCATION	COMP. OR SHEET

44C301216

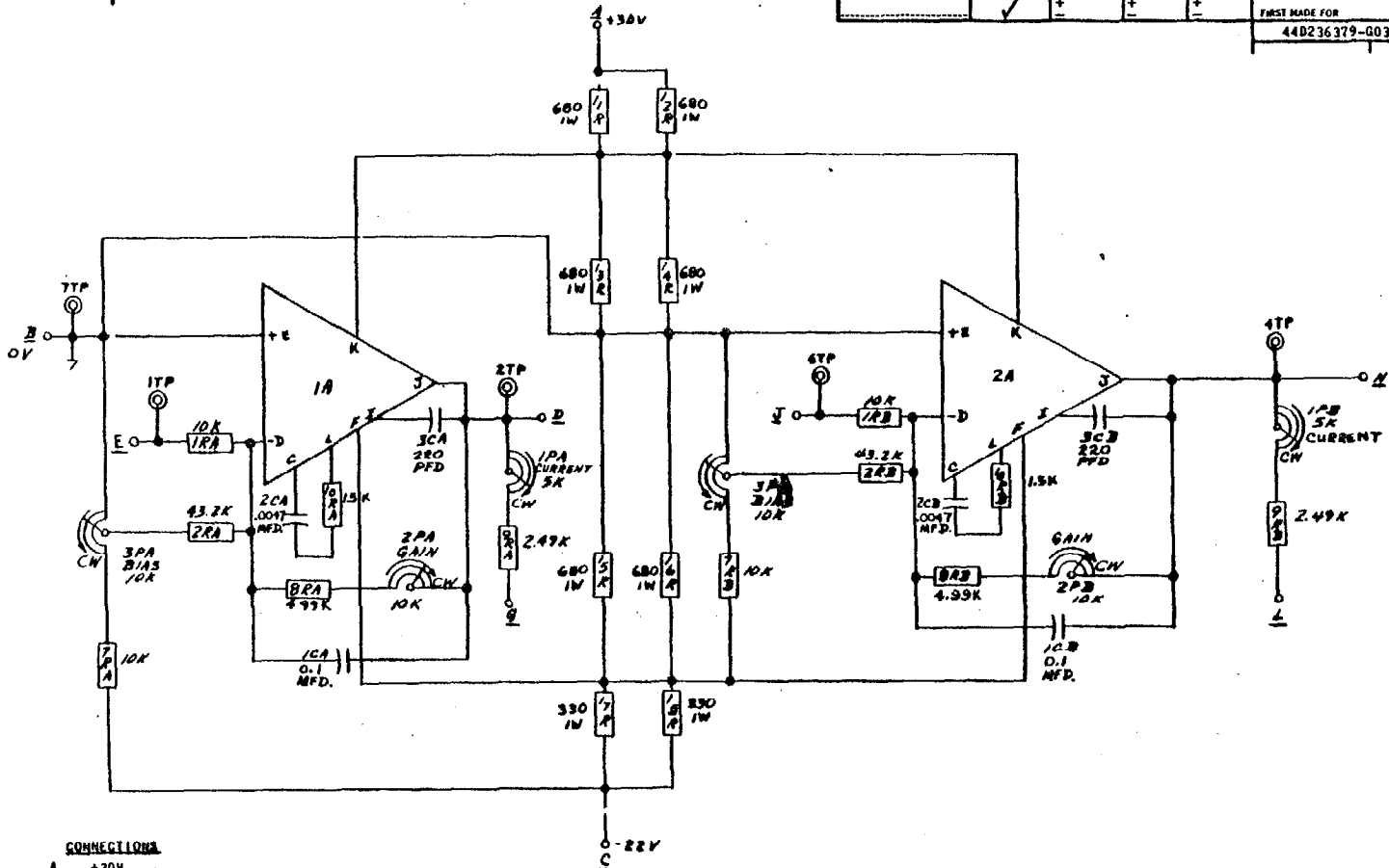
GENERAL ELECTRIC

44C301622

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—
APPLIED PRACTICES | SURFACES ✓ | FRACTIONS | DECIMALS ± | ANGLES ±

TITLE
ANALOG OUTPUT 3A#2

FIRST MADE FOR
44D236379-G03



CONNECTIONS

A	+30V
B	0V
C	-22V
D	1A VOLTAGE OUTPUT
E	1A INPUT
Q	1A CURRENT OUTPUT
H	2A VOLTAGE OUTPUT
J	2A INPUT
L	2A CURRENT OUTPUT

REVISIONS

1	[illegible]	SEP 17 1968
2	[illegible]	Nov 17, 1971
3	[illegible]	Aug 1, 1974

PRINTS TO: 11

DRAWN	SEP 17 1968	
TRACED	SEP 19 1968	MBC

SPECIALTY CONTROL WAYNESBORO, VA

44C301622

GENERAL ELECTRIC

44C301682

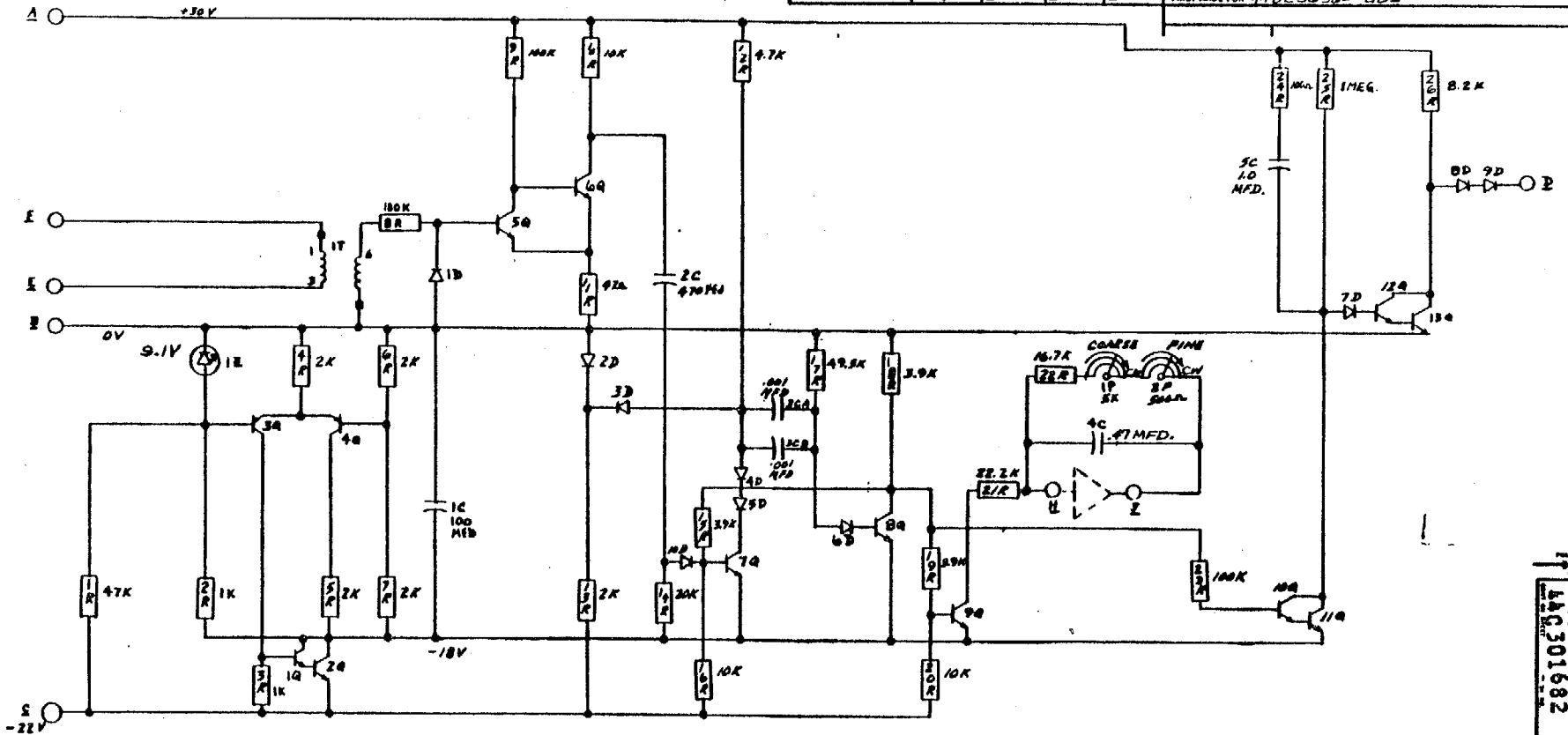
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

TITLE

FREQUENCY-VOLTAGE CONVERTER (2FVC)

FIRST MADE FOR 44D236365-G02



REVISIONS		PRINTS TO
1		
2		12P-DFB
3		FGW

G.E. Tentative NOV 15 1966 — OCT 3 1966 — R.L. R.T. Lee

SPECIALTY CONTROL DEPT. WAYNESBORO, VA.

44C301682

44C301682 REV NO.

GENERAL ELECTRIC

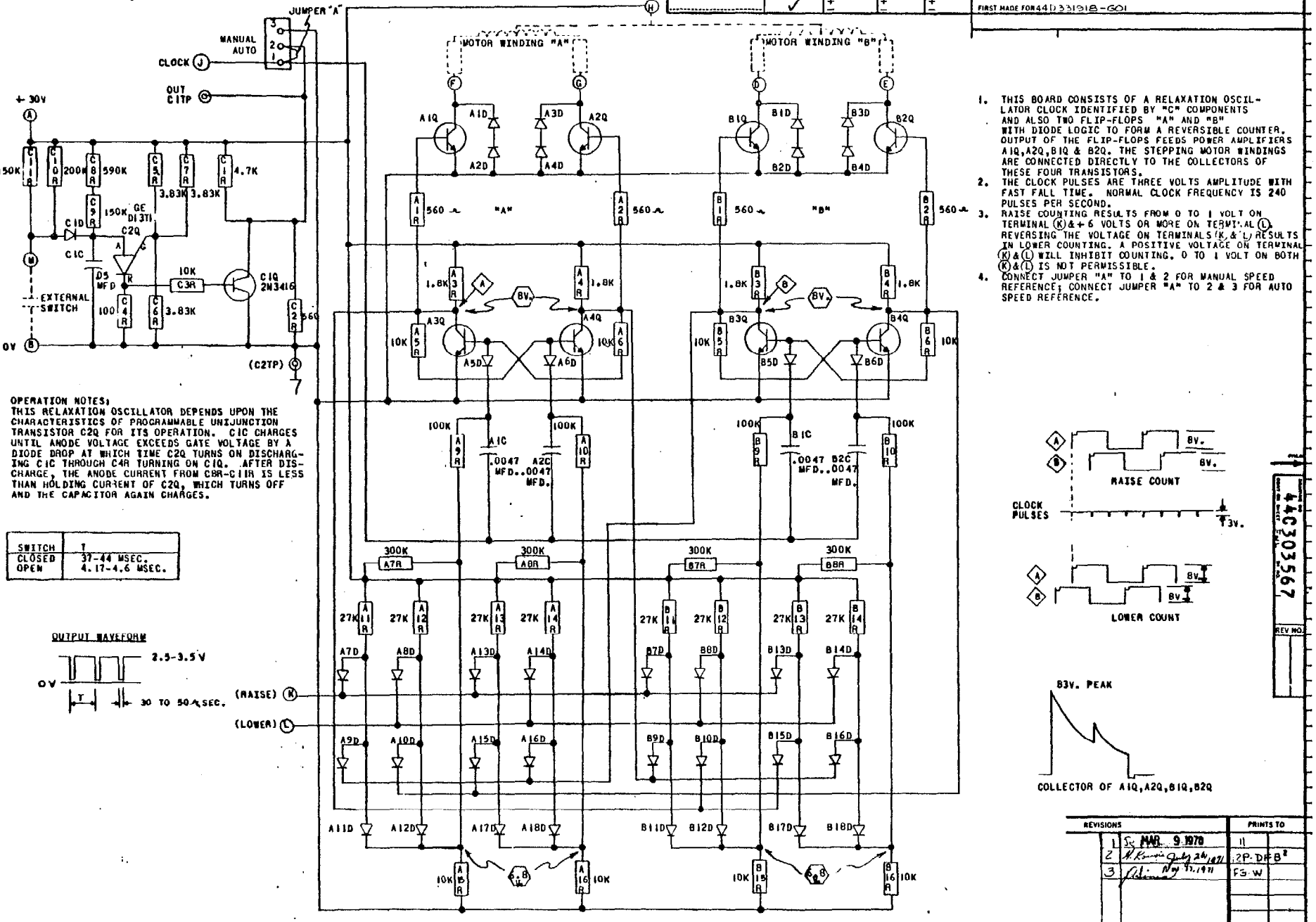
44C303567

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:— APPLIED PRACTICES | SURFACES ✓ | TOLERANCES OR MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE

CLOCK & STEPPING MOTOR DRIVE

FIRST MADE FOR 44D331918-G01



OPERATION NOTES;
THIS RELAXATION OSCILLATOR DEPENDS UPON THE CHARACTERISTICS OF PROGRAMMABLE UNIJUNCTION TRANSISTOR C2Q FOR ITS OPERATION. C1C CHARGES UNTIL ANODE VOLTAGE EXCEEDS GATE VOLTAGE BY A DIODE DROP AT WHICH TIME C2Q TURNS ON DISCHARGING C1C THROUGH C4R TURNING ON C1Q. AFTER DISCHARGE, THE ANODE CURRENT FROM C8R-C11R IS LESS THAN HOLDING CURRENT OF C2Q, WHICH TURNS OFF AND THE CAPACITOR AGAIN CHARGES.

SWITCH	T
CLOSED	37-44 MSEC.
OPEN	4.17-4.6 MSEC.

OUTPUT WAVEFORM

1. THIS BOARD CONSISTS OF A RELAXATION OSCILLATOR CLOCK IDENTIFIED BY "C" COMPONENTS AND ALSO TWO FLIP-FLOPS "A" AND "B" WITH DIODE LOGIC TO FORM A REVERSIBLE COUNTER. OUTPUT OF THE FLIP-FLOPS FEEDS POWER AMPLIFIERS A1Q,A2Q,B1Q & B2Q. THE STEPPING MOTOR WINDINGS ARE CONNECTED DIRECTLY TO THE COLLECTORS OF THESE FOUR TRANSISTORS.
2. THE CLOCK PULSES ARE THREE VOLTS AMPLITUDE WITH FAST FALL TIME. NORMAL CLOCK FREQUENCY IS 240 PULSES PER SECOND.
3. RAISE COUNTING RESULTS FROM 0 TO 1 VOLT ON TERMINAL (K) & + 6 VOLTS OR MORE ON TERMINAL (L). REVERSING THE VOLTAGE ON TERMINALS (K) & (L) RESULTS IN LOWER COUNTING. A POSITIVE VOLTAGE ON TERMINAL (K) & (L) WILL INHIBIT COUNTING. 0 TO 1 VOLT ON BOTH (K) & (L) IS NOT PERMISSIBLE.
4. CONNECT JUMPER "A" TO 1 & 2 FOR MANUAL SPEED REFERENCE; CONNECT JUMPER "A" TO 2 & 3 FOR AUTO SPEED REFERENCE.

COLLECTOR OF A1Q,A2Q,B1Q,B2Q

REVISIONS		PRINTS TO
1	S. MAR. 9 1970	11
2	M. Brown July 24, 1971	2P-DFB
3	[illegible] Nov 11, 1971	FS-W

DRAWN BY [illegible] 2-26-70 ISSUED MAR 3 1970 | APPROVED MAC | COMM. & CONT. DEVICES WAYNESBORO, VA. | 44C303567

REV NO.

GENERAL ELECTRIC

44C304462

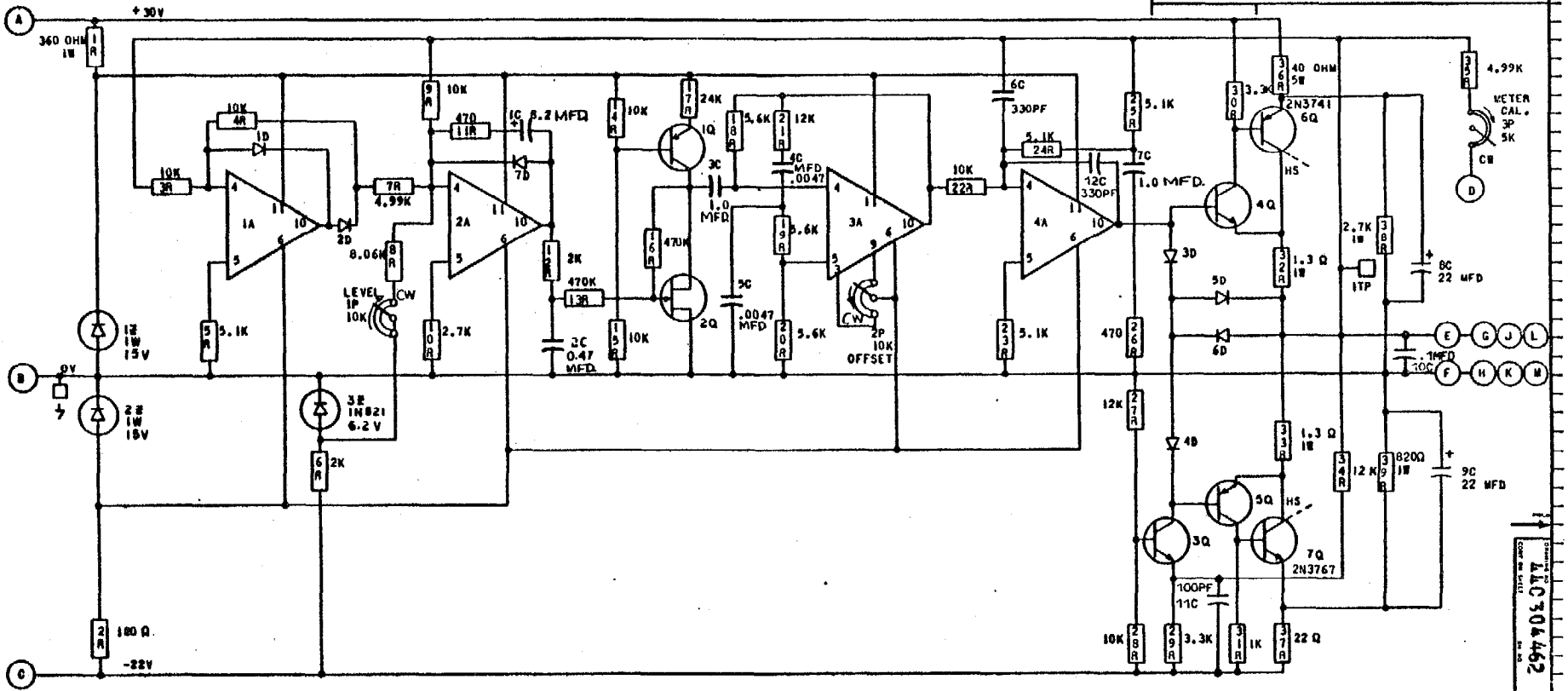
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

TITLE

3KHZ OSCILLATOR

FIRST MADE FOR EHC 6-05C 44B331717-G01



FREQUENCY: 3000HZ± 300HZ
AMPLITUDE: 6.00 VRMS (17V PEAK TO PEAK)
DC OFFSET: ADJUSTABLE TO ZERO
MINIMUM LOAD IMPEDANCE: 25Ω
DRIFT: 0.1% FIFTEEN HOURS
AMBIENT TEMPERATURE

		V	%
DISTORTION	2ND HARMONIC	.056	0.93
	3RD HARMONIC	.0083	0.14
	4TH HARMONIC	.0006	0.01
	5TH HARMONIC	.0022	0.04

REVISIONS			PRINTS TO
1	[illegible]	Nov 13, 1968	11
2	E.A. Carver	Oct. 8, 1970	12-DEC
3	C.D. Cook	Nov. 9, 1971	F6-W
4	[illegible]	May 24, 1973	

ISSUED MAY 20 1974

COMM. & CONT. DEVICES WAYNESBORO, VA.

44C304462

GENERAL ELECTRIC

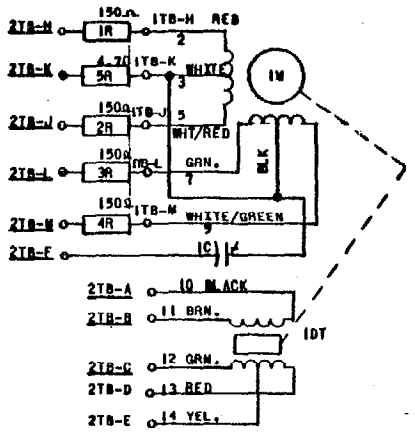
44C304865

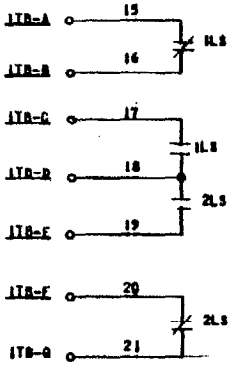
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—
APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE
DIFFERENTIAL TRANSFORMER DRIVE

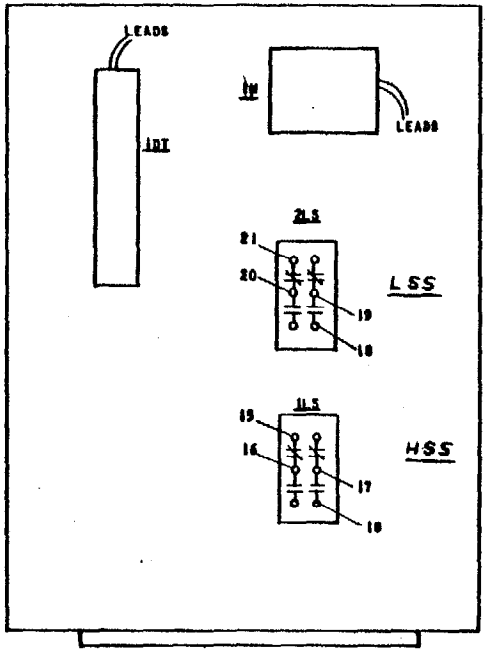
FIRST MADE FOR 440315178-G02

ELEMENTARY



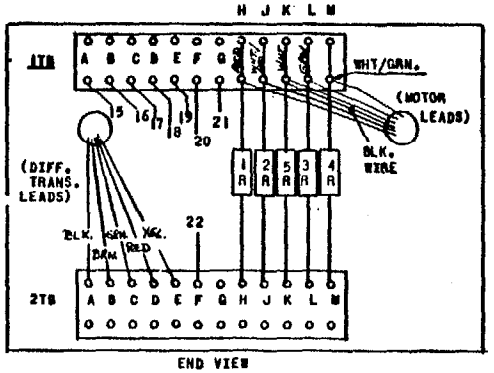


CONNECTION

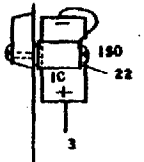


TOP VIEW

TOP VIEW



END VIEW



WIRE TABLE

CIRCUIT	CONNECTION
15	1TB-A, 1LS
16	1TB-B, 1LS
17	1TB-C, 1LS
18	1TB-D, 1LS
18	1TB-D, 2LS
19	1TB-E, 2LS
20	1TB-F, 2LS
21	1TB-G, 2LS
22	IC1, 1S0, 2TB-F
3	IC2, 1TB-K

IM - MOTOR LEADS

CIRCUIT	COLOR	CONNECTION
2	RED	1TB-H
3	WHITE	1TB-K
5	WHITE/RED	1TB-J
7	GREEN	1TB-L
9	WHT/GRN	1TB-M
8	BLACK	1TB-K

IDT - DIFFERENTIAL TRANS. LEADS

CIRCUIT	COLOR	CONNECTION
10	BLACK	2TB-A
11	BROWN	2TB-B
12	GREEN	2TB-C
13	RED	2TB-D
14	YELLOW	2TB-E

FACTORY NOTES:
1) USE #20 WIRE, M9729494P1618
2) RESISTORS 1R,2R,3R,4R 150 OHM 10 WATT
   RESISTOR 5R 4.7Ω 10 WATT
3) GUM LABELS:
   1TB
   2TB
   LSS
   HSS

REVISIONS

PRINTS TO

DRAWN		CHECKED	
ISSUED	JUN 3 1971	APPROVED	

GENERAL ELECTRIC CO., WAYNESBORO, VA.

44C304865

GENERAL ELECTRIC

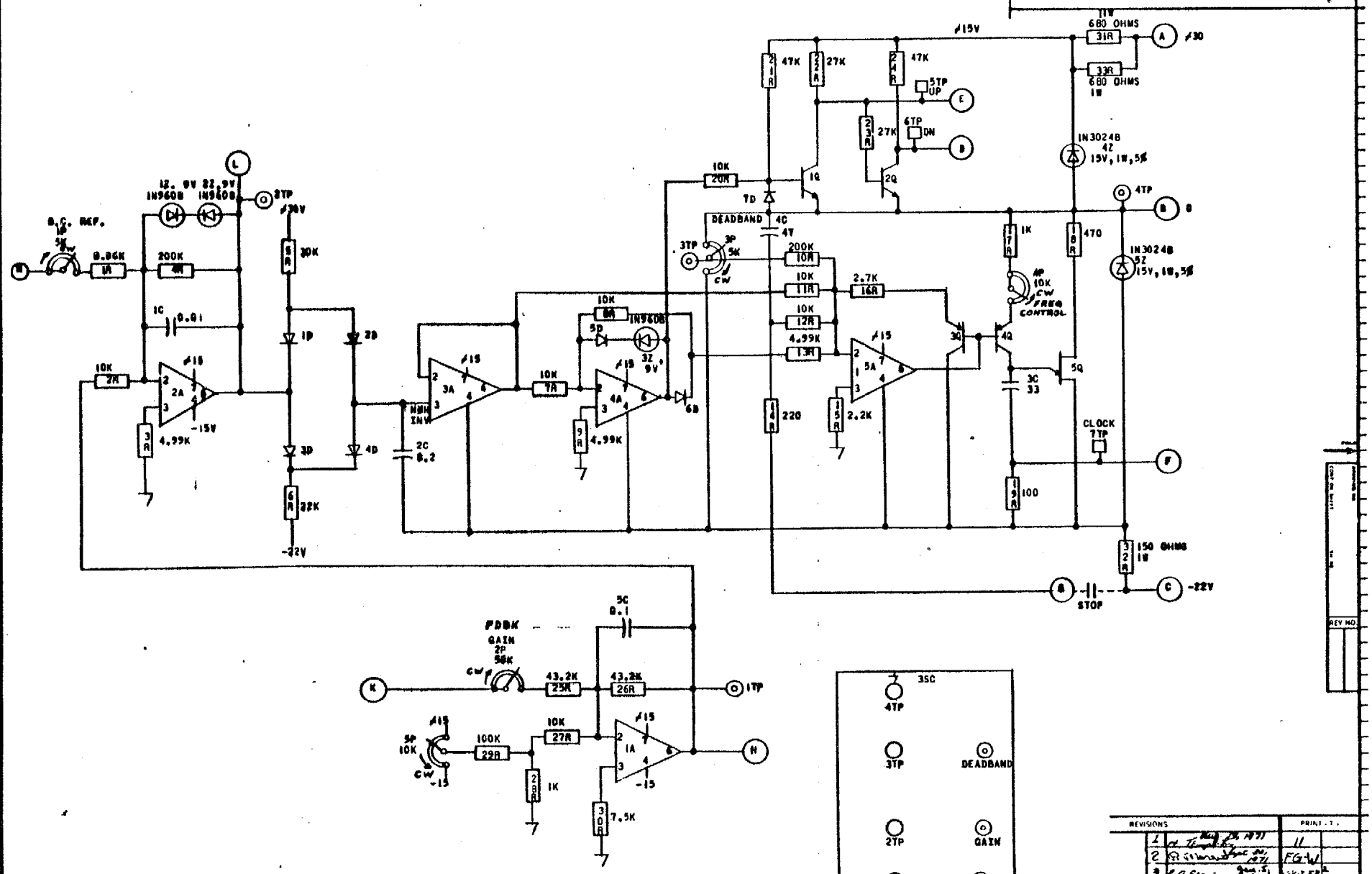
44C304898

UNLESS OTHERWISE SPECIFIED PRACTICES | SURF. | ✓

TITLE

EHC SIGNAL CONVERTER

FIRST MADE FOR



REVISIONS

1		
2		
3		

PRINT - T

11

FG-W

MADE BY Linda Zinkle July 27 1971

CHECKED BY Al Patterson July 27 1971

COMM. & CONT. DEVICES WAYNESBORO, VA.

44C304898

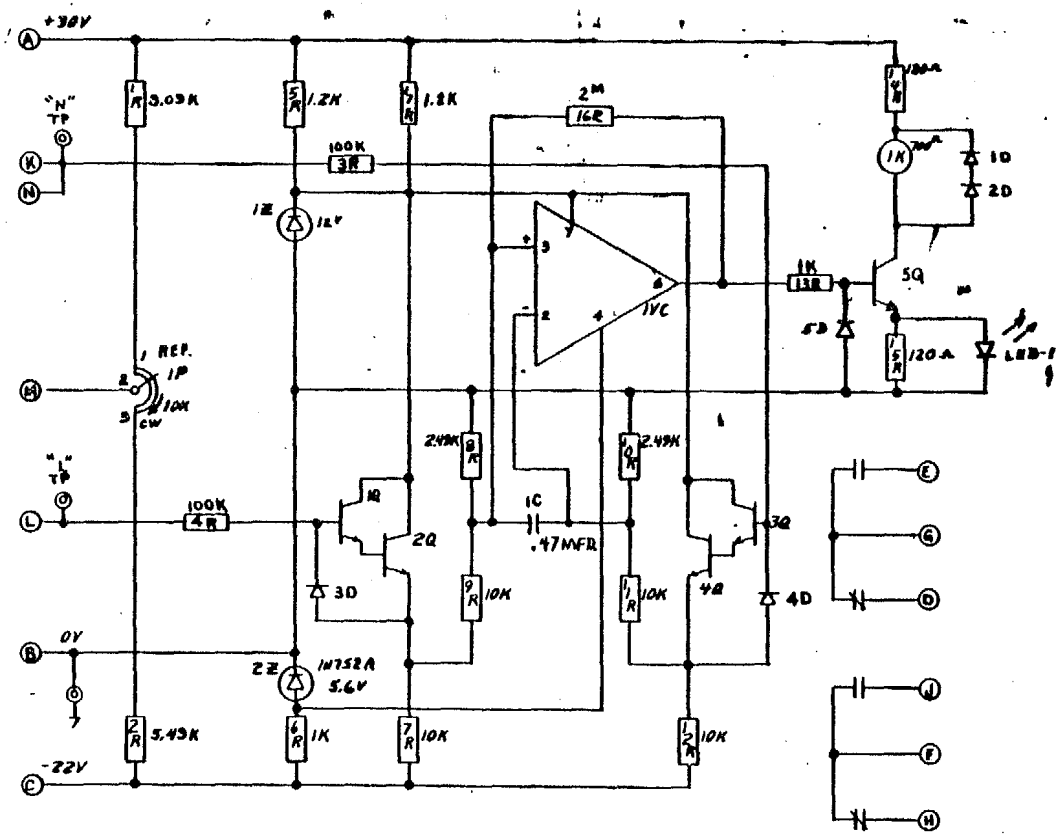
44C305259

GENERAL ELECTRIC

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—
APPLIED PRACTICES | SURFACES | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE: VOLTAGE COMPARATOR — 1VC

FIRST MADE FOR 44B331761-G01



CONNECTIONS

A	+30V	N, K — INPUT
B	0 V	L — SEE NOTE 2
C	-22V	M
D		
E		
F		
G		
H		
J		

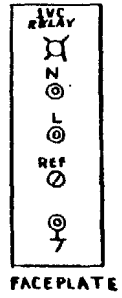
NOTES:

1. IN NORMAL OPERATION THE RELAY IS ENERGIZED. IT CAN BE ADJUSTED TO DROPOUT ON EITHER INCREASING OR DECREASING SIGNALS BETWEEN PLUS AND MINUS 10 VOLTS.
2.

DROPOUT ON	INPUT	JUMPER	REF. TP
INCREASING SIGNAL	N	L-B	L
DECREASING SIGNAL	L	N-B	N

DROPOUT N>L

3. RELAY CONTACTS RATED 2A, 28VDC.
4. RELAY ENERGIZED WHEN LIGHT IS ON.



FACEPLATE

44C305259 REV NO.

REVISIONS
1 V.O. Cox Aug. 7, 1973

PRINTS TO
11
12P DPR
FG-INV

SPECIALTY CONTROL DEPT. WAYNESBORO, VA.

44C305259

GENERAL ELECTRIC

44C304444

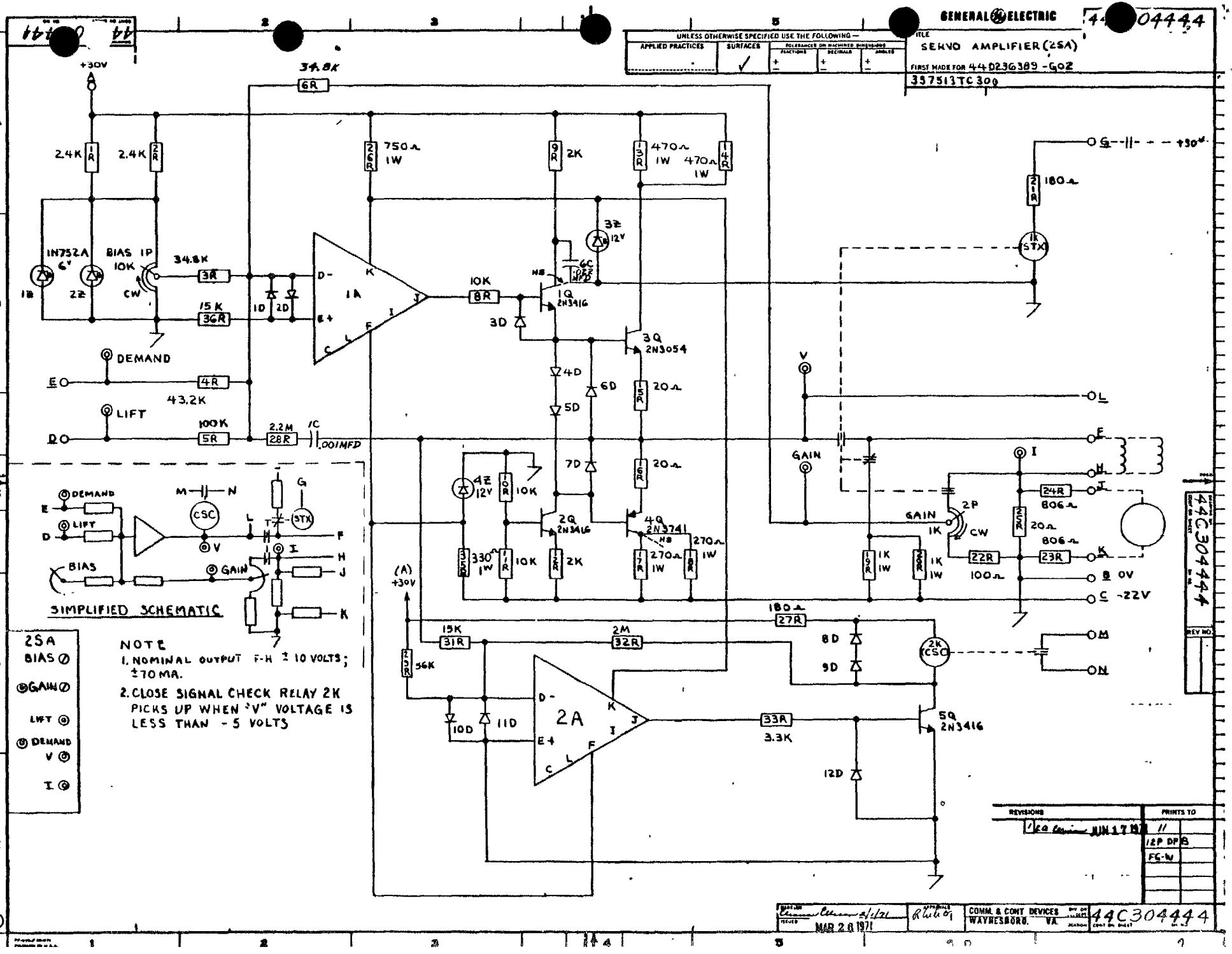
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ±	DECIMALS ±	ANGLES ±
.	✓			

TITLE
SERVO AMPLIFIER (2SA)

FIRST MADE FOR 44D236389-G02

337513TC300



NOTE
1. NOMINAL OUTPUT F-H ± 10 VOLTS; ±70 MA.
2. CLOSE SIGNAL CHECK RELAY 2K PICKS UP WHEN "V" VOLTAGE IS LESS THAN -5 VOLTS

REVISIONS

PRINTS TO
12P DP8
FG-W

44C304444 REV NO.

ISSUED MAR 28 1971		COMM & CONT DEVICES WAYNESBORO, VA	44C304444

GENERAL ELECTRIC

44C306404

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING—
APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE
ELECTRONIC BOILER FEEDPUMP TURBINE CONTROL

FIRST MADE FOR 367513TC380A12 TB122808-809

ELEMENTARY DIAGRAM

REQN. 091-40786 ARKANSAS P&L RUSSELVILLE

PRINTED CIRCUIT BOARD LOCATIONS

RACK 1

LP STOP VALVE		HP STOP VALVE		CONTROL VALVE		SPEED SIGNAL		SPEED SIGNAL		OVERSPEED TEST		REF. LVG.				LOCAL	MANUAL REFERENCE		AUTO REFERENCE							
1DI		1DI		1AØ2		3AØ2		3AØ2		1ØS1		1ØA	1ØA	1ØA	2LVG	1LV	CL & SMD	1DI	7BCI		3SC		CL & SMD	1DI	
X	2	X	4	X	6	X	8	X	10	X	12	13	14	15	16	17	18	X	20	X	22	X	24	25	26	X	28

RACK 2

PRI SPEED			SEC. SPEED			SIGNAL LVG				REGULATION & COMPENSATION		SERVO VALVE LOOP			30HZ OSC	SPEED SENSING 10%	105%	110%	200 RPM	SPEED INDICATION AND SWITCHING			LOW SPEED SWITCH		LSS PWR SUP	
4BA	2FVC		4ØA	2FVC		1ØA	1ØA	1LVG		1ØA	10R	20A		1BE	60SC	1VC	1VC	1VC	1VC	1ØA	2FVC		LSS		LPS	
1	X	3	4	X	6	7	8	9	X	11	12	X	14	X	16	17	18	19	20	21	22	X	24	X	X	27	X

RACK 3

30 VOLT SUPPLY										32 VOLT SUPPLY								34 VOLT SUPPLY									
				LINE A			LINE B					LINE A			LINE B					LINE A			LINE B				
1BB	1BB	1BB		1REC		1SR		1SR		1REC		1REC		1SR		1SR		1REC		1REC		1SR		1SR		1REC	
1	2	3	X	X	6	X	8	X	10	X	12	X	14	X	16	X	18	X	20	21	22	X	24	X	26	X	28

FRONT VIEW

PRINTED CIRCUIT BOARD LOCATIONS

44C306404 REV NO.

REVISIONS	
1	E.A. Clemmons Nov 15 1974
2	E.C. Allison Dec 1 1975

PRINTS TO

11	
12P-D	2
FG-U	
MP-FS	2

CONTROL & CONT. DEVICES WAYNESBORO, VA.

44C306404

GENERAL ELECTRIC — 44C306404

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —
APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE: ELECTRONIC BOILER FEEDPUMP TURBINE CONTROL
FIRST MADE FOR 357513TC300A12 TB122808-609
ELEMENTARY DIAGRAM
REQN. 091-40786 ARKANSAS P&L RUSSELLVILLE

PRINTED CIRCUIT BOARD AND SUB. ASM. LIST

PRINTED CIRCUIT BOARD

SYMBOL	TITLE	ELEM. DIAG.	ORDER NO.	QTY.
1A92	ANALOG OUTPUT BOARD	44C301246	44D236379-G01	1
CL& SMD	CLOCK & STEPPING-MOTOR DRIVE	44C303567	44D331918-G01	2
1CR	COMPENSATION & REGULATION	44A301400	44D236388-G01	1
1DB	DIODE BOARD	44A300243	44D236370-G01	3
1DI	DETECTOR INDICATOR	44C300609	44D381585-G01	5
3A92	ANALOG OUTPUT BOARD	44C304622	44D236379-G03	2
1LVG	SIGNAL LOW VALUE GATE	44A304763	44D236372-G02	1
2FVC	FREQUENCY TO VOLTAGE CONVERTER	44C304682	44D236365-G02	3
1REG	SERIES REG. BD. #1	44A306392	44B317827-G01	6
1SR	SERIES REG. BD. #2	44B306197	44D236371-G04	6
1#A	OPERATIONAL AMPLIFIER	44C300610	44D381589-G01	9
1#SI	OVERSPEED INTEGRATOR	44A302692	44D236373-G02	1
2SA	SERVO AMPLIFIER	44C304444	44D236389-G02	1
3SC	SIGNAL CONVERTER	44C304898	44B331743-G01	1
LPS	LOW SPEED SWITCH POWER SUPPLY	44B306144	44B337328-G01	1
LSS	LOW SPEED SWITCH	44C306444	44B337327-G01	1
1VC	VOLTAGE COMPARATOR	44C305259	44B331761-G01	1
7BCI	BOILER CONTROL INTERFACE	44C306492	44B331746-G03	1
2LVG	REF. LOW VALUE GATE	44A300281	44D236378-G01	1
60SC	3KHZ. OSC. BOARD	44C304462	44B331717-G04	1

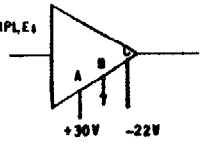
SUB. ASSEMBLIES - PANELS -

TITLE	ORDER NO.	CONNECTION DIAGRAM
CONTROL PANEL	357513KL300A9	44D305974
POWER PANEL	357513KF300A5	44C306526
METER PANEL	357513KF306A9	44C305834
METER PANEL	357513KF306A11	44C306262
RELAY PANEL	357513K307A13	44D305975
DIFF. XFMR. DRIVE	44D315178-G01	44C300882 (MANUAL)
DIFF. XFMR. DRIVE	44D315178-G02	44C304865 (AUTO) 4 SEC STROKE

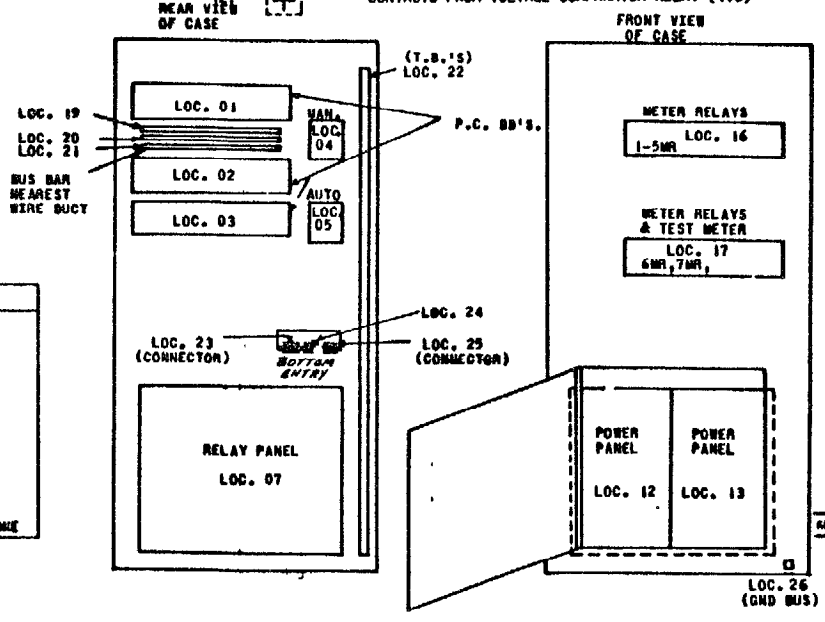
CUSTOMER NOTES:

1. ⏚ INDICATES ZERO REFERENCE POTENTIAL, NOT CASE GROUND.
2. ⏚ INDICATES STATION GROUND.
3. ALL OPERATIONAL AMPLIFIERS HAVE ADDITIONAL CONNECTIONS NOT SHOWN ELSEWHERE

EXAMPLE:



4. THE LOCATION OF THE PRINTED CIRCUIT BOARDS IS SHOWN BY THE NUMBERS APPEARING IN OR NEAR THE SQUARE WHICH REPRESENTS THE P.C. BD. EXAMPLE 0215 WOULD BE RACK #2 AND SLOT #15.
5. 1MR TO 7MR (SHEET 10) ARE METER RELAYS.
6. * INDICATES ITEMS NOT FURNISHED BY WAYNESBORO.
7. INDICATES SHIELDED CABLE.
8. INDICATES TWISTED WIRES.
9. INDICATES TERMINALS AT EHC CASE FOR CUSTOMER CONNECTION
10. INDICATES TERMINALS AT TURBINE
11. INDICATES SUB-PANEL & PC BOARD TERMINALS IN EHC CASE
12. CONTACTS FROM VOLTAGE COMPARATOR RELAY (1VC)



REVISIONS			PRINTS TO
1	C.A. Cline	8/30/78	11
2	C.A. Cline	9/1/78	12P-D46
3	C.A. Cline	8/31/78	FG-8
			12P-16

DRAWN R. Brown Aug 30, 1978	CHECKED R.C. Sullivan	ENGRG. & CONT. DEVICES WAYNESBORO, VA	44C306404	

GENERAL ELECTRIC

44C306404

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —

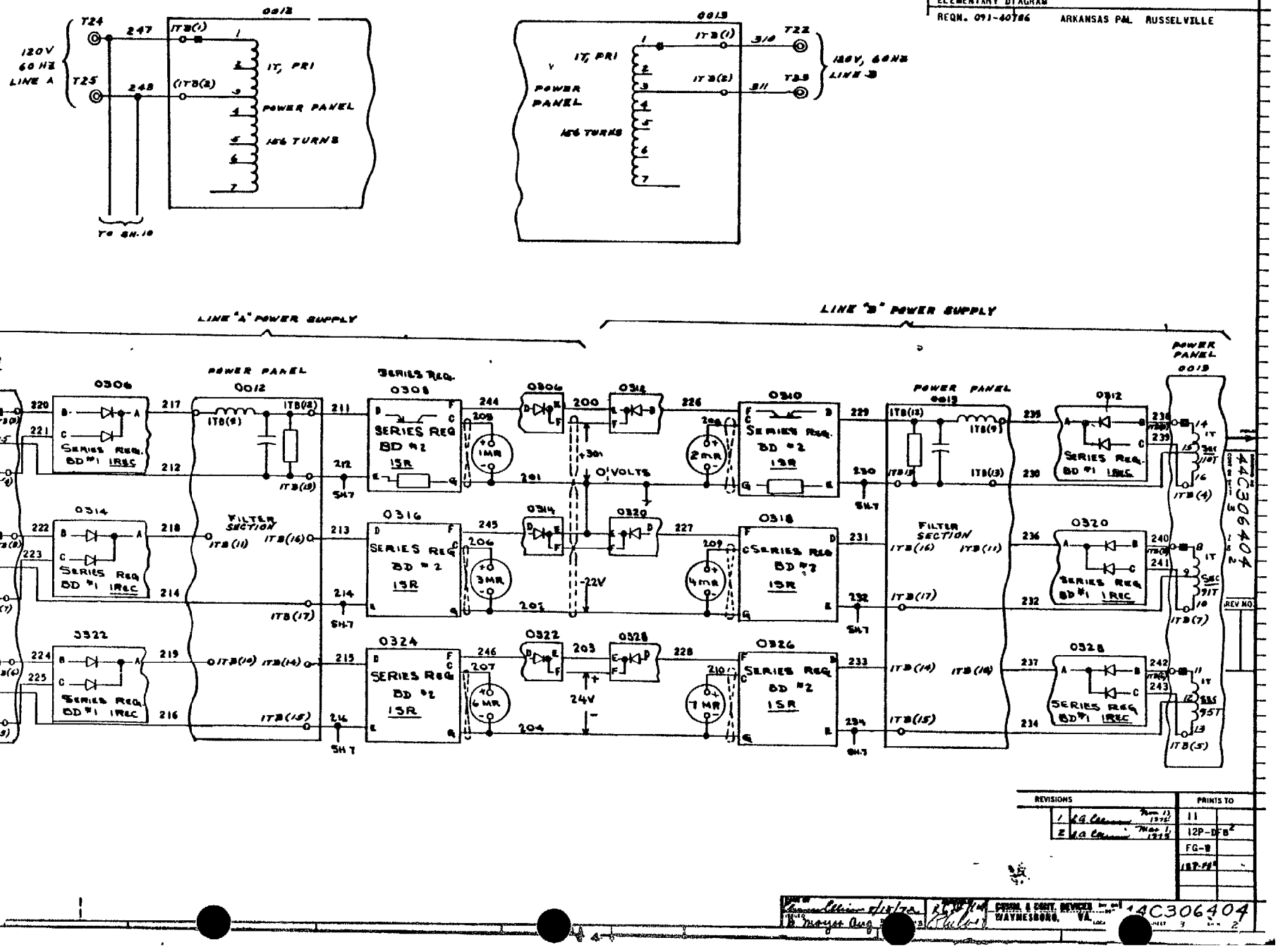
APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

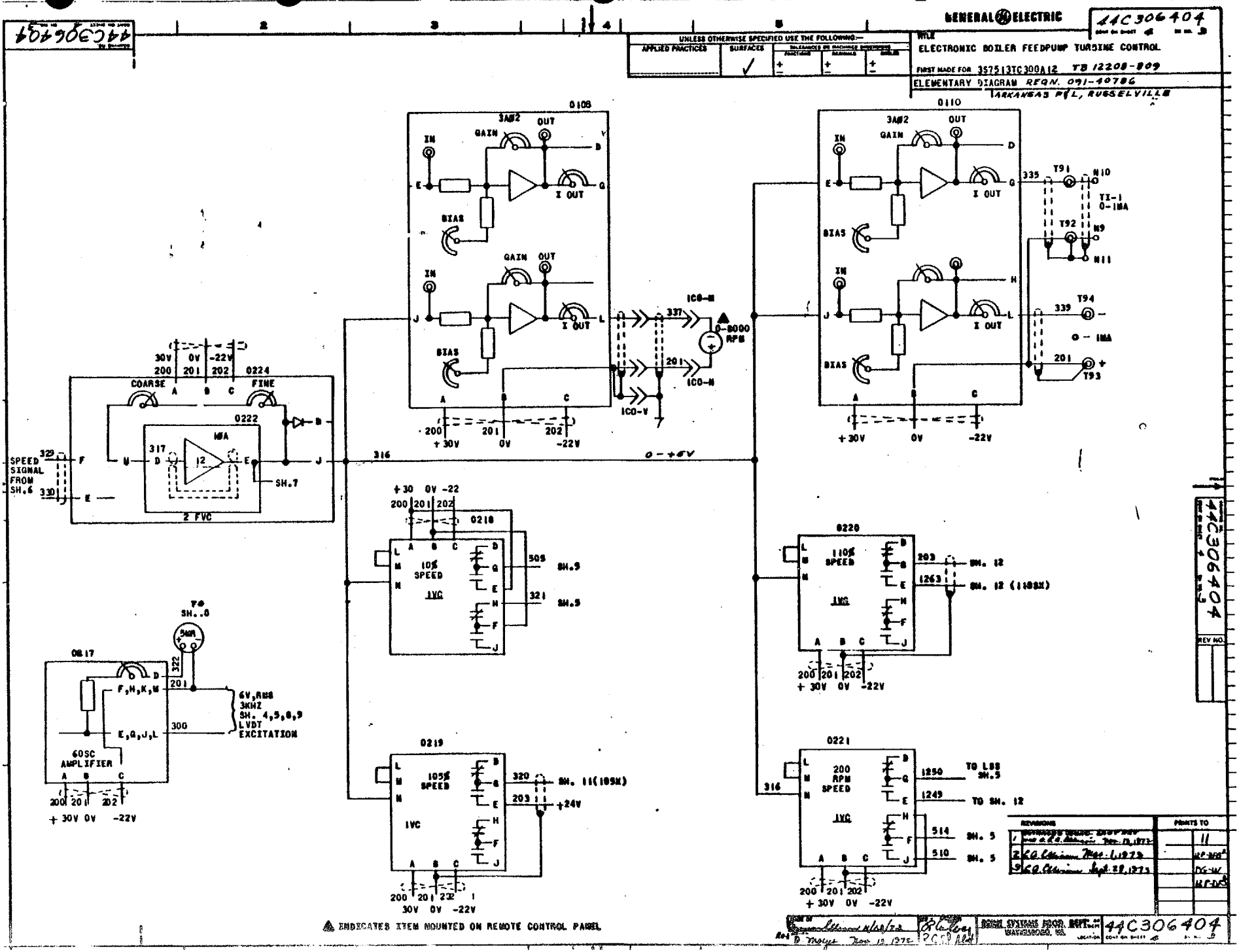
TITLE
ELECTRONIC BOILER FEEDPUMP TURBINE CONTROL

FIRST MADE FOR 357513TC300A12 TB122808-809

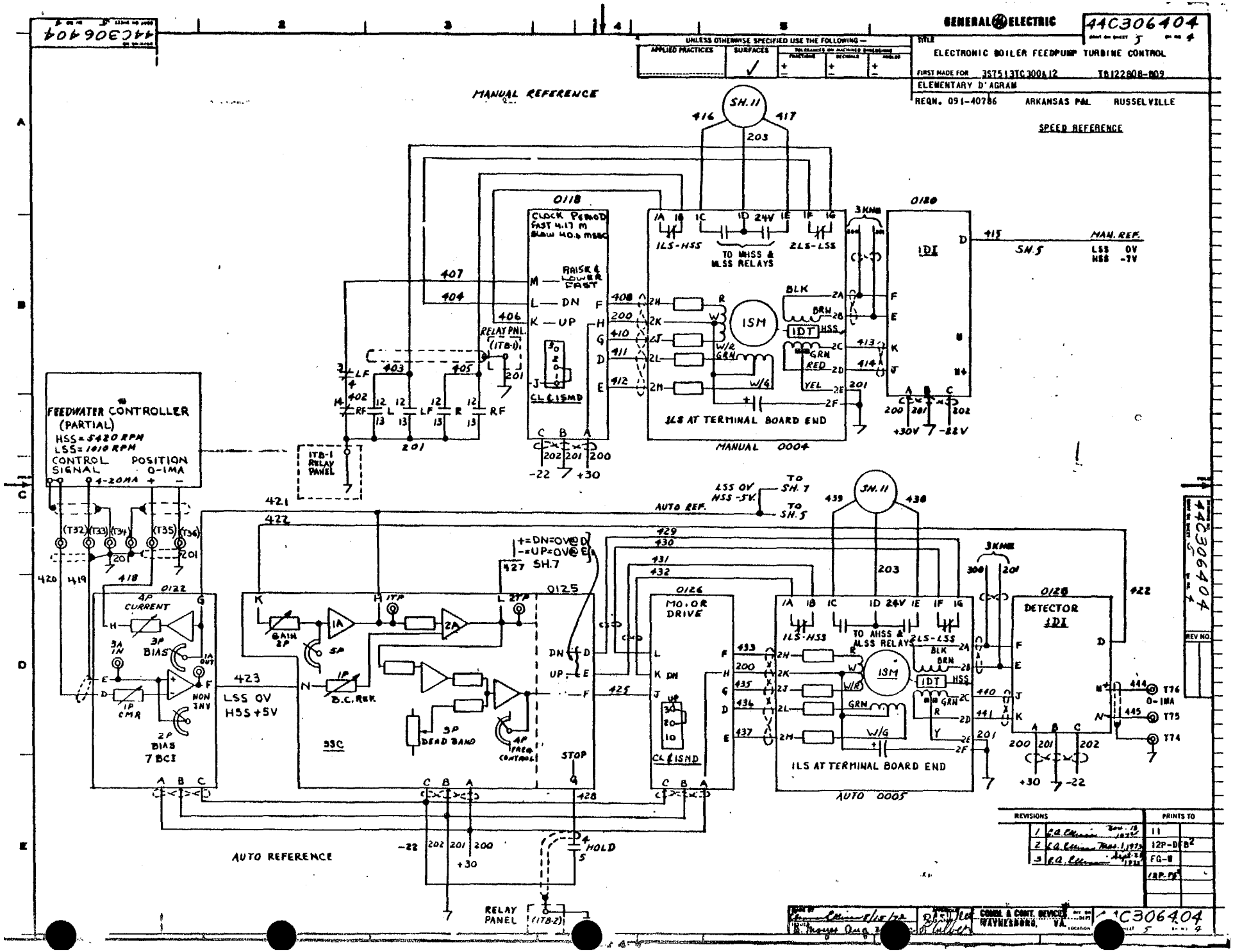
ELEMENTARY DIAGRAM

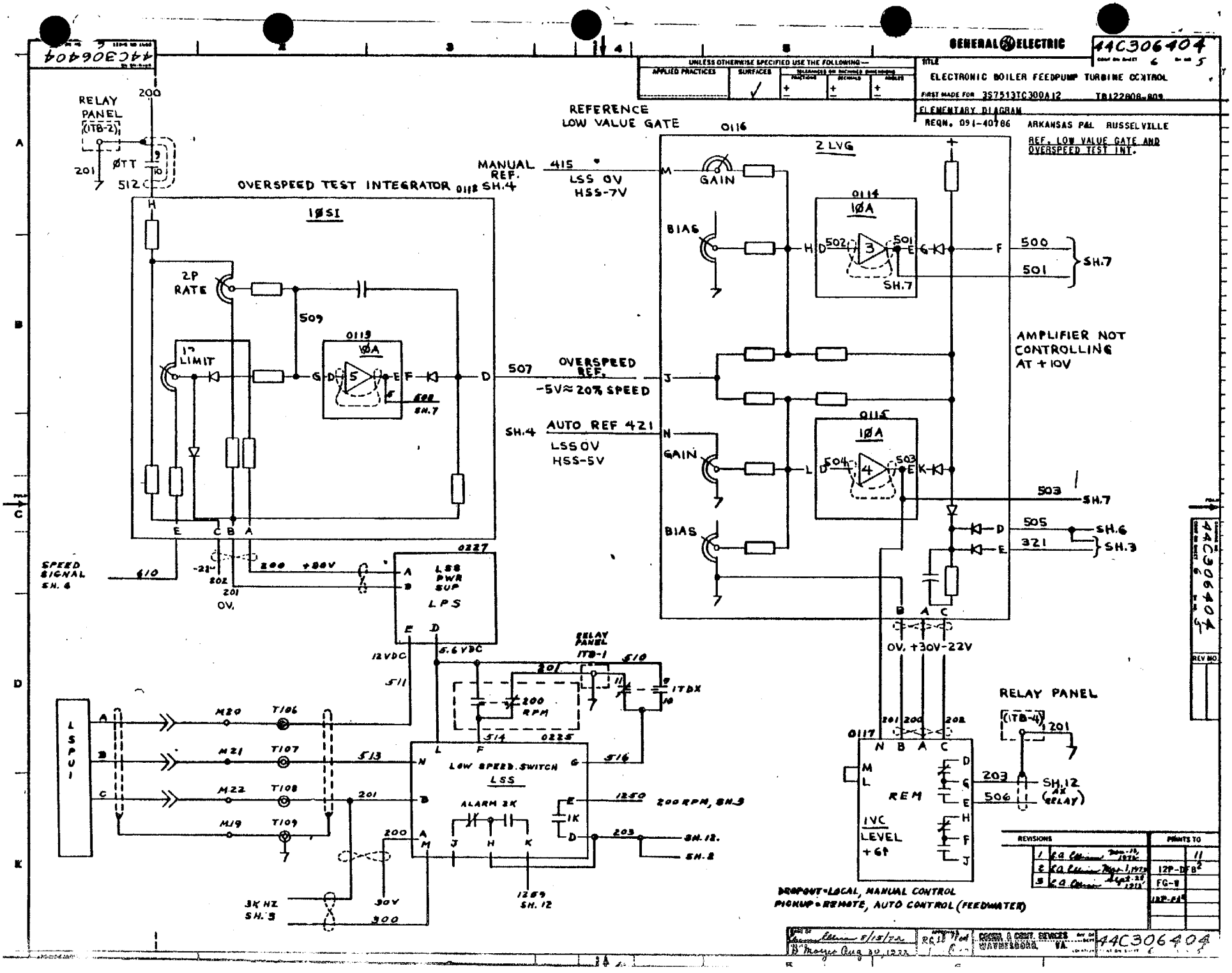
REQN. 091-40786 ARKANSAS P&L RUSSELLVILLE

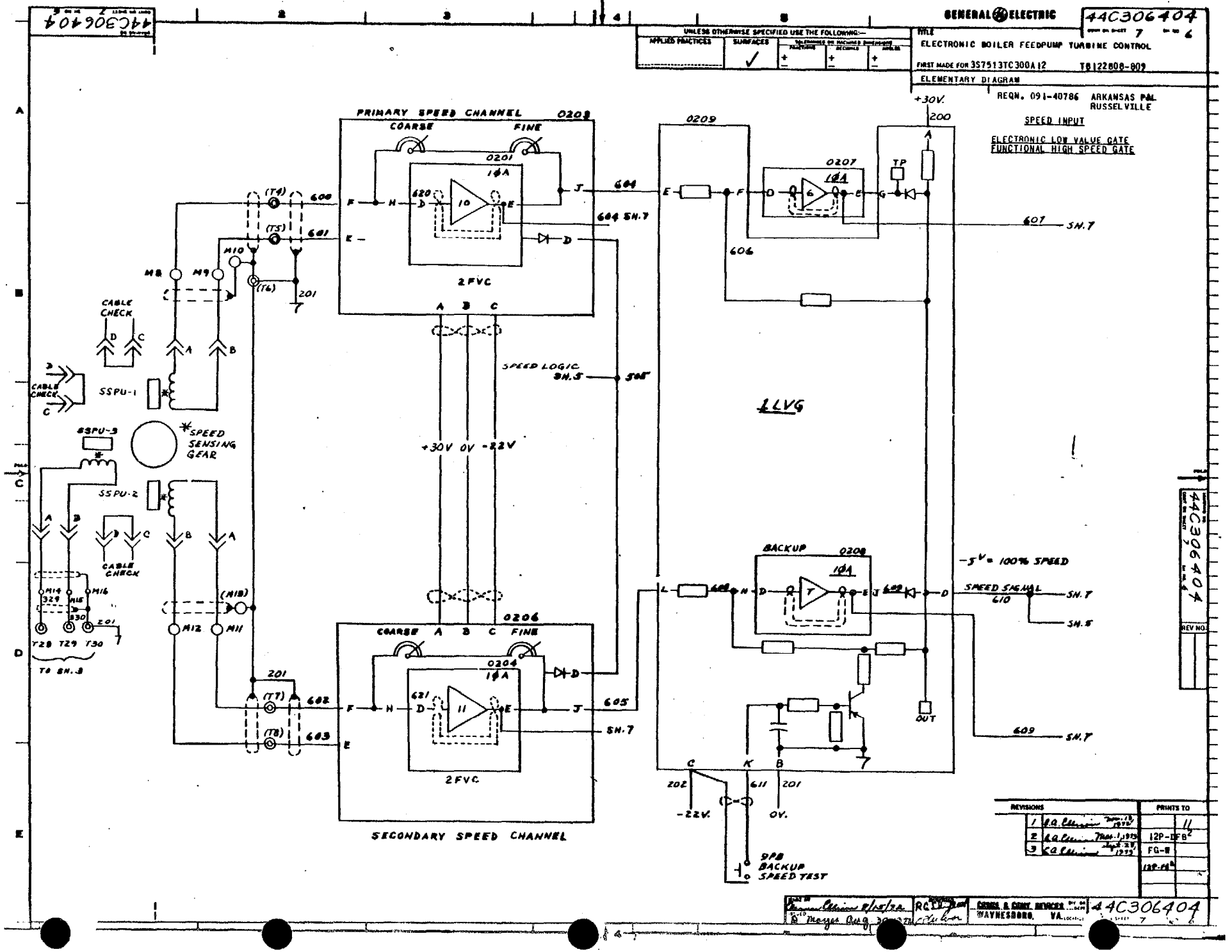




⚠ INDICATES ITEM MOUNTED ON REMOTE CONTROL PANEL


GENERAL ELECTRIC
44C306404
ELECTRONIC BOILER FEEDPUMP TURBINE CONTROL
ELEMENTARY D'AGRAM
REQN. 091-40786 ARKANSAS P&L RUSSELLVILLE
MANUAL REFERENCE
SPEED REFERENCE
FEEDWATER CONTROLLER (PARTIAL)
HSS=5420 RPM
LSS=1010 RPM
CONTROL SIGNAL
POSITION 0-1MA
4-20MA
ITB-1 RELAY PANEL
CLOCK PERIOD FAST 4.17 M SLOW 40.6 MSEC
RAISE & LOWER FAST
DN
UP
RELAY PNL. (ITB-1)
CL & ISMD
ILS-HSS
ILS-LSS
TO HHSS & MLSS RELAYS
ILS AT TERMINAL BOARD END
MANUAL 0004
ISM
IDT
HSS
LSS 0V
HSS -7V
+30V
-22V
AUTO REF.
LSS 0V HSS +5V
B.C. REF.
DEAD BAND
MOTOR DRIVE
DETECTOR IDI
ILS AT TERMINAL BOARD END
AUTO 0005
AUTO REFERENCE
HOLD
RELAY PANEL (ITB-2)
+=DN=OVED
-=UP=OVEE
REVISIONS
PRINTS TO
COMB. & CONT. DEVICES WAYNESBORO, VA.
44C306404


GENERAL ELECTRIC
44C306404
ELECTRONIC BOILER FEEDPUMP TURBINE CONTROL
ELEMENTARY DIAGRAM
REFERENCE
LOW VALUE GATE
0116
2 LVG
MANUAL REF.
415
LSS 0V
HSS-7V
GAIN
BIAS
0114
1ØA
SH.7
500
501
AMPLIFIER NOT CONTROLLING AT +10V
OVERSPEED TEST INTEGRATOR 0118 SH.4
1ØSI
2P RATE
509
I² LIMIT
0119
507
OVERSPEED REF.
-5V≈20% SPEED
SH.4
AUTO REF 421
LSS 0V
HSS-5V
0115
503
505
321
SH.6
SH.3
RELAY PANEL (TB-2)
200
201
512
SPEED SIGNAL SH. 4
510
202
0V.
0227
LSS PWR SUP
LPS
12VDC
5.6VDC
511
RELAY PANEL (TB-1)
201
200 RPM
1TDX
514
0225
LOW SPEED SWITCH
LSS
ALARM 3K
513
516
1250
200 RPM, SH.3
203
SH.12.
SH.2
1259
SH. 12
3K HZ
SH. 3
30V
300
LSPU-1
M20
M21
M22
M19
T106
T107
T108
T109
0V, +30V-22V
RELAY PANEL (TB-4)
0117
REM
IVC LEVEL +6t
203
506
SH.12 (2K RELAY)
DROPOUT=LOCAL, MANUAL CONTROL
PICKUP=REMOTE, AUTO CONTROL (FEEDWATER)
REF. LOW VALUE GATE AND OVERSPEED TEST INT.
ARKANSAS P&L RUSSELLVILLE
REVISIONS
PRINTS TO
12P-D
FG-U
12P-FA

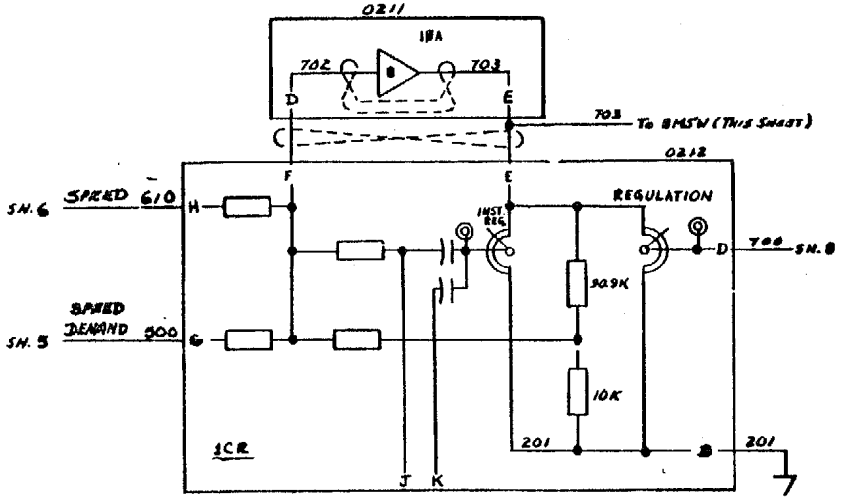

GENERAL ELECTRIC
44C306404
ELECTRONIC BOILER FEEDPUMP TURBINE CONTROL
ELEMENTARY DIAGRAM
REQN. 091-40786 ARKANSAS P&L RUSSELVILLE
SPEED INPUT
ELECTRONIC LOW VALUE GATE
FUNCTIONAL HIGH SPEED GATE
PRIMARY SPEED CHANNEL 0208
COARSE
FINE
0201
1ΦA
2FVC
SECONDARY SPEED CHANNEL
0206
0204
0209
0207
BACKUP 0208
+30V.
200
TP
CABLE CHECK
SSPU-1
SSPU-2
SSPU-3
SPEED SENSING GEAR
SPEED LOGIC SH.5
+30V 0V -22V
LLVG
-5V = 100% SPEED
SPEED SIGNAL
OUT
-22V.
0V.
DPB BACKUP SPEED TEST
TO SH.3
REVISIONS
PRINTS TO
RCTV
GENER. & CONT. DEVICES WAYNESBORO, VA.

GENERAL ELECTRIC

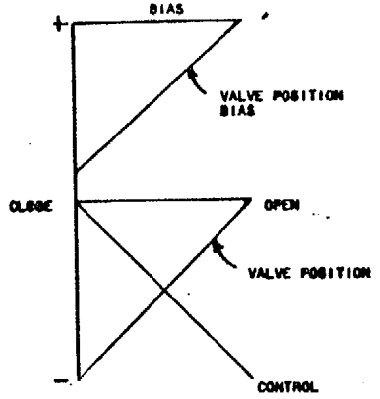
44C306404

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:
APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON DECIMALS: FRACTIONS ± | ANGLES ± | DECIMALS ±

TITLE
ELECTRONIC BOILER FEEDPUMP TURBINE CONTROL
FIRST MADE FOR 357513TC300A12 TB122808-809
ELEMENTARY DIAGRAM
REQN. 091-40786 ARKANSAS P&L RUSSELVILLE



· REGULATION AND STABILIZATION

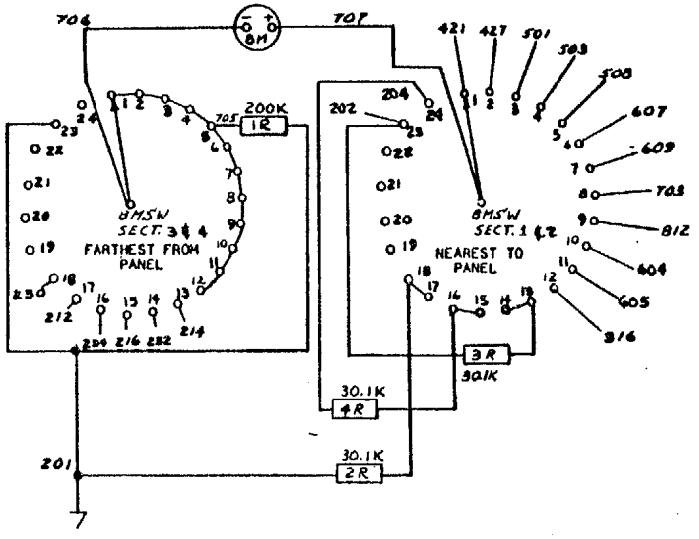


AMPLIFIER

SWITCH POSITION	NUMBER	AMP USED WITH	FUNCTION	RACK LOCATION	APPROXIMATE OUTPUT VOLT RANGE	SHEET NO.
1	1	3SC	SIG. CONV.	0125	0 TO −5VDC	4
2	2	3SC	SIG. CONV.	0125	10VDC TO −10VDC	4
3	3	2LVG	MAN. REF.	0114	10VDC TO −5VDC	5
4	4	2LVG	AUTO REF.	0115	10VDC TO −5VDC	5
5	5	0'SPD	OVERSPEED	0113	10VDC TO −5VDC	5
6	6	1LVG	PRI. SPD.	0207	10VDC TO −7VDC	6
7	7	1LVG	SEC. SPD.	0208	10VDC TO −7VDC	6
8	8	1CR	COMP. & REG.	0211	10VDC TO −10VDC	7
9	9	2SA	SERVO AMP.	0214	7VDC TO −7VDC	8
10	10	2FVC	PRI. SPD.	0201	10VDC TO −7VDC	6
11	11	2FVC	SEC. SPD.	0204	0 TO −7VDC	6
12	12	2FVC	SPD. IND.	0222	0 TO −7VDC	3

POWER SUPPLY

					APPROXIMATE OUTPUT CURRENT RANGE	
13	LINE A	1SR	−22VDC	0316	5 AMPS	2
14	LINE B	1SR	−22VDC	0318	↓	2
15	LINE A	1SR	+24VDC	0322	↓	2
16	LINE B	1SR	+24VDC	0328	↓	2
17	LINE A	1SR	+30VDC	0308	↓	2
18	LINE B	1SR	+30VDC	0310	5 AMPS	2



REVISIONS

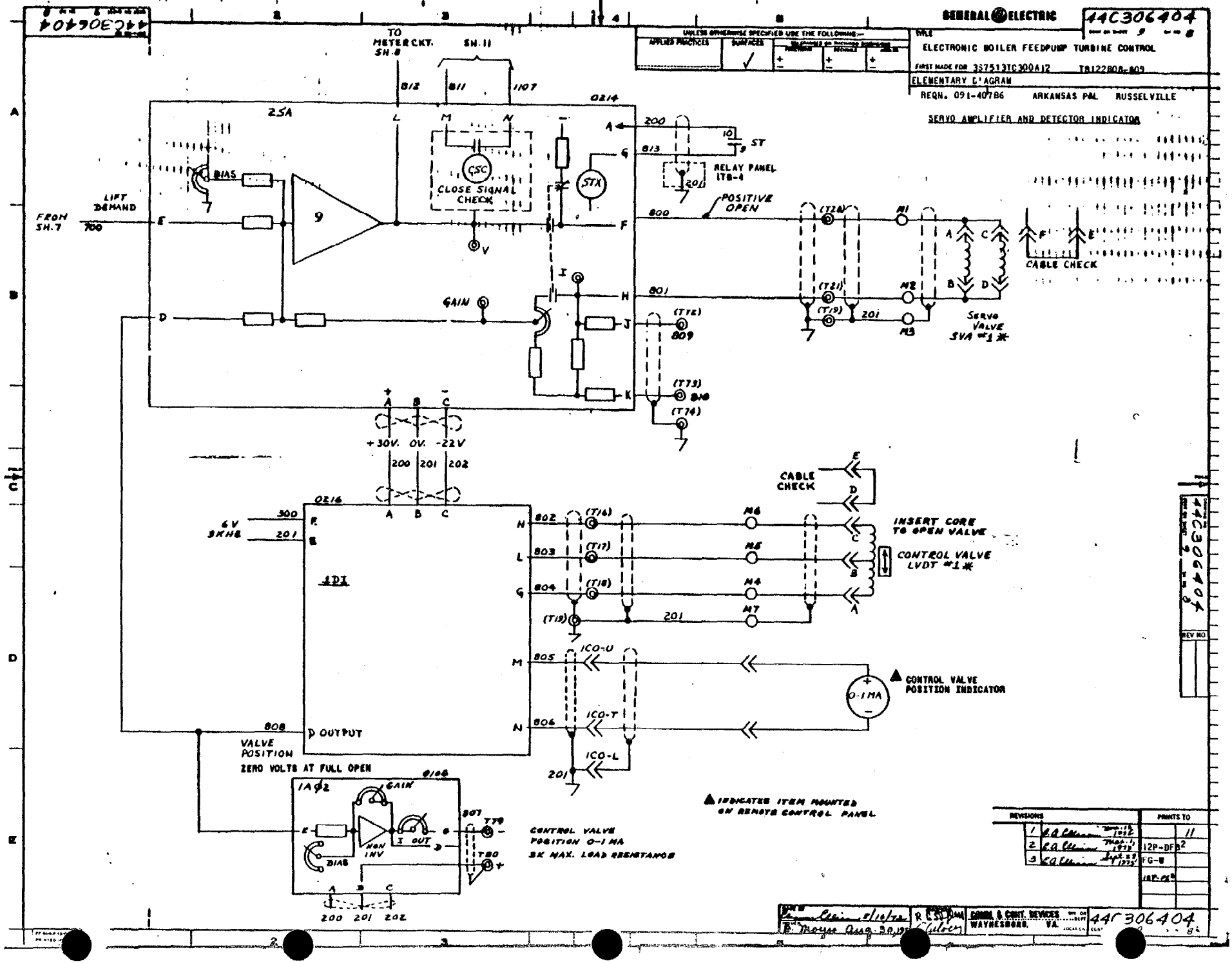
1	K.O. Clem Nov. 15 1972	
2	K.O. Clem May 15 1974	12P-D-B2
3	K.O. Clem Sept. 30 1975	FG-B
		18P-N

PRINTS TO

11	

Reproduction of this drawing is prohibited. Property of General Electric Co. Any reproduction or use without permission is prohibited. If found, please return to General Electric Co., Waynesboro, Va.

44C306404


GENERAL ELECTRIC
44C306404
ELECTRONIC BOILER FEEDPUMP TURBINE CONTROL
ELEMENTARY DIAGRAM
REQN. 091-40786 ARKANSAS P&L RUSSELLVILLE
SERVO AMPLIFIER AND DETECTOR INDICATOR
TO METER CKT. SH.8
SH.11
812
811
1107
0214
25A
L
M
N
A
G
F
H
J
K
I
V
200
10
ST
813
RELAY PANEL 1TB-4
20
STX
CSC
CLOSE SIGNAL CHECK
BIAS
LIFT DEMAND
FROM SH.7
700
E
D
9
GAIN
POSITIVE OPEN
800
801
(T24)
(T21)
(T19)
201
M1
M2
M3
A
B
C
D
F
E
CABLE CHECK
SERVO VALVE SVA *1 *
(T76)
809
(T73)
810
(T74)
+30V.
0V.
-22V
200
201
202
CABLE CHECK
INSERT CORE TO OPEN VALVE
CONTROL VALVE LVDT *1 *
300
6V SKHE
201
0214
4DI
802
803
804
805
806
(T16)
(T17)
(T18)
(T19)
M6
M5
M4
M7
ICO-U
ICO-T
ICO-L
CONTROL VALVE POSITION INDICATOR
0-1MA
808
D OUTPUT
VALVE POSITION
ZERO VOLTS AT FULL OPEN
0106
1A Ø2
GAIN
BIAS
NON INV
I OUT
807
TP9
TB0
INDICATES ITEM MOUNTED ON REMOTE CONTROL PANEL
CONTROL VALVE POSITION 0-1 MA
3K MAX. LOAD RESISTANCE
REVISIONS
PRINTS TO
12P-DF2
FG-B
CONTROL & CONT. DEVICES
WAYNESBORO, VA.
44C306404

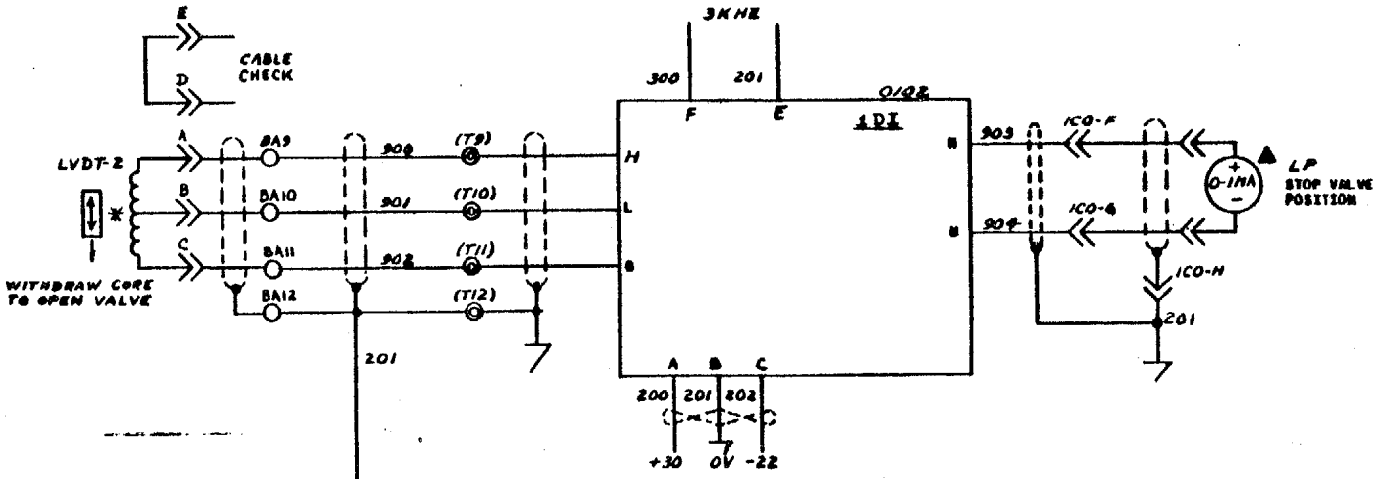
GENERAL ELECTRIC

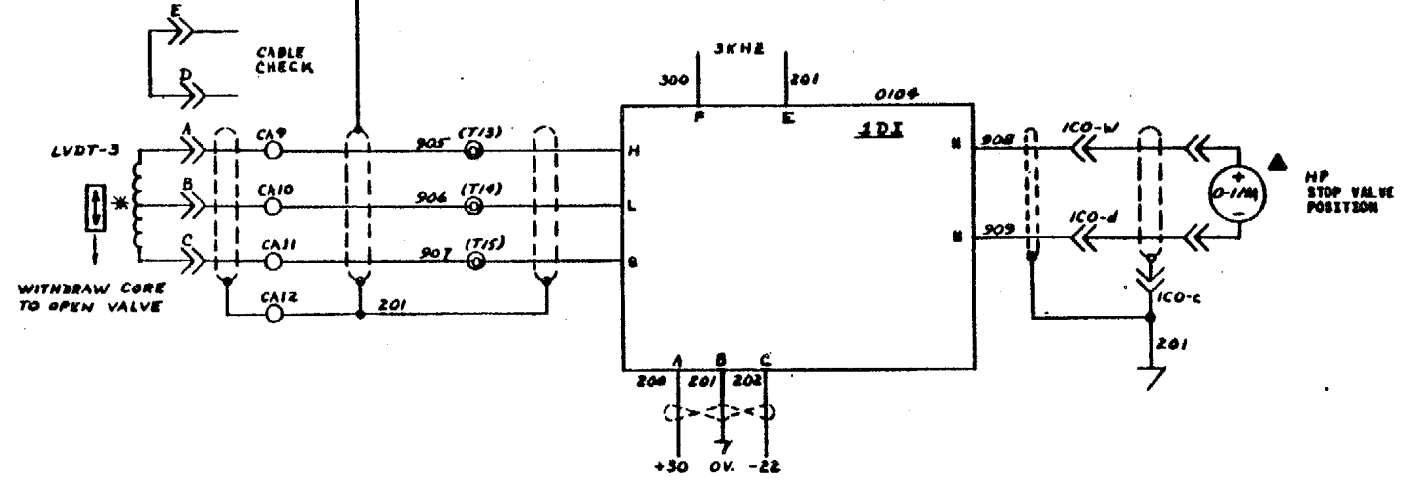
44C306404

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING—
APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON DECIMALS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE
ELECTRONIC BOILER FEEDPUMP TURBINE CONTROL
FIRST MADE FOR 357513TC300A12 TB122808-809
ELEMENTARY DIAGRAM
REQN. 091-40786 ARKANSAS P&L RUSSELVILLE

STOP VALVE POSITION





▲ INDICATES ITEMS MOUNTED ON REMOTE CONTROL PANEL

44C306404

REV NO

PRINTS TO
12P-B[illegible]2
FG-B
[illegible]

DRAWN [illegible] 8/16/72
CHECKED [illegible]
APPROVED [illegible]
GSD&C ENGR. SERVICES WAYNESBORO, VA.
44C306404

GENERAL ELECTRIC

44C306404 11 10

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

TITLE

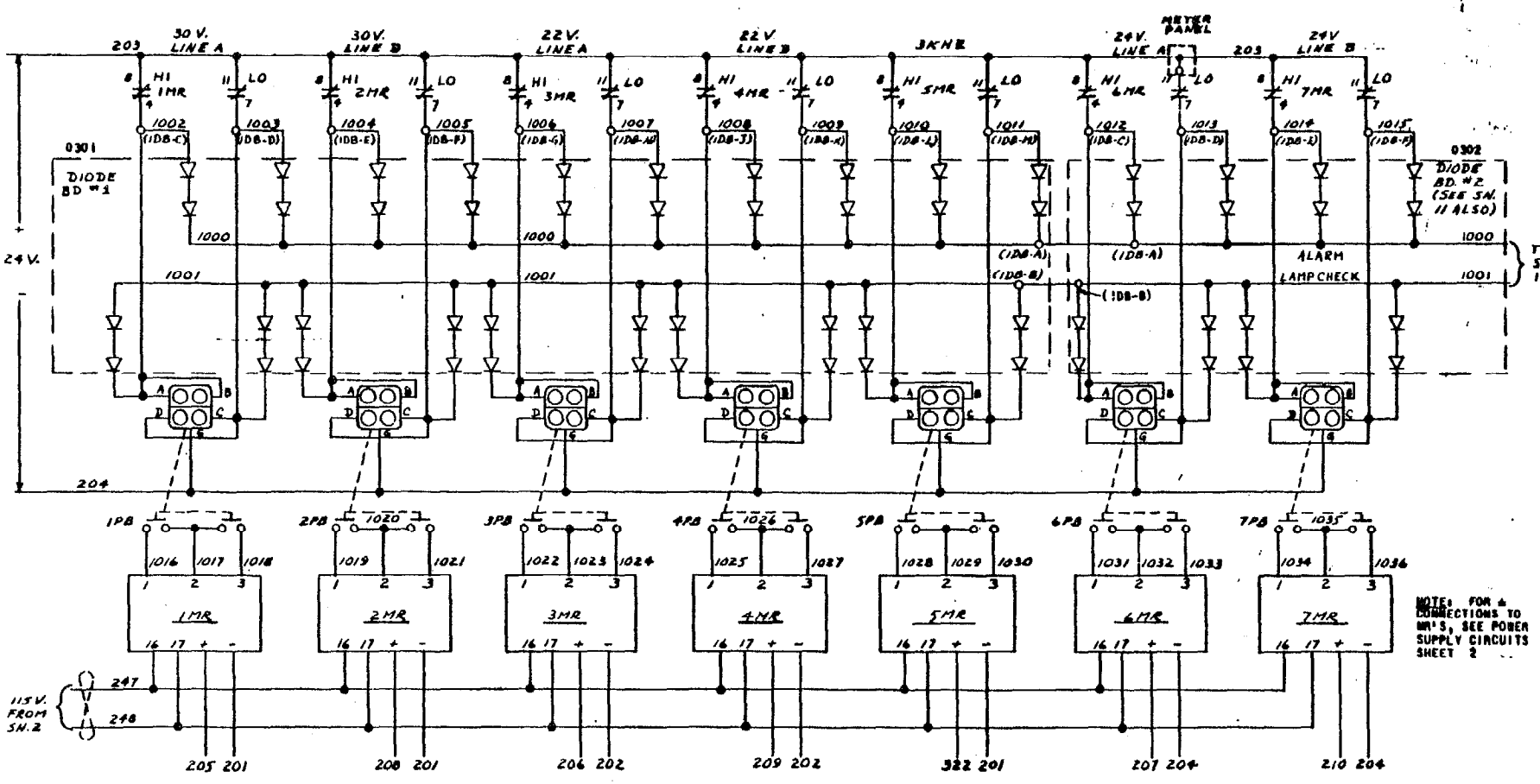
ELECTRONIC BOILER FEEDPUMP TURBINE CONTROL

FIRST MADE FOR 3S7513TC300A12 TB122808-809

ELECTRONIC DIAGRAM

REQN. 091-40786 ARKANSAS P&L RUSSELVILLE

MONITOR



NOTE: FOR ▲ CONNECTIONS TO MR'S, SEE POWER SUPPLY CIRCUITS SHEET 2

44C306404 11 10

REV NO

REVISIONS		PRINTS TO
1	E.C. [illegible]	11
2	E.C. [illegible]	12P-DFB2
3	E.C. [illegible]	FG-W
		12P-FB

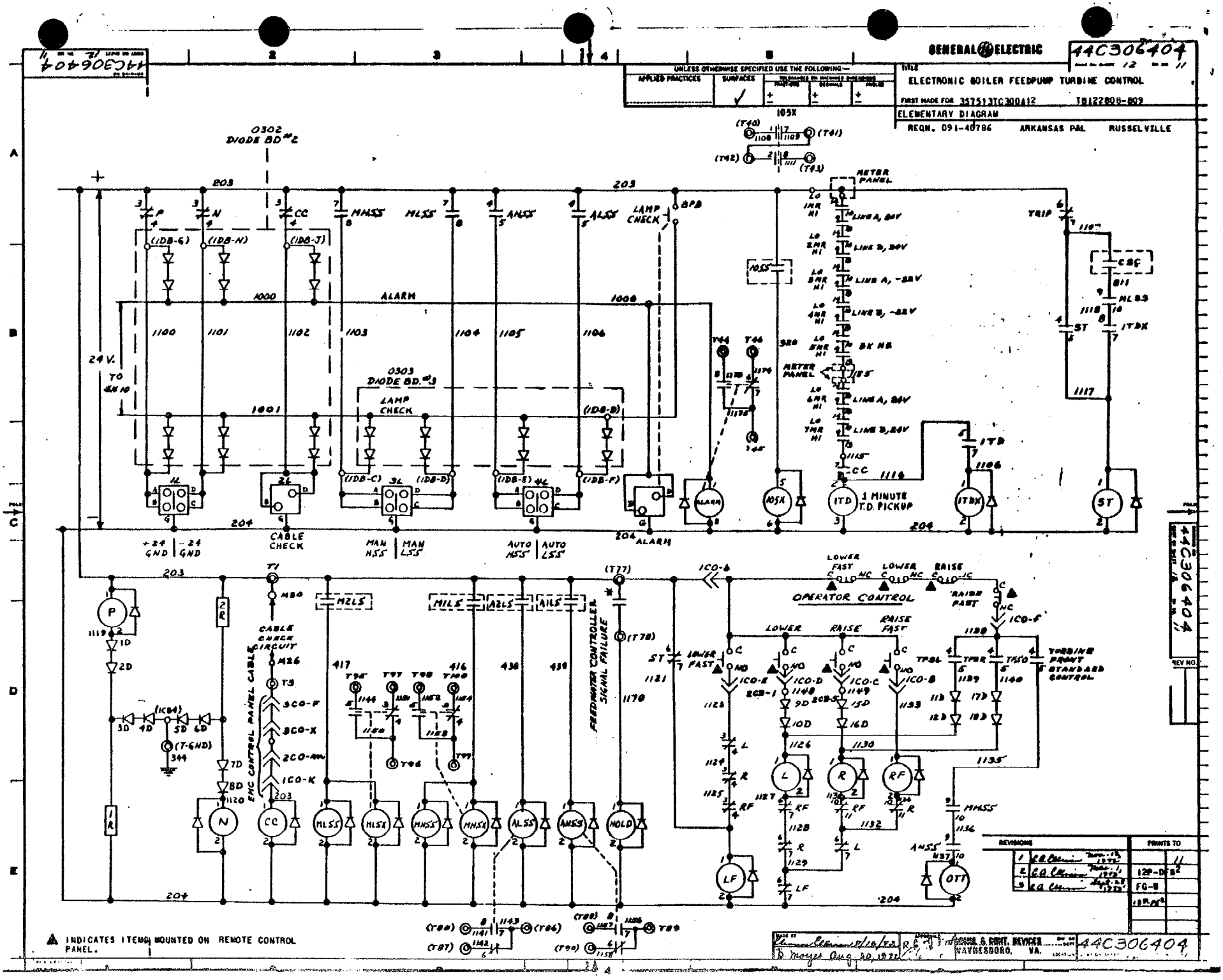
DRAWN [illegible] DATE [illegible]

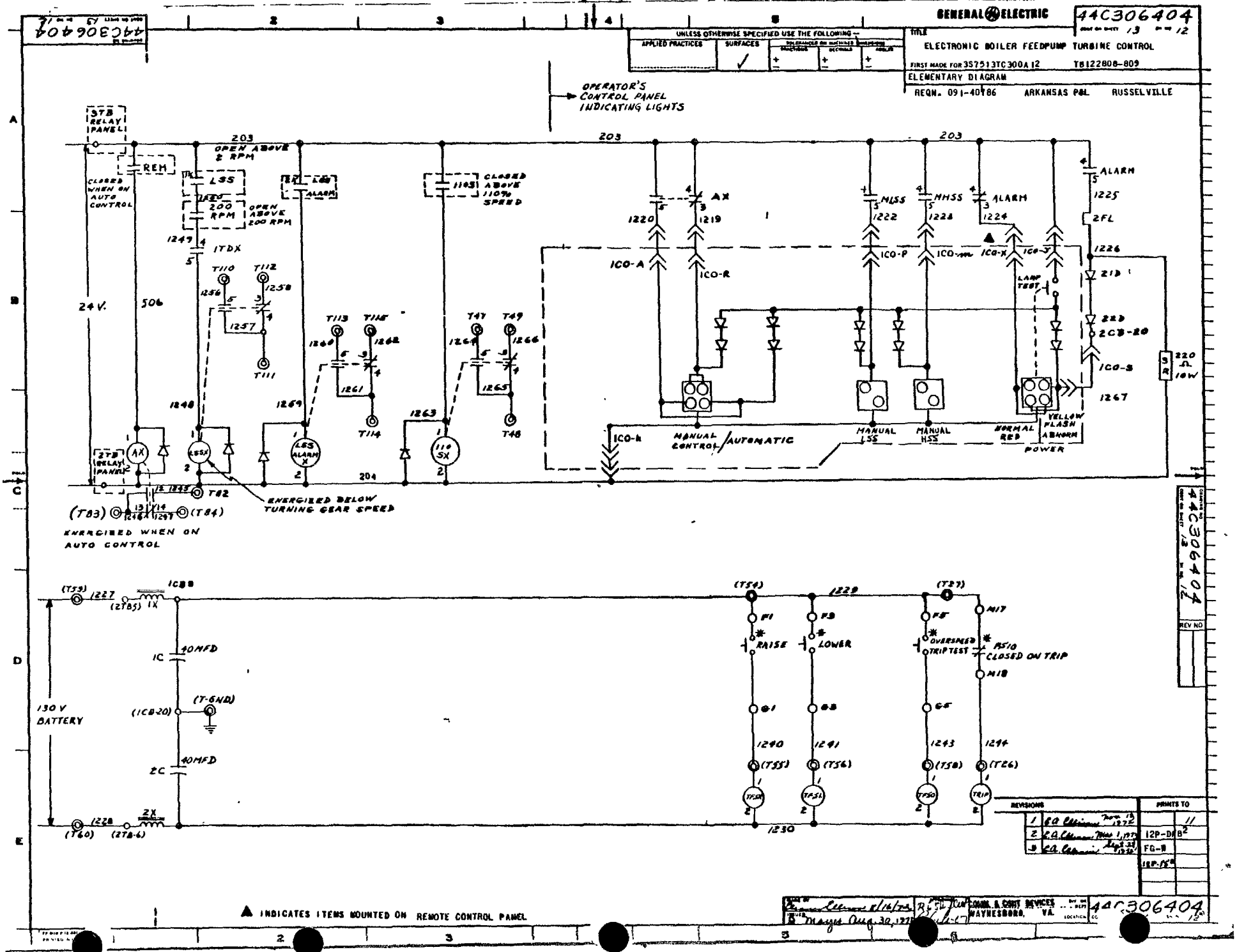
R. Morgan Aug 30, 1[illegible]

RG [illegible]

CONTROL & COMM. DEVICES WAYNESBORO, VA.

44C306404 11


GENERAL ELECTRIC
44C306404
ELECTRONIC BOILER FEEDPUMP TURBINE CONTROL
ELEMENTARY DIAGRAM
REQN. 091-40786 ARKANSAS P&L RUSSELLVILLE
0302 DIODE BD. #2
0303 DIODE BD. #3
24 V. TO 4KV
+24 GND
-24 GND
CABLE CHECK
MAN HSS
MAN LSS
AUTO HSS
AUTO LSS
ALARM
LAMP CHECK
METER PANEL
1 MINUTE T.D. PICKUP
OPERATOR CONTROL
LOWER FAST
LOWER
RAISE
RAISE FAST
CABLE CHECK CIRCUIT
EHC CONTROL PANEL CABLE
FEEDWATER CONTROLLER SIGNAL FAILURE
TURBINE FRONT STANDARD CONTROL
INDICATES ITEMS MOUNTED ON REMOTE CONTROL PANEL.


GENERAL ELECTRIC
44C306404
TITLE
ELECTRONIC BOILER FEEDPUMP TURBINE CONTROL
ELEMENTARY DIAGRAM
REQN. 091-401186 ARKANSAS P&L RUSSELLVILLE
OPERATOR'S CONTROL PANEL INDICATING LIGHTS
STB RELAY PANEL
REM
CLOSED WHEN ON AUTO CONTROL
LSS
OPEN ABOVE 200 RPM
OPEN ABOVE 2 RPM
LSS ALARM
CLOSED ABOVE 110% SPEED
1TDX
T110
T112
T111
T113
T114
T147
T148
T149
24 V.
506
AX
LSSX
LSS ALARM X
110 SX
ENERGIZED BELOW TURNING GEAR SPEED
ENERGIZED WHEN ON AUTO CONTROL
(T83)
(T84)
T82
ICO-A
ICO-R
ICO-P
ICO-M
ICO-X
ICO-Y
ICO-S
ICO-L
MANUAL/AUTOMATIC
MANUAL LSS
MANUAL HSS
NORMAL RED
YELLOW FLASH ABNORM POWER
LAMP TEST
MLSS
MHSS
ALARM
220 Ω 10W
2CB-20
130 V BATTERY
ICB
IC 40MFD
2C 40MFD
(T-GND)
(ICB-20)
1X
2X
(T59)
(T60)
(T54)
(T27)
RAISE
LOWER
OVERSPEED TRIP TEST
MS10 CLOSED ON TRIP
TFSR
TFSL
TFSO
TRIP
(T55)
(T56)
(T58)
(T26)
▲ INDICATES ITEMS MOUNTED ON REMOTE CONTROL PANEL
GOVERNOR & CONT DEVICES WAYNESBORO, VA.
PRINTS TO
12P-DRB
FG-B

GENERAL ELECTRIC

44C306404

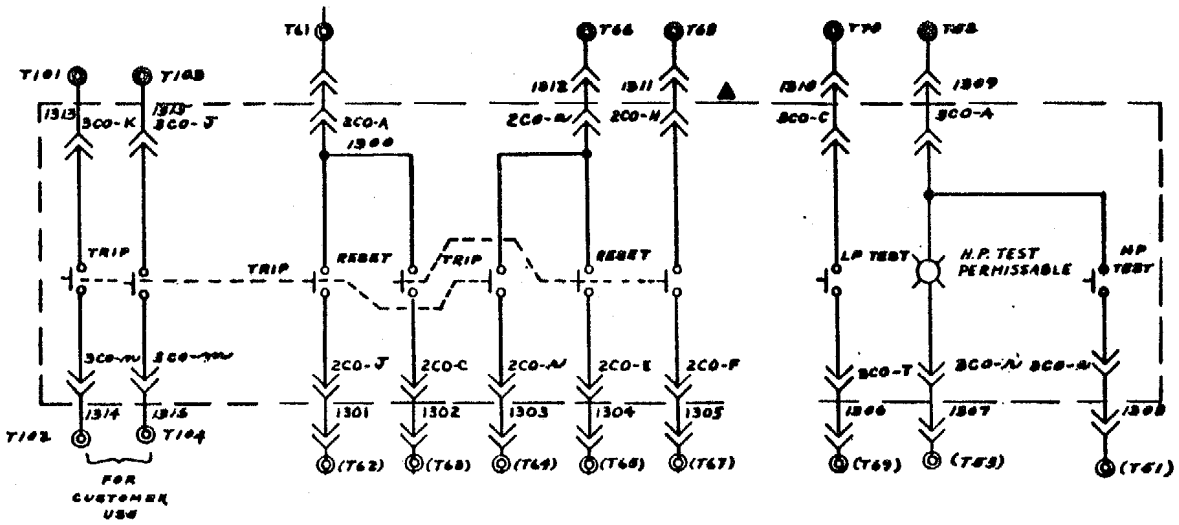
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —

APPLIED PRACTICES	SURFACES	DIMENSIONS ON MACHINED SURFACES: FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

TITLE
ELECTRONIC BOILER FEEDPUMP TURBINE CONTROL

FIRST MADE FOR 357513TC300A12 122808, 809 091-40785

ELEMENTARY DIAGRAM ARKANSAS P&L RUSSELVILLE



▲ INDICATES ITEMS MOUNTED ON REMOTE CONTROL PANEL

REVISIONS	PRINTS TO
1 K.R. Clemens Jan 15 1972	11
	12P-DF82
	FG-W
	IRPF82

DRAWN K.R. Clemens 9/24/70
CHECKED R. Moyer Aug 20, 1970
RELEASED Fulton

GENERAL ELECTRIC CONTROL & COMM. DEVICES DEPT. WAYNESBORO, VA. LOCATION

44C306404

CONT ON SHEET F SH NO 13

GENERAL ELECTRIC

44C306405

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —

APPLIED PRACTICES	SURFACES			
	✓			

TITLE

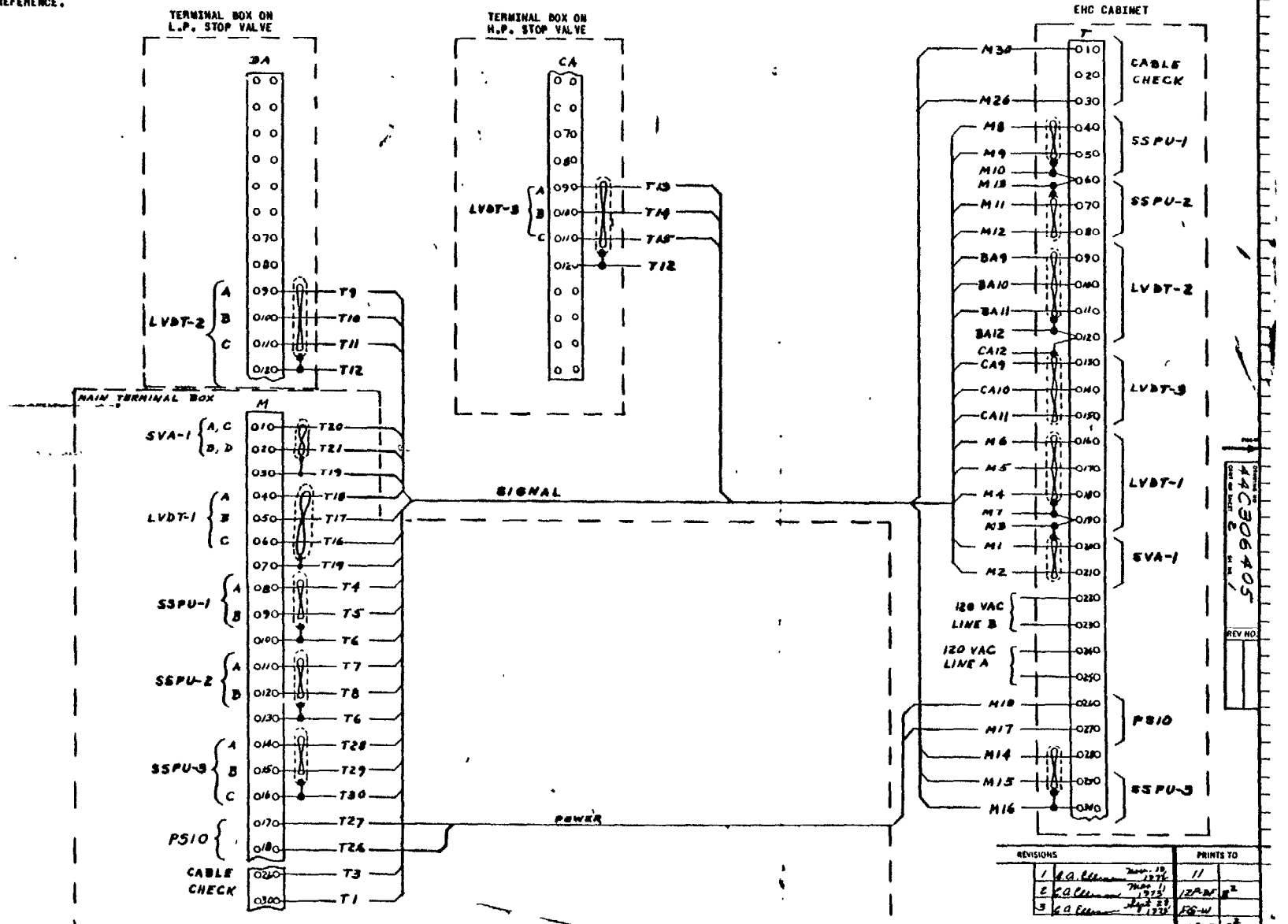
ELECTRONIC FEEDPUMP TURBINE CONTROL

FIRST MADE FOR 3575 13TC300A12 TURBINE: 122808,809 691-40706

INTERCONNECTION DIAG. ARKANSAS P&L RUSSELLVILLE

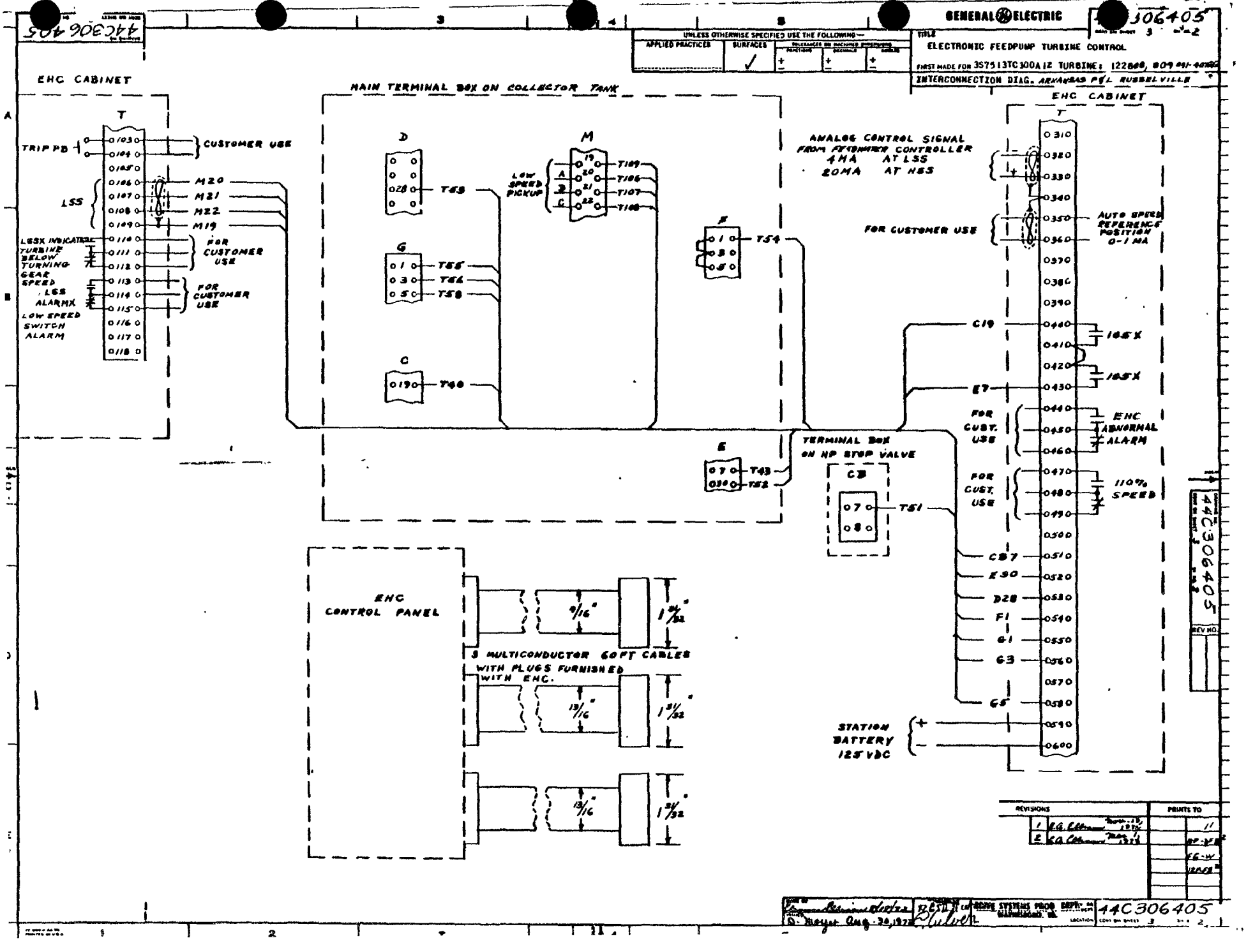
NOTE:

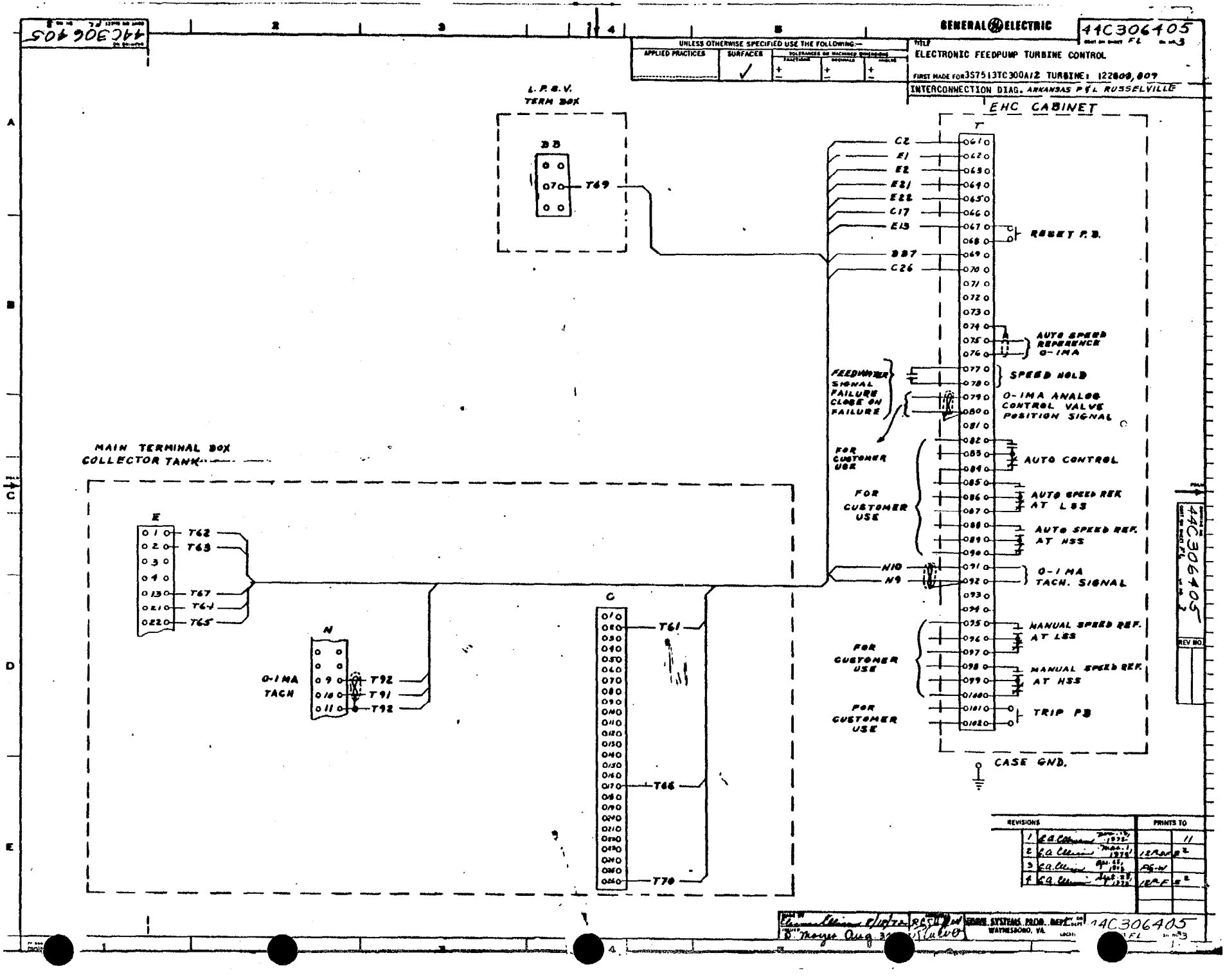
1. REFER TO 44C306404 (EHC ELEMENTARY) AND 1348550BB (TURBINE WIRING DRG.) FOR REFERENCE.



REVISIONS			PRINTS TO	
1	K.A. Ellen	Nov. 19 1976	11	
2	K.A. Ellen	Mar. 1 1977	12A-M	B2
3	K.A. Ellen	Sept. 28 1977	FS-M	
			12D-F	52

44C306405


GENERAL ELECTRIC
44C306405
ELECTRONIC FEEDPUMP TURBINE CONTROL
EHC CABINET
MAIN TERMINAL BOX ON COLLECTOR TANK
TRIP PB
CUSTOMER USE
LSS
M20
M21
M22
M19
LSSX INDICATING TURBINE BELOW TURNING GEAR SPEED
LSS ALARMX
FOR CUSTOMER USE
LOW SPEED SWITCH ALARM
LOW SPEED PICKUP
ANALOG CONTROL SIGNAL FROM FEEDWATER CONTROLLER
4MA AT LSS
20MA AT NSS
FOR CUSTOMER USE
AUTO SPEED REFERENCE POSITION 0-1 MA
C19
E7
185X
185X
FOR CUST. USE
EHC ABNORMAL ALARM
FOR CUST. USE
110% SPEED
TERMINAL BOX ON HP STOP VALVE
CB7
E30
D28
F1
G1
G3
G5
STATION BATTERY 125 VDC
EHC CONTROL PANEL
3 MULTICONDUCTOR 60FT CABLES WITH PLUGS FURNISHED WITH EHC.
9/16"
13/16"
1 31/32"
REVISIONS
PRINTS TO
GE-W
44C306405


GENERAL ELECTRIC
44C306405
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—
APPLIED PRACTICES
SURFACES
TOLERANCES ON MACHINED DIMENSIONS
FRACTIONS
DECIMALS
ANGLES
TITLE
ELECTRONIC FEEDPUMP TURBINE CONTROL
FIRST MADE FOR 3S7513TC300A12 TURBINE: 122800, 809
INTERCONNECTION DIAG. ARKANSAS P & L RUSSELLVILLE
L.P.B.V. TERM BOX
BB
T69
EHC CABINET
C2
E1
E2
E21
E22
C17
E13
RESET P.B.
BB7
C26
AUTO SPEED REFERENCE 0-1MA
SPEED HOLD
FEEDPUMP SIGNAL FAILURE CLOSE ON FAILURE
0-1MA ANALOG CONTROL VALVE POSITION SIGNAL
FOR CUSTOMER USE
AUTO CONTROL
AUTO SPEED REF AT LSS
AUTO SPEED REF. AT HSS
N10
N9
0-1 MA TACH. SIGNAL
MANUAL SPEED REF. AT LSS
MANUAL SPEED REF. AT HSS
TRIP PB
CASE GND.
MAIN TERMINAL BOX COLLECTOR TANK
E
T62
T63
T67
T64
T65
N
0-1MA TACH
T92
T91
T92
C
T61
T66
T70
REVISIONS
PRINTS TO
GENERAL ELECTRIC SYSTEMS PROD. DEPT. WAYNESBORO, VA.
44C306405

GENERAL ELECTRIC

44C306444

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

TITLE: LOW SPEED SWITCH

FIRST MADE FOR 3.5 25/8 TC 3004 A.V.

CLOCK

COUNTER

ALARM

SPEED DETECTOR

SPEED INHIBIT

STANDSTILL

SPEED

ALARM RELAY

LSS RELAY

DIGITAL OUTPUT

REVISIONS

PRINTS TO

DRIVE SYSTEMS PROD. DEPT. WAYNESBORO, VA.

44C306444

GENERAL ELECTRIC

44C3C6444

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:— APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE: LOW SPEED SWITCH

FIRST MADE FOR 357513TC300A11

Z

4042

355PU

20

19

8V

2CL

Z

$\overline{Z}$

SET PULSE

+5.6V

5.6V

1/2 W

9R

7.5K

1C

150MF

2C

150MF

20

20

SET

0V

0V

SHAFT SPEED PICK-UP PULSE

TOOTH WHEEL PICKUP

3C .0022MF

0V

21

22

22

23

24

24

24

24

5.6V

2CL

ZTP

18

18

18

18

17

200 RPM

155PU

355PU

255PU

55PU

REVISIONS

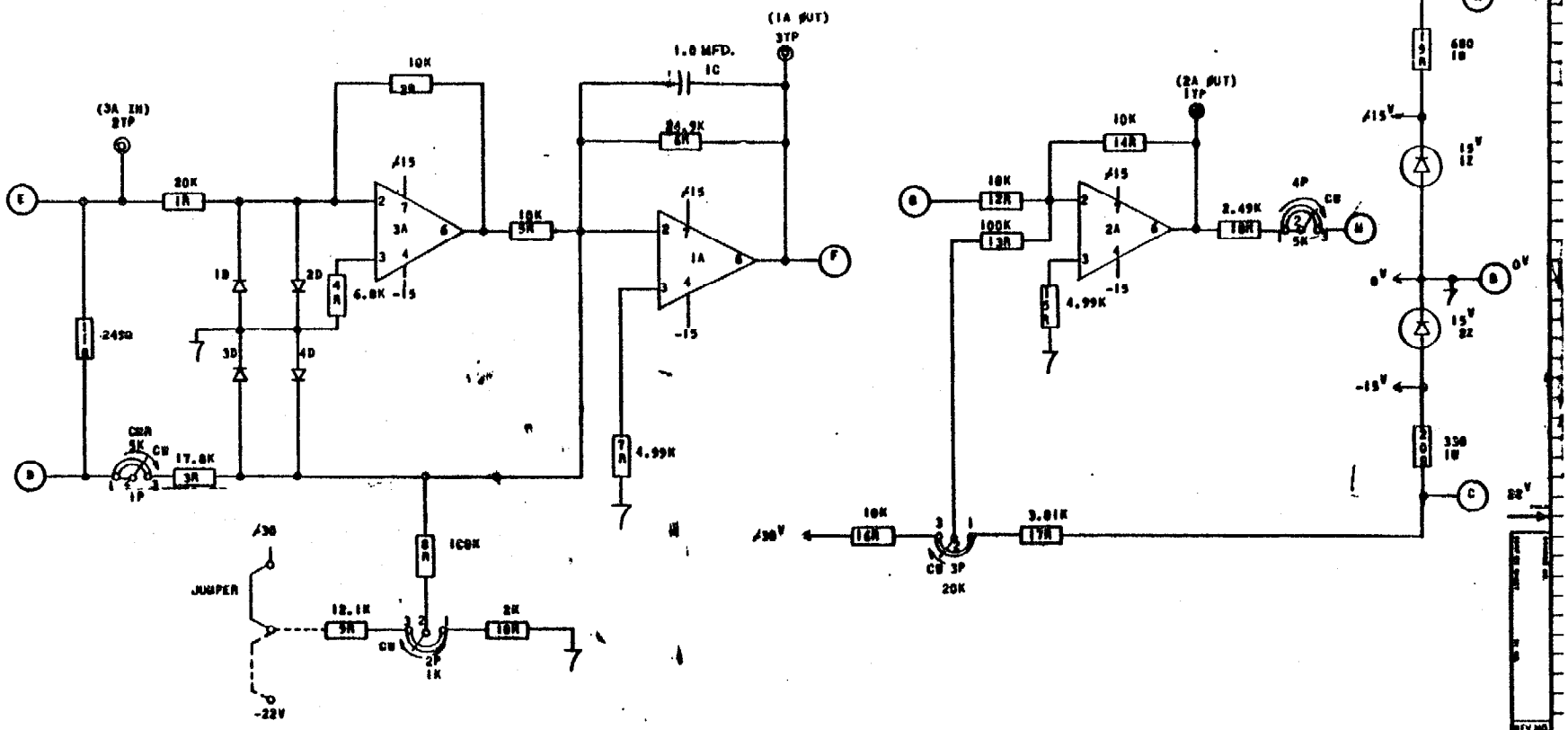
PRINT TO

D. Troyer Oct. 6, 1972

GENERAL ELECTRIC 44C306492

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—
APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE
7BCI BOILER CONTROL INTERFACE
4-20 MA INPUT
FIRST MADE FOR 44B331746-G03



NOTES:

1. 2TP INPUT

LSS	HSS
4MA	20MA

3TP OUTPUT

LSS	HSS
0V	5V

2. CONNECTIONS

A /30 VDC
B 0V
C -22 VDC
D BOILER CONTROL COMMON
E INPUT FROM BOILER CONTROL
F OUTPUT
G EHC AUTO SPEED REFERENCE
H AUTO REF POSITION TO BOILER CONTROL

REVISIONS | PRINTS TO

Drawn	11-22-72	GENERAL ELECTRIC COMPANY WATERVLIET, N.Y.	44C306492
Designer	June 7, 1973		

GENERAL ELECTRIC

44C306526

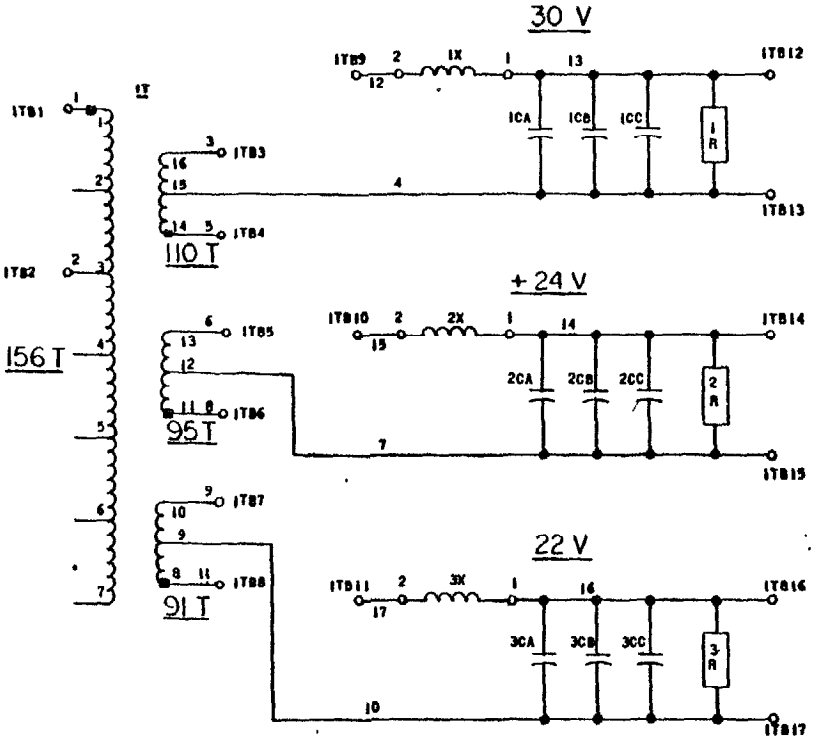
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

TITLE

LINE POWER SUPPLY

FIRST MADE FOR 3S7513KF300A5



CKT. NO. WIRE SEQUENCE, WIRE M9729494P1918 #18 WH.

1	ITB(1),IT(1)
2	ITB(2),IT(3)
3	ITB(3),IT(16)
4	ICC(1),ICB(1),ICA(1),IT(15),1R(1),ITB(13)
5	IT(14),ITB(4)
6	IT(13),ITB(5)
7	2CC(1),2CB(1),2CA(1),2R(1),IT(12),ITB(15)
8	IT(11),ITB(6)
9	IT(10),ITB(7)
10	3CC(1),3CB(1),3CA(1),3R(1),IT(9),ITB(17)
11	IT(8),ITB(8)
13	ICC(2),ICB(2),ICA(2),1R(2),ITB(12)
14	2CC(2),2CB(2),2CA(2),2R(2),ITB(14)
16	3CC(2),3CB(2),3CA(2),3R(2),ITB(16)

1X

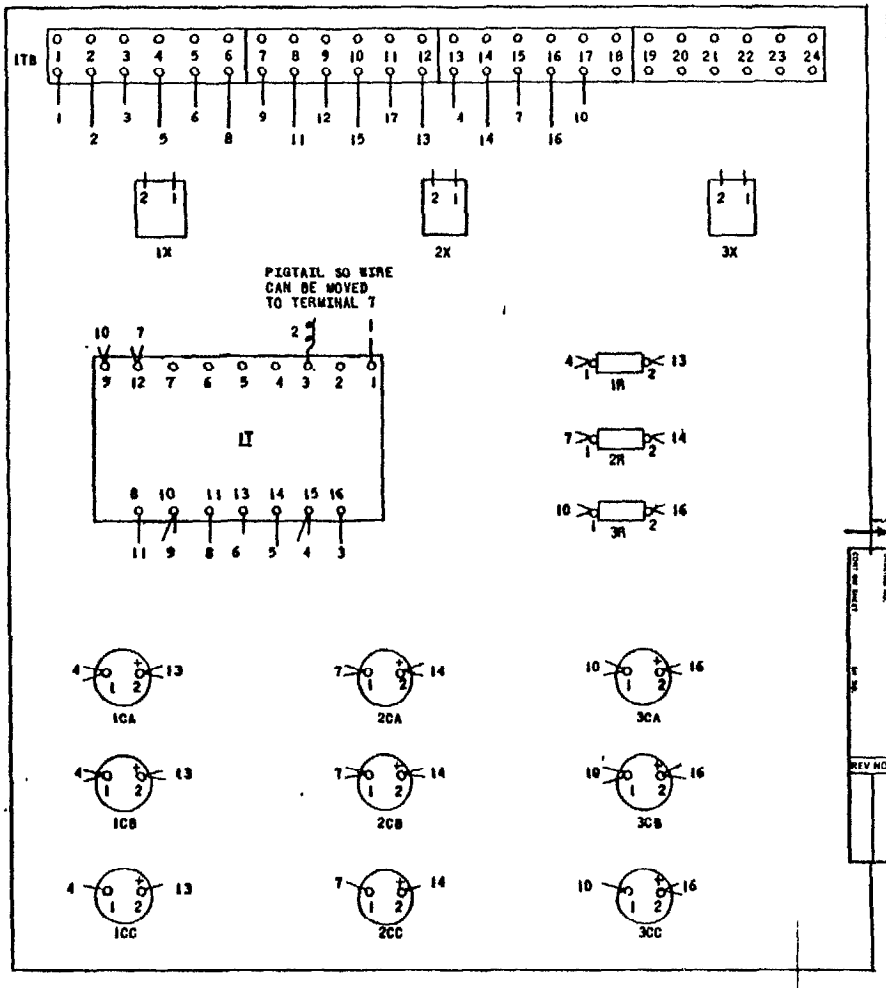
CKT.	LEAD NO.	WIRE TO
13	1	ITB(12)
12	2	ITB(9)

2X

CKT.	LEAD NO.	WIRE TO
14	1	ITB(14)
15	2	ITB(10)

3X

CKT.	LEAD NO.	WIRE TO
16	1	ITB(16)
17	2	ITB(11)



REVISIONS

1	E.A.	Nov. 3, 1971
2	E.A. Collins	May 10, 1972
3	W.H. Chapman	May 30, 1972

PRINTS TO

REV NO

DRAWN

Tompkins 10-18-72

ISSUED

OCT 20 1972

CHECKED

R. Culbertson

DRIVE SYSTEMS PROD. DEPT. WAYNESBORO, VA.

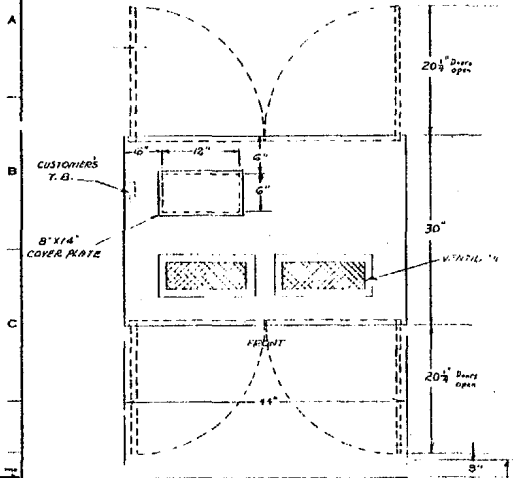
44C306526

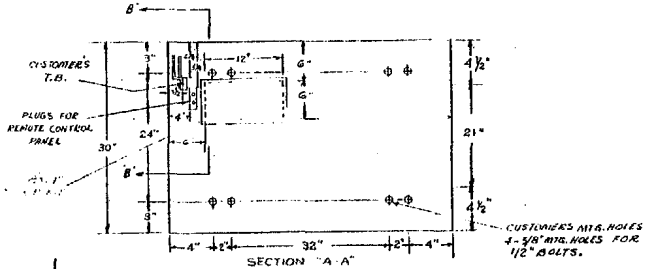
GENERAL ELECTRIC 44D300043

44D300043

ELECTRONIC FEEDPUMP TURBINE CONTROL

CABINET OUTLINE

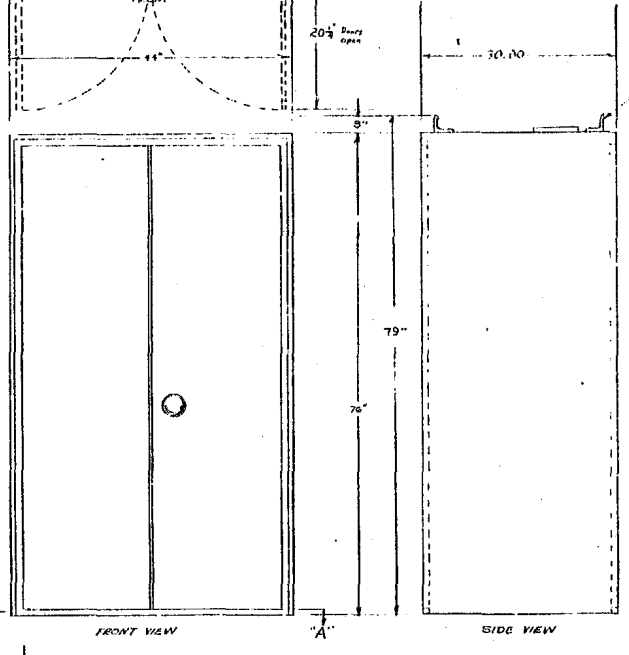


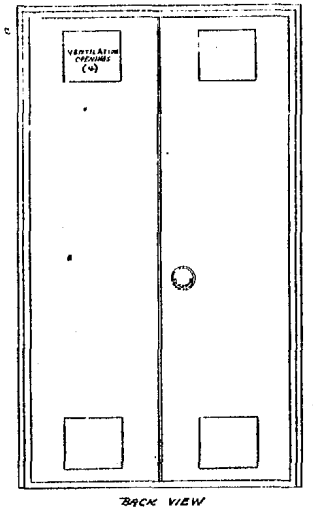


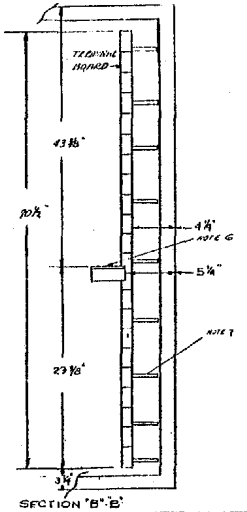
SECTION "A-A"
FRONT OF CASE

NOTES:

1. ONE CABINET REQUIRED FOR EACH FOURTH FEED PUMP TURBINE.
2. ESTIMATED WT. 1500 LBS.
3. MAXIMUM AMBIENT TEMP. 40°C
4. FINISH OUTSIDE: ASA 49 GRAY
   INSIDE : GLOSS WHITE LACQUER
   DOORS : BLUE HAMMERTONE LACQUER
5. CABINET HAS LOOSE PIN REMOVABLE HINGED DOORS.
6. DISCONNECT SWITCH-VENT/FUSE CONTROL CABLE ENTRANCE EITHER FROM TOP ☐ OR BOTTOM ☐ AS CUSTOMER DESIRES. (CUSTOMER TO CHECK ONE BLOCK.)
7. CUSTOMER'S PROVISIONS FOR STRAPPING CABLE RUN NEXT TO CUSTOMER'S TERMINAL BOARDS.
8. IF NON-STANDARD FINISH REQUIRED SPECIFY ON ORDERING DRAWING.
9. POWER CONSUMPTION/DISSIPATION 750W.





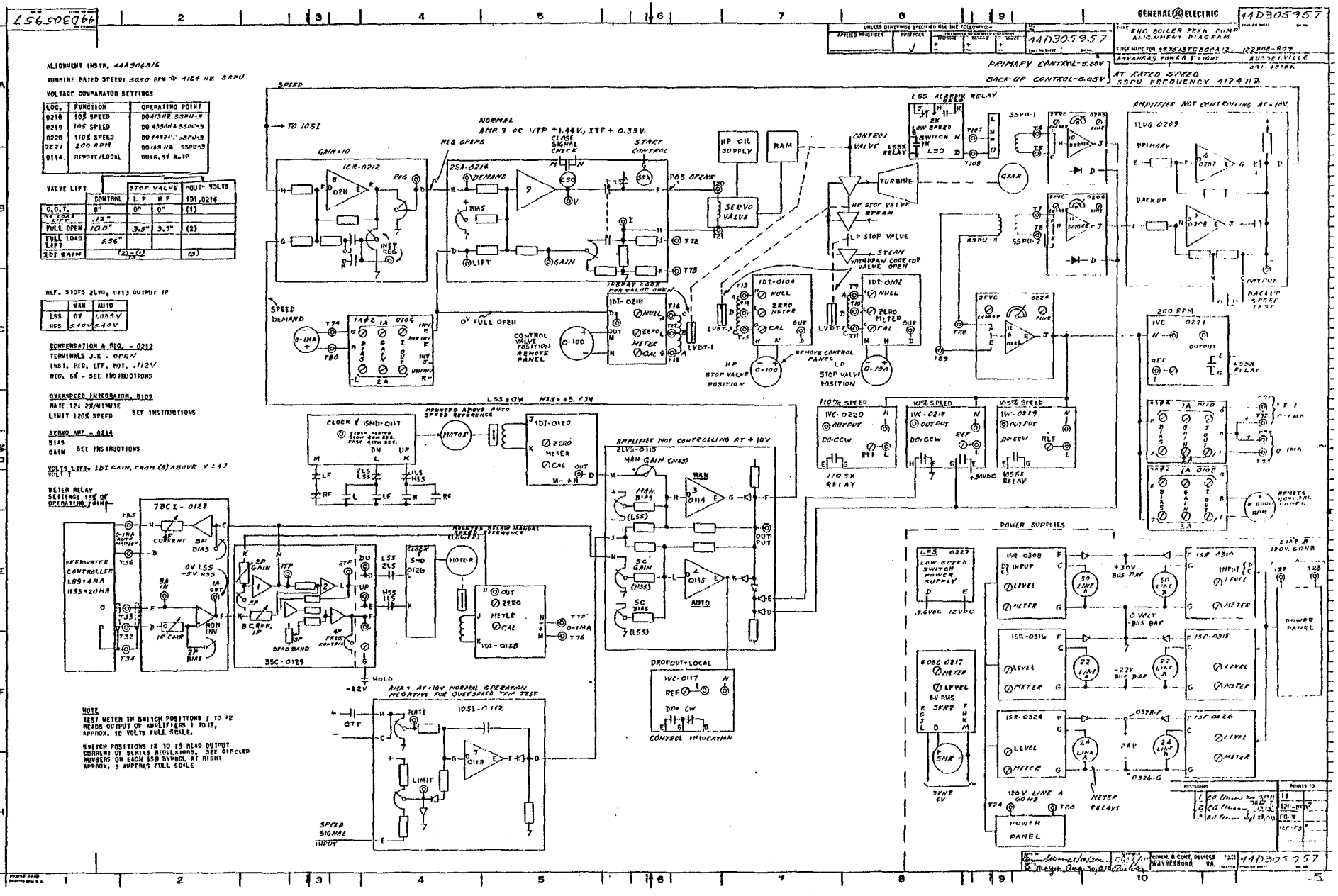


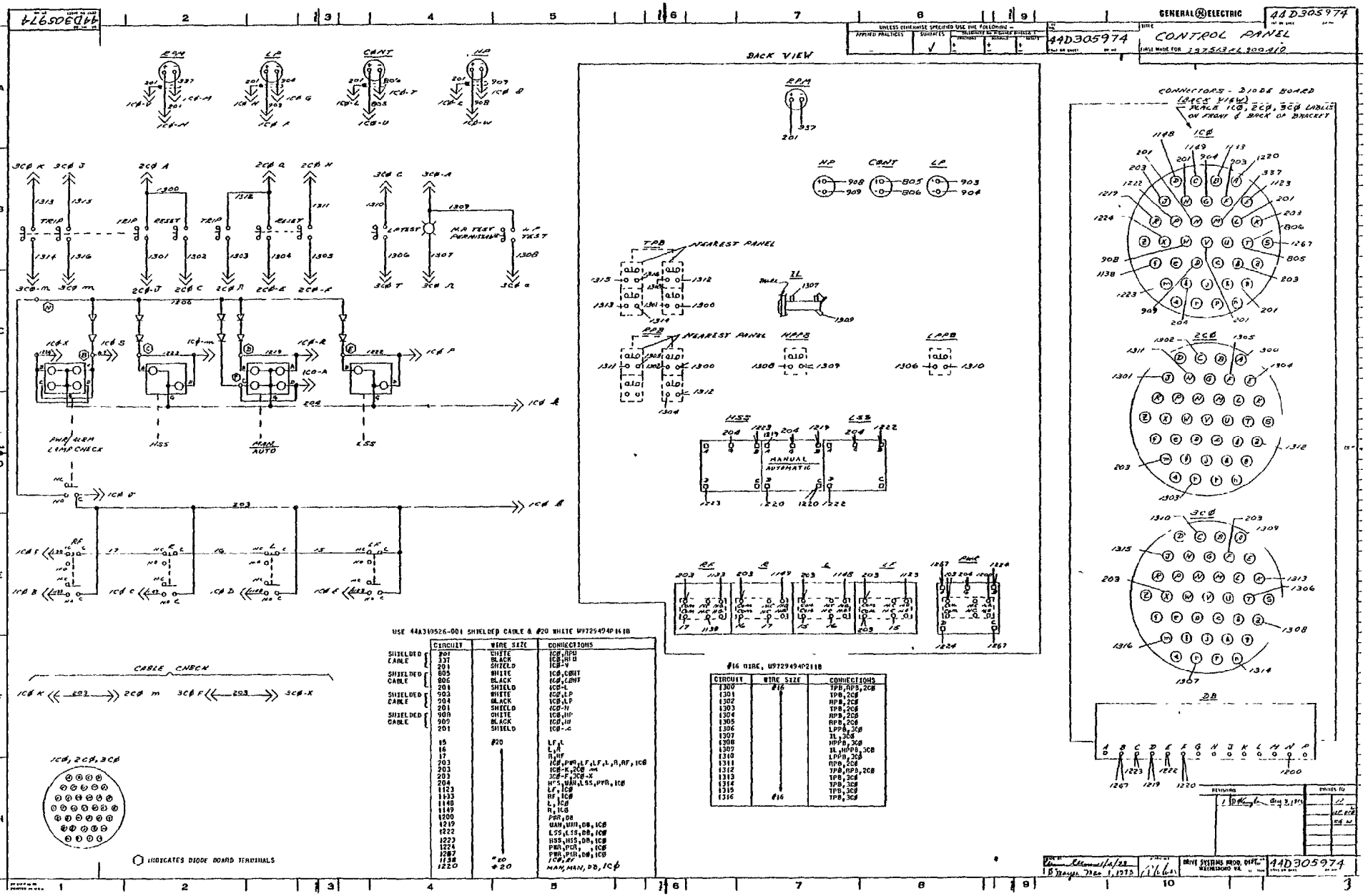
SECTION "B-B"

REVISIONS			PRINTS TO
1	8-1-67	KW	11
2	10-8-68	KW	12-12-8
3	11-21-68	KW	12-12-8
4	12-24-68	KW	12-12-8
5	2-10-69	KW	12-12-8
6	2-24-69	KW	12-12-8
7	7-8-69	KW	12-12-8

SPECIALTY CONTROL WAYNESBORO, VA.

44D300043


GENERAL ELECTRIC
44D305957
ENG. BOILER FEED PUMP ALIGNMENT DIAGRAM
ARKANSAS POWER & LIGHT
RUSSELLVILLE
PRIMARY CONTROL-5.00V
BACK-UP CONTROL-5.05V
AT RATED SPEED
5SPU FREQUENCY 4129 HZ
ALIGNMENT INSTR, 44A306316
TURBINE RATED SPEED: 5050 RPM @ 4129 HZ 5SPU
VOLTAGE COMPARATOR SETTINGS
LOC. FUNCTION OPERATING POINT
0218 105 SPEED
0219 105 SPEED
0220 110% SPEED
0221 200 RPM
0114 REMOTE/LOCAL
VALVE LIFT
STOP VALVE "OUT" VOLTS
CONTROL L P H P 1DI,0216
O.O.T. NO LOAD SETT.
FULL OPEN
FULL LOAD LIFT
1DI GAIN
REF. 530FS 2LV0, 0113 OUTPUT IP
MAN AUTO
LSS
HSS
COMPENSATION A REG. - 0212
TERMINALS J-K - OPEN
INST. REG. EFF. ROT. .112V
REG. 6% - SEE INSTRUCTIONS
OVERSPEED INTEGRATOR, 0102
RATE 12% PER MINUTE
LIMIT 120% SPEED
SEE INSTRUCTIONS
SERVO AMP. - 0214
BIAS
GAIN
SEE INSTRUCTIONS
METER RELAY SETTINGS 105% OF OPERATING POINT
NOTE
TEST METER IN SWITCH POSITIONS 1 TO 12 READS OUTPUT OF AMPLIFIERS 1 TO 12, APPROX. 10 VOLTS FULL SCALE.
SWITCH POSITIONS 12 TO 18 READ OUTPUT CURRENT OF SERIES REGULATORS. SEE CIRCLED NUMBERS ON EACH ISR SYMBOL AT RIGHT APPROX. 5 AMPERES FULL SCALE
SPEED
TO 1051
NORMAL AMP 9 OR VTP +1.44V, ITP + 0.35V.
NEG OPENS
CLOSE SIGNAL CHECK
START CONTROL
GAIN=10
ICR-0212
2SA-0214
DEMAND
BIAS
LIFT
GAIN
POS. OPENS
HP OIL SUPPLY
RAM
SERVO VALVE
CONTROL VALVE
TURBINE
GEAR
HP STOP VALVE STEAM
LP STOP VALVE
STEAM WITHDRAWN CORE FOR VALVE OPEN
LSSX ALARM RELAY
LSS
SSPU-1
SSPU-2
SSPU-3
AMPLIFIER NOT CONTROLLING AT +1V.
1LV6 0209
PRIMARY
BACK UP
OUTPUT
BACKUP SPEED TEST
SPEED DEMAND
0-1MA
TRIPPED 0202X FOR VALVE OPEN
0V FULL OPEN
CONTROL VALVE POSITION REMOTE PANEL
0-100
1DI-0216
1DI-0104
1DI-0102
NULL
ZERO METER
CAL
LVDT-1
LVDT-2
LVDT-3
HP STOP VALVE POSITION
REMOTE CONTROL PANEL
LP STOP VALVE POSITION
3FVC 0224
200 RPM
1VC 0221
OUTPUT
REF
LSSX RELAY
110% SPEED
105% SPEED
1VC-0220
1VC-0218
1VC-0219
DO:CCW
110 SX RELAY
105SX RELAY
+30VDC
LSS=0V HSS=+5.43V
CLOCK & 1SMD-0117
MOTOR
MOUNTED ABOVE AUTO SPEED REFERENCE
1DI-0120
ZERO METER CAL
DN UP
LF RF
ZLS LSS
LLS HSS
AMPLIFIER NOT CONTROLLING AT +10V
2LV6-0115
MAN GAIN CNES
MAN BIAS
SC GAIN
SC BIAS
AUTO
OUTPUT
7BCX-0122
FEEDWATER CONTROLLER
LSS=4MA HSS=20MA
IC CHR
B.C.REF. IP
DEAD BAND
35C-0125
MOUNTED BELOW MANUAL SPEED REFERENCE
SMD 0126
1DI-0128
0-1MA
-22V
HOLD
AMP= AT+10V NORMAL OPERATION NEGATIVE FOR OVERSPEED "TRIP TEST"
1051-0112
RATE
LIMIT
OTT
SPEED SIGNAL INPUT
DROPOUT+LOCAL
1VC-0117
DO: CW
CONTROL INDICATION
POWER SUPPLIES
LPS 0227
LOW SPEED SWITCH POWER SUPPLY
-5.6VDC 12VDC
6SC-0217
6V BUS
SMR2
SENS 6V
ISR-0308
ISR-0316
ISR-0324
ISR-0310
ISR-0318
ISR-0326
+30V BUS BAR
0 VOLT BUS BAR
-22V BUS BAR
24V
0328-F
0326-G
120V LINE A GONE
METER RELAYS
POWER PANEL
LINE A 120V, 60HZ
POWER PANEL
SCHEN. & CONT. DEVICES WAYNESBORO, VA
44D305957



CIRCUIT	WIRE SIZE	CONNECTIONS
201	SHIELD	1CØ-C
15	#20	LF,L
16		L,R
17		R,RF
203		1CØ,PWR,LF,LF,L,R,RF,1CØ
203		1CØ-K,2CØ-m
203		3CØ-F,3CØ-X
204		HSS,MAN,LSS,PWR,1CØ
1123		LF,1CØ
1133		RF,1CØ
1148		L,1CØ
1149		R,1CØ
1200		PBR,DB
1219		MAN,MAN,DB,1CØ
1222		LSS,LSS,DB,1CØ
1223		HSS,HSS,DB,1CØ
1224		PWR,PWR,1CØ
1267		PWR,PWR,DB,1CØ
1138	*10	1CØ,RF
1220	*20	MAN,MAN,DB,1CØ

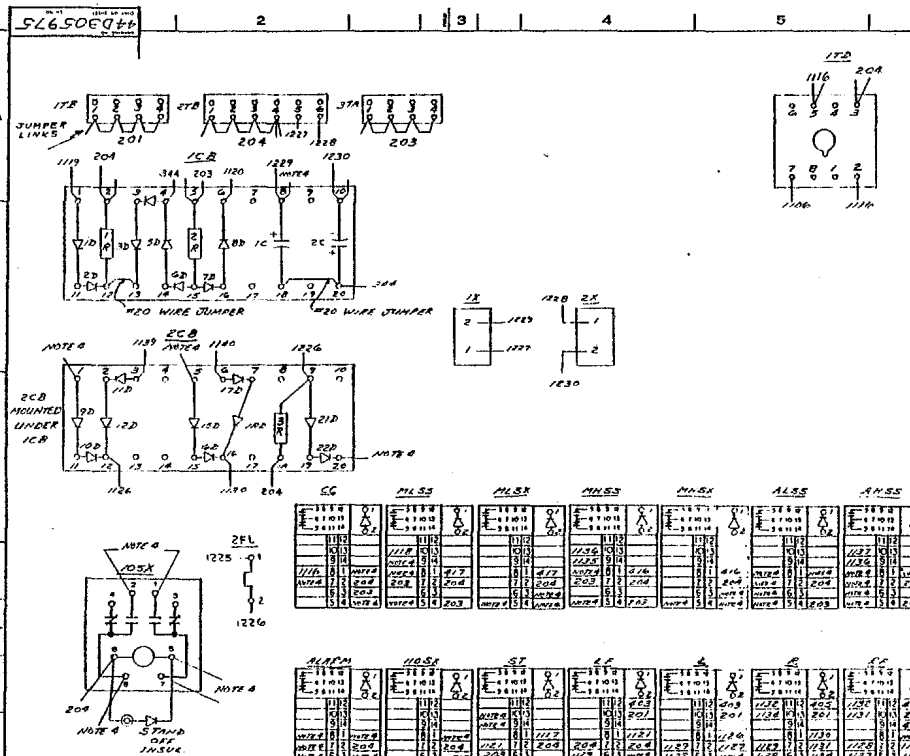
CIRCUIT	WIRE SIZE	CONNECTIONS
1300	#16	TPB,RPB,2CØ
1301		TPB,2CØ
1302		RPB,2CØ
1303		TPB,2CØ
1304		RPB,2CØ
1305		RPB,2CØ
1306		LPPB,3CØ
1307		IL,3CØ
1308		HPPB,3CØ
1309		IL,HPPB,3CØ
1310		LPPB,3CØ
1311		RPB,2CØ
1312		TPB,RPB,2CØ
1313		TPB,3CØ
1314		TPB,3CØ
1315		TPB,3CØ
1316	#16	TPB,3CØ

GENERAL ELECTRIC — 44D305975

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —						
APPLIED FRACTIONS ±	CONTACTS	ANGLES ±	DECIMALS ±			
	√	±	±	±	44D305975	

TITLE: RELAY PANEL

FIRST MADE FOR 345 73/19 AT LOU 2413



WIRE TABLE

CIRCUIT	WIRE SEQUENCE
201	RF,R,L,LF,1TDX,1TB
203	CC,MLSS,MLSX,MHSS,MHSX,ALSS,AHSS,ST, ALARM,P,2F,N,AX,TRIP, FCR,3TB
204	105X,CC,MLSS,MLSX,MHSS,MHSX,ALSS,AHSS, HOLD,OTT,LF,LF,ST,1TDX,ALARM,N,1TDX, AX,LSS ALARMX,LSSX,1CR,1TD,2TB
204	2CR-16, 2TB
344	1CR-4,1CR-20
402	LF,1LF
403	L,LF
405	R,RF
416	MHSS,MHSX
417	MLSS,MLSX
1106	1TD,1TDX
1107	TRIP,ST
1116	1TD,1TB,CC
1117	ST,ST,1TDX
1118	1TDX,ALSS
1119	1CR,P
1120	N,1CR
1121	RF,LF,ST
1124	L,R
1125	R,RF
1126	L,2CR
1127	L,RF
1128	RF,R
1129	R,L,LF
1130	R,2CR
1131	R,RF
1132	RF,R,L
1134	RF,R
1135	TFSD,AHSS
1136	AHSS,AHSS
1137	AHSS,OTT
1138	TFSD,TFSR,TFSL
1139	TFSL,2CR
1140	TFSR,2CR
1225	ALARM,2FL
1226	2FL,2CR
1230	TRIP,TFSD,TFSR,TFSL,1CR
1248	LSSX,1TDX

REACTOR WIRE TABLE

CIRCUIT	CONNECTIONS
1227	1X, 2TB
1228	2X, 2TB
1229	1X, 1CR
1230	2X, 1CR

FACTORY NOTES:
1. USE #20 WIRE, MP22949AP16AR
2. DO NOT CORD-TIE WIRES, INTERCONNECTING WIRES TO BE ADDED AFTER PANEL IS MOUNTED IN CASE ASM.
3. CONNECT DIODES TO EACH RELAY SOCKET AS SHOWN IN RELAY SYMBOL.
4. WIRE CONNECTION SHOWN ON MASTER WIRE TABLE.

REVISIONS	SHEET NO.
LTR D Design Chge 3-1-73	11
	11206
	LGW

DRAWN 4/4/73	CHECKED	DWNY SYSTEMS PROD. DEPT. WASHINGTON, PA.	44D305275
D. Design Chge 3-1-73			

GENERAL ELECTRIC CO.
WAYNESBORO, VA.

REGULATOR SUMMARY 44X303674

COMPILED BY: RJ Culver **DATE:** Sept 1, 1972 **SHEETS; SHEET** 1

CUSTOMER: ARKANSAS POWER & LIGHT (MDTD) Russelville Nuclear #1 & #2

TURBINES: 122808 - 809 REQN.: 091-40786

ITEM	MATERIAL	QUAN.
1	3S7513TC300A12 Electronic Feedpump Turbine Control	2
	Outline Drawing 44D300043	
	Interconnection Dwg. 44A306315	
	Elementary Dwg. 44C306404	
	Connection Dwg. 44C306405	
	Alignment Dwg. 44D305957	
	Alignment Instructions 44A306316	
	Renewal Parts 44A335416	
2	3S7513KL300A10 Control Panel	2
	Outline 44B306127	
3	Cabinet-Control Panel Power Cable	4
	XXXXXXXX 44B306148	
4	Cabinet-Control Panel Control Cable	2
	44B300423	

REVISIONS: Oct 3, 1972

DISTRIBUTION	
11	
12P DFB[2]	
FG-W	
12P-FS[2]	

5